PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

D Speedway: Automotive Repair

A Smart Inventory and Parts Planning System

In Partial Fulfillment of the Requirements

for Capstone 2 and Research Subject

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First Semester, 2026–2027

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CHAPTER I

INTRODUCTION

1.1 Background of the Study

D Speedway was a local automotive repair shop that had been serving vehicle owners in the community for a year. The shop handled everything from routine maintenance and oil changes to parts replacement and engine diagnostics. The owner and staff genuinely took pride in the quality of service they provided, and customers kept coming back because of it.

It was the kind of shop where a customer could walk in not knowing what was wrong with their car and walk out understanding what had been fixed and why. That level of transparency and honesty was not something every repair shop offered, and people in the community had noticed. Word spreads fast in a neighborhood, and when a shop treats its customers right consistently, its reputation builds on its own. D Speedway had earned that kind of standing in just one year, which showed how seriously the team took the work.

The range of services mattered too. Vehicle owners did not have to go to one place for an oil change and another for a more complex repair, because D Speedway handled it all under one roof. That convenience, together with the trust the shop had built, was a big reason the customer base kept growing after it opened. More customers meant more vehicles coming in, more parts being used, and more transactions every day. That growth was good for the business, but it also began to put pressure on the way the shop managed things behind the scenes.

The problem was not the quality of their work. It was how they kept track of everything behind the scenes. All of the shop's sales records and inventory were managed through a collection of Excel files. On the surface this might have sounded manageable, but in practice it caused real headaches every single day. When a part was sold, someone had to remember to update the spreadsheet. When two staff members used the same file, their entries sometimes conflicted. When the owner wanted to know how much revenue had come in the previous month, it took hours of manually sorting through rows of data to find the answer.

Spreadsheets only work as well as the person maintaining them. They do not update on their own, they do not catch mistakes, and they do not stop two people from typing into the same cell at the same time and ending up with corrupted data. Every entry depended on a person remembering to make it, making it correctly, and making it at the right time. In a shop where the main priority was fixing cars and serving customers, that kind of consistent and precise administrative work was always going to be the first thing to slip. This was not because the staff was irresponsible, but because it was genuinely hard to keep up that level of discipline in the middle of a busy workday.

There was also the issue of how many files were involved. It was not just one spreadsheet that tracked everything but a collection of them. Parts inventory lived in one file, sales records in another, and other

inventory records were scattered across different files. Because those files were maintained separately and updated by different people at different times, they started to drift apart. The numbers in one file stopped matching the numbers in another. Figuring out the actual state of things meant opening multiple files, cross-referencing them, and reconciling discrepancies that should never have existed in the first place. For a shop that was supposed to be focused on repairs, spending time on that kind of detective work over spreadsheet data was a real problem.

What made this harder was that the automotive parts market did not sit still. Demand for certain parts rose at certain times of the year, parts prices changed, and new vehicle models shifted what customers needed. A business that could only look at its data after the fact, rather than in real time, was always one step behind. For a shop like D Speedway that depended on having the right parts ready when mechanics needed them, being one step behind carried a real cost.

Consider what it meant in practice when demand for a certain part spiked and the shop did not see it coming. A customer would bring in a vehicle that needed a part the shop usually kept in stock. But demand had been higher than usual over the past few weeks, nobody had noticed in time to reorder, and the part was gone. The customer had to wait. The shop had to obtain the needed part as quickly as possible, often paying more for the urgency. The mechanic who was supposed to be working on that vehicle had to move on to something else or sit idle. All of that disruption traced back to one thing: not having current, reliable data that could have flagged the trend before it became a problem.

Changes in parts costs added another dimension to this. When prices went up, the shop needed to know quickly so it could adjust what it charged customers or decide whether to stock up before the increase took effect. When prices dropped, that was an opportunity to order more and improve the margin. But the person making those decisions was working from spreadsheet data that was days or weeks out of date, so those windows closed before anyone realized they had been open. The shop ended up making decisions based on old information, which was only slightly better than making no decision at all. [1], [5]

This project developed a Smart Inventory and Parts Planning System to address these issues directly. Rather than replacing the way D Speedway operated, the system was designed to give the shop the tools to operate the way it already did, only faster, more accurately, and with much less manual effort.

That distinction was worth holding onto. Many software solutions for small businesses assume that the business has to change how it operates to fit the software, with new workflows, new processes, and new ways of doing things that staff must learn from scratch. That approach tended to create more problems than it solved, especially in a shop where people were already good at their jobs and had little time to spare for learning a completely new system. The goal here was the opposite: to build something that fit around the way the team already worked, not something that asked them to work differently.

In practice, that meant a system that ran in the background and handled the data side of things automatically so staff could stay focused on what they were there to do. When a part was sold, the inventory count updated on its

own. When stock dropped below a certain level, an alert went out so a reorder could be placed before the shelves ran empty. When the owner wanted to see how the business was doing, the numbers were already there, organized, current, and ready to look at without any manual compiling. The technology handled the administrative work, and the team handled the automotive work. That was the division of labor this system was designed to create.

To understand why this mattered, it helped to think about what a typical day at the shop looked like. A customer would bring in a vehicle, and a mechanic would check it and identify the parts needed. Someone then had to find out whether those parts were in stock, which meant either physically checking the storage area or opening one of the Excel files and hoping the number there was still accurate. If the part was available, that was fine. If it was not, the customer had to wait and the shop had to scramble. That kind of situation happened more often than it should have, and it all traced back to not having a reliable, up to date record of what was on hand.

That moment, when someone had to stop what they were doing and go find out whether a part was actually available, was where a lot of time quietly got lost. It did not seem like much on its own, just a few minutes here and a short walk there. But when it happened multiple times a day across multiple staff members, it added up in a way that was hard to ignore. The shop was paying for that time even if nobody had ever sat down and calculated what it cost. Beyond the time, there was also the uncertainty. Even after checking, there was no guarantee that the number in the spreadsheet or the count on the shelf matched reality, because both depended on records that might not have been fully up to date.

When a customer had to wait because a part was not in stock, the effect went beyond that one visit. It changed the experience they had with the shop. Most people were understanding the first time it happened, but if it became a pattern, they started to wonder whether they should go somewhere else. Customer loyalty was not unconditional. It was built through positive experiences and worn down by frustrating ones. D Speedway had worked hard to earn the loyalty it had, and it would have been a shame to see that eroded by an inventory problem that was entirely solvable.

There was also the scramble itself to consider, with its urgent efforts to obtain needed parts, rush orders, and reshuffling of work to cover a vehicle that could not yet be finished. That kind of reactive mode was stressful for staff and expensive for the business. Rush orders almost always cost more than planned orders, and the time spent sorting out the situation was time that could have gone toward other jobs. The whole thing could have been avoided if the shop had known, even a day or two earlier, that the part was running low and needed to be reordered.

It was also worth pointing out that D Speedway had been open for only a year. The problems it was experiencing were only going to get worse as the shop got busier and the number of parts it carried grew. A system that barely held together under the current workload was not going to handle a larger one. Addressing this early, while the shop was still relatively small, was much smarter than waiting until the problems became impossible to manage.

A year in business was still early. The habits and systems a business builds in its first couple of years tend to stay for a long time, for better or worse. If D Speedway kept running on spreadsheets as it grew, those spreadsheets would become more complicated, more fragile, and more expensive to maintain. More parts to track meant more rows, more files, and more chances for something to go wrong. More staff meant more people touching the same data, which meant more conflicts and inconsistencies. The problems that were annoying at that point would become serious later, and serious problems are a lot harder to fix than annoying ones.

There was a version of this story in which the shop waited until things broke down badly enough to force a change. That version would have been more expensive, more disruptive, and more stressful than dealing with the problem early. Data that had been entered incorrectly over the years would need to be cleaned up. Habits that formed around a broken system would need to be unlearned. Staff would need to be trained on a new system in the middle of a busy operation rather than at a manageable pace. None of that was impossible, but all of it was harder than building the right foundation early, while the shop was still at a scale where change was straightforward.

The idea behind this project was simple. Instead of relying on staff to remember to update a spreadsheet after every transaction, the system was built to handle that automatically. Instead of the owner having to compile data from different files to get a clear picture of the business, the system was designed to present that information whenever it was needed. The shop kept doing what it did, and the system took care of the record keeping in the background.

Underneath all the technical details, this project was about giving D Speedway a fair shot at growing into what it had the potential to become. The shop already had the most important thing: a team that cared about doing good work and a customer base that trusted them because of it. That foundation was solid. What it needed was an operational layer that matched the quality of what happened on the shop floor, a system as reliable as the mechanics, as consistent as the service, and built to hold up as the business kept moving forward.

A Smart Inventory and Parts Planning System would not change the soul of D Speedway. The owner would still care about every job. The mechanics would still take pride in their work. The customers would still come back because they trusted the shop. But the stress of not knowing what was in stock, the hours lost to manual data entry, the scramble when something ran out unexpectedly, and the frustration of not getting a clear read on the business without spending half a day on it would all go away. What would be left was a shop that was not just good at what it did, but well equipped to keep doing it for a long time.

1.2 Statement of the Problem

D Speedway was a local automotive repair shop that had served its community for one year. In that short time, the shop had built something most businesses take years to build: a loyal group of customers and a reputation that spoke for itself. The team handled everything from routine oil changes and preventive maintenance to full parts replacements and complex engine diagnostics. Customers came back because the work was done well, and

also because the staff took the time to explain what was wrong, what was done, and why. That kind of openness was rare in the automotive industry, and people in the community had noticed it.

The shop grew steadily during its first year, and almost all of that growth came from word of mouth. More customers meant more vehicles coming in, more parts being used, and more transactions every day. This showed that the business was doing the right things on the shop floor. The trouble was that the systems behind the scenes had not grown along with it. The quality of service stayed high, but the administrative side of the business had started to show cracks, and with each passing month those cracks were harder to overlook.

All sales records and inventory were kept in manually maintained Excel spreadsheets. There was no single platform, no automated tracking, and nothing that connected what happened at the counter to what was on the shelves. Every update depended on a staff member remembering to make it. Every report needed someone to pull data together from several files by hand. Every stock check meant either opening a spreadsheet that might or might not be current, or walking to the storage area and counting parts one by one. For a business growing as fast as D Speedway, this setup could no longer be sustained. [2], [4], [6]

The problems in this section did not come from assumptions about how small businesses usually run. They came from direct contact with the people at D Speedway. The research team visited the shop, spent time watching how daily work was done, and talked with the owner and with the staff who dealt with these issues regularly. What came out of those visits was consistent and specific. Three connected problems were affecting how efficiently the shop worked, how reliable its records were, and how good an experience it could give its customers. Each one is described below.

● **Inefficient Sales and Inventory Tracking.** The most visible problem at D Speedway was that no single system existed for recording sales and monitoring inventory. The shop used several separate Excel files, and each one was maintained on its own for a different purpose. Parts inventory was in one file, sales records were in another, and purchase history was spread across other documents. None of these files were linked. A sale entered in the sales file did not change anything in the inventory file. Someone had to do that second step by hand, usually while handling other tasks at the same time.

Because of this, even routine reporting took far more effort than it should have. Suppose the owner wanted to know how many units of a certain part sold in the past month. Answering that meant opening several files, copying out the figures, adjusting for differences between the files, and checking the result against other records so nothing was counted twice. Staff said this could use up a whole afternoon. The question itself was simple. The way the records were kept made it hard. A business should be able to answer basic questions about its own sales in seconds, not hours.

Time was not the only cost. The setup also meant the business had no single record it could rely on as the final word. When two files showed slightly different numbers for the same item, nobody could say which one was right without looking into it further. The owner could not open one screen and see where the business stood. Decisions about buying stock, scheduling staff, or choosing which services to focus on were made without clean,

combined, easy to reach data. That is a handicap for any business, and a costly one for a shop that was growing quickly and had to make these decisions more often as it got bigger.

● **Inaccurate and Unreliable Records.** The second problem came straight out of the first, and in many ways it was the more serious one. Every entry depended on a person updating a spreadsheet at the right time, in the right file, with the right information, so the records could only be as reliable as the conditions in which people made those entries. In a busy shop where the main job was fixing vehicles and serving customers, keeping up careful paperwork at all times was very hard. The staff were not careless. The work environment made it difficult to be consistent.

The mistakes that resulted were varied, but they kept happening. A part would be used on a repair, and the inventory count would not be updated until the end of the day, or sometimes never. A sale would be written in the sales file, and the matching entry in the inventory file would be missed because two things needed attention at once. A part would be counted twice because it was logged when it arrived and again during a stock check. Each of these errors was small. Over weeks and months, though, they piled up, and the figures in the spreadsheets stopped matching what was really on the shelves.

What happened after that made the problem worse. Once staff saw that the records could not be fully trusted, they stopped treating them as their main source. To find out if a part was in stock, a staff member walked to the storage area and counted it. To judge how sales were going, the owner relied on memory and general impressions instead of a report. The records were supposed to cut down on guesswork, but the team ended up working around them. Any report or purchasing decision based on those files carried some hidden error, and the people using them did not always know how large it was. That uncertainty built up quietly and weakened every decision that followed.

● **Absence of Real Time Inventory Visibility and Proactive Stock Alerts.** The third problem was the one customers felt most directly, even though they never saw the cause. D Speedway had no way to watch inventory levels in real time and nothing that warned staff when a part was running low. To know the status of an item, a person had to check a spreadsheet that could be out of date or go to the storage area and look. Neither method was dependable, and neither helped the shop plan ahead as it grew.

The effects were already visible on the shop floor, and they touched both operations and customer experience. Mechanics sometimes started a repair and found out halfway through that the part they needed was gone. The whole job had to stop. The customer had to wait while the shop got the missing part, often on short notice, which usually meant paying more for faster delivery. In the meantime, the mechanic either sat idle or moved to another vehicle, which threw off the shop’s workflow and delayed other customers.

Customers felt this frustration, and it added up. Most people could accept one unexpected delay. If it happened more than once, or if a customer heard it had happened to someone they knew, their opinion of the shop began to change. Loyalty was not unconditional. Good experiences built it, and bad ones wore it down. D Speedway had put a lot of effort into earning its customers’ trust, and shortages that could have been avoided kept putting

that trust at risk. There was a financial side as well. Every job put on hold because of a shortage was a job not bringing in revenue, and every emergency order paid at a higher price was money that planned buying would have saved.

What stood out about this problem was how easily it could have been prevented. An automatic alert that told staff when a part's stock fell below a set level would have given the shop several days of warning before a shortage caused trouble. The reorder could have been placed during a normal purchasing cycle, at the regular price, with enough time for the part to arrive before anyone needed it. The repair would go on without interruption, the customer would leave on time, and the mechanic would keep working. None of this called for complicated technology. It called for a system built around how the shop actually worked, instead of one that left important information sitting in files nobody checked regularly.

How These Problems Were Connected

It was important to understand each problem on its own, but it was just as important to see how they were related. That relationship explained why fixing only one of them would not have made a real difference at D Speedway. These were not three separate issues that happened to exist in the same business. They were three stages of one problem that kept growing. It started with how data was entered, moved on to how reliable that data became over time, and ended with how much useful information the business could get out of it.

The tracking problem was where everything began. When sales and inventory were handled in disconnected spreadsheets that people updated by hand, there was room for error at every transaction. Nothing caught a missed entry. Nothing checked whether a recorded sale matched the current stock. Nothing warned anyone when two staff members entered conflicting information about the same item. Each update was done by hand, and each one could go wrong. Over time, some of them did. This was not because anyone was careless. A busy shop simply made perfect record keeping an unrealistic thing to expect.

The errors from the tracking problem then led directly to the accuracy problem. Small differences grew into bigger ones. Files that had been fairly current slowly drifted away from reality. Staff who used to trust the records began to doubt them, and in the end they stopped relying on them. The system that was meant to cut down guesswork ended up adding to it. Reports built from those files could contain mistakes the reader would not notice, and buying decisions based on those reports may have assumed stock levels that did not match the shelves. The whole administrative side of the business became unreliable, and that affected every decision the owner made.

The visibility problem followed from the accuracy problem and completed the cycle. Even if the shop had wanted real time stock monitoring, the data behind it was not reliable enough to act on. An alert based on wrong records could warn about a shortage that did not exist, or, worse, miss one that did. Visibility is only useful when the data behind it can be trusted. As long as manual entry errors and scattered files affected the records, no amount of monitoring would have given dependable results.

This chain was the reason the project dealt with all three problems together. Centralizing and automating data entry fixed the tracking problem and made accurate records possible. Accurate records made real time monitoring meaningful instead of misleading. Meaningful monitoring gave the owner and staff the visibility to plan ahead, react early to changes in stock, and base purchases on current, verified information instead of old figures and educated guesses. Each part of the solution supported the others, and together they covered everything D Speedway was dealing with, which no single fix could have done alone.

Why This Needed to Be Addressed Then

One thing to keep in mind throughout is that D Speedway had been open for only one year. These problems were already present and already causing disruption at the shop's current size. That alone called for urgency, because problems that were visible at that stage would only get worse as the business grew. More customers meant more transactions, and more transactions meant more data to track. More data meant more chances for the manual system to fail and more time spent maintaining it, even if it was maintained poorly. A system that was already struggling with the current workload could not handle a heavier one.

Habits were another concern. The systems a business uses in its early years tend to become part of everyday routine, both in the tools themselves and in how the people using them think about their work. If D Speedway kept building on spreadsheets as it grew, those spreadsheets would get more complicated, harder to maintain, and more tightly tied into how the team worked. At some point, replacing them would cost much more than it would have earlier, counting data migration, retraining of staff, and disruption to daily operations. The longer the shop waited, the harder the change would be.

The other option was to leave things alone until the problems got bad enough to force a crisis. That would have cost much more and caused far more disruption than acting early. Data entered wrongly over months or years would have to be checked and corrected. Workflows built around a broken system would have to be redesigned from scratch. Staff would have to learn a new system in the middle of normal operations, without the room to adjust that early adoption allowed. That would be hard and expensive in any situation, and harder still with the daily demands of running a repair shop.

Dealing with these problems while D Speedway was still young, and while the change was still small enough to manage, was the sensible choice both in practical terms and for the long run. The shop already had the most important things: a team that took pride in its work, customers who trusted it, and a reputation built on honest, quality service. What it lacked was an operational base as solid as the service it offered. A Smart Inventory and Parts Planning System would not change what made customers want to come back to D Speedway. It would give the business the tools to keep and build on everything it had already worked to earn.

1.3 Objectives of the Study

1.3.1 General Objective

The goal of this project was to design and develop a Smart Inventory and Parts Planning System for D Speedway that would bring order to how the shop handled its sales data and stock levels. The system was meant to make it easier for staff to record transactions, give the owner a real time view of what was happening in the business, and take the guesswork out of deciding when to reorder parts. In simple terms, the shop needed one dependable place where its sales and inventory information could live, and this system was built to be that place.

This goal came from the actual needs of the shop, not just from general good practice. The features that were built responded directly to the problems described in Section 1.2, which were the scattered Excel files, the records that could not be trusted, and the lack of any warning before a part ran out. Every decision made during development, whether about what to include, how it should work, or how it should look, went back to one question: did it make things easier for the people at D Speedway? If a feature did not help the owner or the staff in their daily work, there was no reason to include it.

It was also important that staff could use the system without a lot of training. Not everyone at the shop was comfortable with technology, and a system that needed technical knowledge to operate would not have helped anyone. The people at D Speedway were mechanics and service staff, and their time was better spent on vehicles and customers than on learning complicated software. The design had to be simple and clear enough that someone using it for the first time could figure out what to do without step by step instructions.

For the owner, the main benefit was being able to see what was happening in the business at any moment. Getting that picture used to take a lot of manual effort, such as opening several files, comparing numbers, and sorting out differences. With this system, the information was simply there when it was needed, including current stock levels, recent sales, and which parts were running low. That kind of visibility made it much easier to run the business well, because the owner could make decisions from what was actually happening and not from memory or rough guesses.

1.3.2 Specific Objectives

To achieve this, the study:

● Built a module for recording and tracking sales on a daily, monthly, and yearly basis, replacing the old multi file Excel setup with one centralized tool.

● Implemented inventory monitoring that updated stock levels automatically after every transaction and sent alerts when parts fell below a set minimum.

● Integrated a forecasting feature that looked at past sales patterns to predict which parts the shop would likely need in the weeks ahead.

● Developed a reorder recommendation feature that flagged low stock parts and helped the owner and manager make timely restocking decisions.

● Tested the finished system with D Speedway staff and evaluated it against criteria for usability, accuracy, and performance.

The first objective, the sales tracking module, was the foundation of the whole system, since everything else depended on whether sales data was recorded accurately and consistently. D Speedway had relied on several Excel files that were updated separately, which made tracking sales harder than it needed to be. Some records were kept in one file while other details sat somewhere else, and because updates were done by hand, there was always a chance that something was missed, entered wrongly, or updated late. Each of those small slips made the records a little less reliable than they were the day before.

The centralized sales tracking module simplified this by putting all transaction records into one system. Staff no longer had to move between several spreadsheets just to record a sale or look at past transactions, because everything was stored on a single platform that was easier to reach. This also meant that two staff members could work in the same system without overwriting each other's entries, which had been a real problem with shared files. Daily, monthly, and yearly sales reports were generated automatically, so the owner no longer had to spend hours sorting and compiling data to see how the business was doing.

This objective mattered a great deal because sales data affected nearly every other part of the system. If transactions were not recorded correctly, inventory counts became inaccurate, reports could not be trusted, and forecasting results became less useful. The system could not make good inventory decisions if the sales information going into it was incomplete or wrong. A forecast built on bad numbers would only produce bad suggestions. That was why getting this module right was treated as one of the most important parts of the whole project.

The module also improved visibility. Once sales records were in one place, the owner could see which automotive parts sold often, which services were most common, and which times of the month or year brought in the most business. Before, this kind of information was buried in separate files and mostly lived in the owner's memory. Having it organized made decisions easier, since the owner no longer had to depend only on recollection or rough estimates when planning purchases, staffing, or services.

The second objective, automatic inventory monitoring, dealt with one of the shop's biggest operational problems, which was the lack of reliable, current stock information. Inventory tracking had depended heavily on manual updates and on physically checking parts. In a busy repair shop where mechanics and staff were focused on repairs and customer service, updating spreadsheets accurately after every transaction was hard to keep up, so stock counts often became unreliable over time. The longer this went on, the wider the gap grew between what the records said and what was actually on the shelves.

The system took away much of that manual work by updating inventory records automatically whenever a transaction was completed. Every time a part was sold or used in a repair, the system adjusted the stock count

right away without any extra encoding from staff. Nobody had to remember a second step after recording a sale, because the system handled it. This gave a more accurate, real time view of inventory levels and reduced the gaps between recorded stock and the actual stock on the shelves.

This objective was important for more than record accuracy. It also improved how work flowed through the shop. Mechanics and staff no longer had to stop what they were doing just to check physically whether a part was available. They could look at the system for the latest stock information instead. Those few minutes saved each time, repeated across many checks in a day and across several people, meant fewer interruptions and more time for actual repair work. It let the shop run more smoothly from opening to closing.

Another major feature under this objective was the low stock alert, which notified staff whenever the quantity of a part reached a minimum level set by the shop. This helped prevent parts from running out in the middle of a repair. Instead of finding a shortage only after a customer's vehicle was already being serviced, the shop could spot low inventory earlier and reorder before the part became unavailable. Because the minimum level was set by the shop, it could reflect how quickly each part was actually used.

This approach was much more practical than the old reactive process. In the past, shortages caused unnecessary stress because staff only learned a part was missing once it was urgently needed, and by then the only choices were to make the customer wait or to pay extra for a rush order. With early warnings, the shop could plan ahead, avoid those delays, and keep operations running more smoothly.

The third objective was to add a forecasting feature to the system, meant to help D Speedway prepare for future inventory needs by looking at earlier sales and inventory movement. Instead of depending entirely on guesswork or personal judgment, the system used historical data to estimate which automotive parts were likely to be needed more often in the coming weeks or months. The idea was to give the owner a starting point based on real numbers.

Forecasting mattered because demand in automotive repair shops was rarely steady. Some parts were needed more depending on the season, the types of vehicles commonly serviced, or repair trends in the community. Without a reliable way to see these patterns, businesses often reacted to shortages after they had already happened instead of preparing ahead. A pattern that is easy to miss day by day can become clear once the sales history is laid out over several weeks.

The forecasting feature gave the shop a more organized way to plan its inventory purchases. By reviewing past sales records, the system showed which items moved quickly, which parts were used steadily, and which products might need restocking soon. This allowed the owner to base purchasing decisions on actual trends and not on assumptions alone. It also meant that the owner did not have to dig through old records to see how a part had been selling, since that history was already organized by the system.

Forecasting could not guarantee perfect predictions, but it still offered a real advantage over having no planning support at all. Even a reasonable estimate based on past sales patterns helped reduce stock shortages, improve

preparation, and make purchasing more efficient. Instead of always reacting, the shop could begin making more proactive decisions about its inventory.

The forecasting feature also supported better financial planning. Overstocking parts that were not needed tied up money that could have been used elsewhere in the business, while understocking important items caused delays and lost opportunities. Both mistakes cost the shop something. By helping the shop understand how its inventory moved, forecasting improved both day to day operations and the practical use of business resources.

The fourth objective was to create a reorder recommendation tool that made restocking easier. Low stock alerts told staff that inventory levels were becoming critical, and the reorder recommendation feature went a step further by helping decide what to do next. An alert alone tells you there is a problem. The recommendation helped answer what to do about it.

When the system detected that a part was running low, it suggested reorder quantities based on earlier purchasing history and inventory movement. This reduced the need for staff to go through old records by hand just to estimate how much to buy. Instead of relying on memory or rough calculations, the owner could make decisions using organized, up to date information produced by the system. The suggestion was only a guide, and the owner or manager still made the final call.

The system displayed a reorder list showing all parts that had reached or fallen below their minimum stock level, sorted by urgency so the most critical shortages appeared first. This let the owner or manager see at a glance which parts needed attention right away and which could wait a little longer. From this list, the owner or manager could immediately record a restock transaction for any flagged item. This updated the inventory ledger and cleared the low stock status once enough stock had been added.

This objective was especially useful for cutting down delays during restocking. In many small businesses, shortages created stressful situations where employees had to review inventory needs quickly and decide how much to buy in a short time. Rushed decisions like that were more likely to lead to mistakes. A system that already offered reorder suggestions simplified that process and helped the shop respond faster whenever stock needed replacing.

The reorder recommendation tool also supported better inventory control overall. Instead of ordering parts at random or purely by estimate, the shop could make more consistent purchasing decisions using actual inventory movement data. This helped reduce both overstocking and understocking and made inventory planning better organized from one order to the next.

Finally, the fifth objective was to test and evaluate the completed system using actual D Speedway staff. This stage was just as important as development itself, because a system could not be called successful if it worked technically but was hard or confusing for people to use under real working conditions. A program that runs without errors on a developer’s computer can still fail in a busy shop if the people using it find it slow or unclear.

The testing looked at the system in terms of usability, accuracy, functionality, and overall performance. Staff used the system during their normal tasks so the researchers could see whether the features were easy to understand, responsive, and practical in the real shop setting. The researchers watched how staff handled the system and listened to their comments. That feedback helped show whether the system really addressed the operational problems found in the study.

Usability mattered especially because not all employees were experienced with technology or computerized systems. The system had to stay simple enough for staff to use with confidence and without extensive technical training. A complicated system could have added frustration instead of solving existing problems, which was why user experience was treated as a major part of the evaluation.

Accuracy testing was also done to make sure that inventory updates, stock alerts, forecasting outputs, and sales reports worked correctly. Since the system was meant to replace manual spreadsheet tracking, the reliability of the information it produced was critical. If the numbers could not be trusted, staff would go back to counting parts by hand, and the system would lose its purpose. The shop needed to be able to rely on the records the system gave it for the system to become a practical daily tool.

Performance evaluation looked further at how efficiently the system handled transactions, retrieved records, and processed inventory updates. Since automotive repair shops move at a fast pace, the system had to stay responsive and handle the demands of a normal day without causing delays. A screen that took too long to load would have been a problem when a customer was waiting at the counter.

In the end, the testing phase confirmed whether the proposed Smart Inventory and Parts Planning System was not only technically functional but also realistic, reliable, and suited to the actual needs of D Speedway. The goal of the project was not simply to create software. It was to build a practical system that genuinely improved how the shop managed inventory, sales, and planning in everyday situations.

1.4 Scope and Delimitation

This study focused on the design and development of a Smart Inventory and Parts Planning System for D Speedway, a local automotive repair shop that had managed its sales and inventory records through manually maintained Excel spreadsheets. The project addressed the operational difficulties the shop experienced in tracking sales, monitoring stock levels, planning future inventory needs, and organizing purchasing decisions. The system was developed specifically around the actual workflow and daily operations observed at D Speedway, with the goal of making inventory management more accurate, organized, and easier to handle without adding unnecessary complexity to staff's work.

The study covered the development of a centralized system combining sales recording, inventory monitoring, forecasting, and reorder planning into a single platform. Instead of maintaining separate files for different records, the system stored and managed information in one connected database, so updates made in one part of

the system were automatically reflected throughout the rest of it. This removed the need for repetitive manual updating and reduced the risk of inconsistencies between records.

One of the main areas covered by the study was sales tracking. The system allowed manager to record daily transactions involving automotive parts and services performed at the shop. These records were then organized automatically into daily, monthly, and yearly summaries the owner could access whenever needed, eliminating the previous process of manually reviewing and compiling spreadsheet data just to generate basic business reports. Through the system, the owner could monitor recent sales activity, identify frequently purchased parts, and review the overall performance of the business more efficiently.

The study also included inventory monitoring and stock management. Every time a part was sold or used during a repair, the system automatically updated the inventory count in real time, reducing reliance on physical stock checking and preventing situations where records no longer matched the actual quantity of parts on hand. The system also included low-stock notifications that alerted staff whenever a part reached a predefined minimum quantity, giving the shop enough time to reorder needed parts before they ran out completely.

Another major scope of the study was the forecasting and reorder planning feature. The system analyzed previous sales data to identify usage patterns for automotive parts commonly used by the shop, and based on these patterns, generated simple forecasts estimating which parts might be needed in the coming weeks or months. The forecasting process was not intended to provide perfect predictions, but rather to help the owner make more informed purchasing decisions using available historical data instead of relying entirely on memory or estimation.

Alongside forecasting, the study also covered reorder recommendations. When a part reached a low-stock level, the system not only issued an alert but also provided suggested reorder quantities based on previous purchasing activity and stock movement. Historical inventory and purchasing records could also be referenced during this process to helped the owner and manager identify which parts needed to be restocked based on current stock levels in the past, simplifying the restocking process and reducing the time spent manually reviewing records before placing orders.

The study further included the testing and evaluation of the developed system using criteria such as usability, accuracy, functionality, and performance. Actual staff members from D Speedway participated in the testing process to determine whether the system was understandable, practical, and suitable for real shop operations. Feedback gathered during testing helped identify areas that needed adjustments or improvements before final implementation.

Although the study covered several important operational areas, there were also clear limitations to what the system was designed to do. The project focused only on inventory management, sales recording, parts forecasting, and reorder planning within D Speedway. It did not cover every possible aspect of running an automotive repair business.

The system did not include advanced accounting functions such as payroll processing, tax computation, full financial auditing, or complete bookkeeping services. While the system generated sales summaries and inventory-related reports, it was not intended to replace professional accounting software or the services of an

accountant. Financial management beyond sales and inventory monitoring remained outside the scope of the study.

The project was also limited to the operations of a single repair shop branch. The system was designed specifically for the workflow, size, and operational structure of D Speedway as observed during the researchers visits and interviews, and was not intended to function as a large-scale enterprise system capable of managing multiple branches or large-scale inventory operations.

In addition, the forecasting feature was based only on historical sales records entered into the database. The system did not analyze external market factors such as economic conditions, inventory shortages, seasonal industry disruptions, or changes in automotive trends outside the available data. As a result, forecasting results were understood as decision-support recommendations rather than guaranteed predictions.

The study was likewise limited by the accuracy of the data entered into the system. Although automation reduced many of the problems associated with manual spreadsheets, the system still depended on staff correctly recording transactions and inventory information. If inaccurate or incomplete data was entered, the reports, forecasts, and stock monitoring generated by the system could also become inaccurate. The system improved record management, but it could not completely eliminate human error.

The project also did not include online customer booking, mobile application development, online payment processing, or direct integration with external ordering platforms.. The focus of the study was strictly on improving the shop's internal inventory and sales management operations rather than developing a full customer management or e-commerce system.

Another limitation of the study was the time frame available for development and testing. The evaluation of the system was conducted within the period allowed for the completion of the research project, so long-term operational testing across several years of business activity was not included. The study instead evaluated whether the system performed properly under the conditions and sample data available during the testing phase.

Despite these limitations, the researchers believed the proposed Smart Inventory and Parts Planning System could still provide significant improvements to the way D Speedway handled its daily operations. The study centered on solving the specific problems affecting the shop, particularly the inefficiency of spreadsheet-based tracking, the unreliability of manually updated records, and the absence of real-time inventory visibility. By focusing directly on these operational concerns, the project aimed to provide D Speedway with a more organized, dependable, and practical foundation that could support the continued growth of the business in the years ahead.

1.5 Significance of the Study

The people and groups who stood to benefit from this project included:

- **D Speedway (Business Owner and Management):** The business owner and management of D Speedway Auto Repair Shop were the primary beneficiaries of the proposed Smart Inventory and Parts Planning System. The shop had relied heavily on manually maintained Excel spreadsheets to track sales, inventory, and stock

movement. That method may have worked for basic monitoring, but it became harder to keep accurate as daily transactions continued to grow. What was manageable in the first few months of operation was no longer enough once the shop became busier.

Because inventory updates depended entirely on manual encoding, records could easily become outdated, inconsistent, or incomplete, especially during busy hours when staff were focused on customers and repairs. A single missed entry was enough to throw off the count for a part, and nobody would know about it until someone needed that part. By then, the mistake had already caused a delay.

With the proposed system in place, the owner gained a centralized platform that recorded sales, monitored inventory, and generated reports in real time. Instead of spending hours going through spreadsheets and verifying stock counts, management could see updated information right away. This gave a clearer and more reliable picture of how the shop was actually running, including which parts were selling quickly and which ones sat unused for long periods. The owner no longer had to guess which items deserved attention.

The system also improved decision making within the business. Since inventory records and sales data were stored accurately and consistently, the owner could base purchasing decisions on actual inventory movement and not on assumptions or incomplete information. This lowered the risk of buying too much stock that would sit on the shelves. It also reduced the chance that an important automotive part would suddenly be unavailable in the middle of a repair, which was the kind of situation that cost the shop both time and money.

In the long term, having organized and reliable business data was worth more than solving the current inventory problems. As D Speedway grew, the system could serve as a base for better operational planning, budgeting, forecasting, and business expansion. Accurate records made it easier to spot business trends, watch seasonal demand, and judge how the shop was performing over time. This kind of information had existed before only in scattered spreadsheets that were hard to organize and analyze. The proposed system turned that raw data into useful information that management could use to support future growth and improvement.

Another important benefit for management was less administrative work. Repetitive tasks such as updating inventory files by hand, cross checking records, and preparing reports had taken up a good amount of time that could have gone to running the business. By automating these tasks, the system let the owner put more effort into improving services, handling customers, and expanding the business, and less into correcting inventory mismatches. Time that used to go into paperwork could now go into decisions that moved the shop forward.

● **Shop Staff:** The employees and staff of D Speedway also benefited greatly from the proposed system. Many daily tasks had depended on manual processes, such as recording sales transactions, updating inventory counts, and checking stock availability. Each of these tasks may have seemed manageable on its own. But in a busy repair shop where many repairs and transactions happened throughout the day, even small delays and repeated paperwork slowly became tiring and frustrating for employees.

The Smart Inventory and Parts Planning System reduced these daily difficulties by simplifying how inventory was managed. Recording transactions became faster and better organized because staff no longer had to switch

between several spreadsheets or update separate files after every sale or repair. Once a transaction was recorded, inventory levels updated automatically within the system, which lowered the chance of forgotten entries or mismatched stock counts. Staff could finish the task and move on to the next one without carrying a second step in their heads.

This automation cut down on human error and eased the pressure on employees to keep double checking their records. Staff no longer had to spend unnecessary time looking for the correct spreadsheet, confirming whether another employee had already updated a file, or walking to the storage area just to check if a part was there. These small inconveniences looked minor one by one, but together they had used up a large part of the workday and slowed the whole operation.

The system also improved coordination among employees. Since everyone used the same centralized database, inventory information became more consistent and easier to monitor. Employees could quickly check stock availability, watch low stock items, and see which parts were needed without relying on word of mouth or handwritten notes. This made the workplace more organized and let tasks get done faster and with fewer interruptions. Everyone was looking at the same information, so there was less room for confusion about who had updated what.

Reducing repetitive administrative work also let staff focus on their actual responsibilities, especially customer service and vehicle repairs. Mechanics and service personnel could concentrate on finishing repairs efficiently instead of worrying about whether the inventory records were correct. This improved productivity and also eased the stress that came from disorganized inventory and missing stock information. A mechanic who could trust the system did not have to stop in the middle of a job to double check the shelf.

Another important benefit was that the system was designed to be user friendly and practical for everyday use. It was not meant to complicate operations or require advanced technical skills from employees. It was built around the actual workflow of the repair shop so staff could adjust to it with little difficulty. This made it more likely that the system would be adopted and used consistently in the business, since people tend to keep using tools that make their work easier and drop the ones that make it harder.

● **Vehicle Owners (Customers):** Customers may never have used the proposed system directly, but they were still among the most important beneficiaries, because better inventory management directly affected the quality and speed of the service they received. In automotive repair, customer satisfaction depended heavily on how quickly repairs were finished and whether the needed parts were on hand during servicing.

Before, inventory inaccuracies and delayed updates sometimes meant that mechanics found a required part was unavailable only after a repair had already started. These situations delayed repairs, interrupted the workflow, and made customers wait longer than expected while replacement parts were obtained. In some cases, customers even had to return on another day because the needed parts were not in stock. For someone who had planned their day around getting the vehicle back, this was a real inconvenience.

With real time inventory monitoring and automated stock updates, the proposed system helped reduce these disruptions significantly. Mechanics and staff could confirm stock availability before repairs began, which made unexpected shortages during service less likely. Low stock alerts and reorder recommendations also helped keep commonly used automotive parts available when they were needed, so the shop was better prepared before the customer even arrived.

As a result, customers benefited from faster service, more accurate repair scheduling, and fewer delays caused by inventory problems. Repairs became better organized because the shop could prepare the needed parts in advance instead of reacting to shortages after they happened. This made the overall experience smoother for customers and made the shop more dependable. When a shop could give a realistic estimate and then meet it, customers had more reason to trust it.

Customers may not have noticed the technical system working behind the scenes, and that was perfectly fine. What they noticed was that service was more consistent. They were less likely to face delays, cancellations, or interruptions caused by missing parts. Over time, this reliability built customer trust and made it more likely that customers would keep coming back for future services.

In a competitive automotive service industry, customer satisfaction and reputation were extremely important. A repair shop known for finishing work efficiently and on time was more likely to keep loyal customers and attract new ones through good recommendations. For this reason, better inventory management indirectly supported the long term growth and reputation of the business in the local community. In a neighborhood where word spreads quickly, every smooth visit helped the shop's name, and every avoidable delay could hurt it.

● **Future Researchers:** The proposed study also served as a useful reference for future researchers, particularly students and developers interested in inventory management systems, automotive business operations, and business process automation. Many small businesses, especially local automotive shops, kept relying on manual or spreadsheet based systems despite the problems these methods caused. Because of this, there was still a strong need for studies on practical and affordable technology solutions made for small and medium sized businesses.

This project offered a real example of how inventory management problems could be studied and addressed in an actual automotive repair shop. Unlike theoretical projects that dealt only with imagined scenarios, this study was based on the real operational experiences of D Speedway Auto Repair Shop. The challenges it identified, including delayed updates, disconnected records, inconsistent inventory counts, and inefficient stock monitoring, were real problems that had affected the business. Researchers who read it could see where these problems came from and how the system answered each one.

Future researchers could use this study as a starting point for building similar systems for other industries, such as retail stores, hardware shops, motorcycle repair businesses, or other service businesses that also struggled with inventory management. The study also helped researchers better understand why centralized data management, automation, forecasting, and real time inventory monitoring mattered in small business operations.

These ideas were not limited to one kind of shop, so the lessons could carry over to other settings with similar needs.

The project also added useful information about system usability and practical implementation. Many research studies focused heavily on technical development and gave less attention to whether the system was actually suited for everyday business use. This study showed why it was important to design systems that matched real operational workflows and stayed understandable for users with limited technical knowledge. Future developers could use these insights to create systems that were not only technically functional but also practical, user friendly, and sustainable in real business settings.

The study could also serve as a reference for students doing research on forecasting systems, reorder planning, sales tracking, inventory automation, and decision support systems. Since the project combined several business management functions into one platform, future researchers could look for ways to improve or expand these features based on the needs of other businesses or industries. Each of those functions could be studied further on its own, or in a different setting from the one covered here.

Overall, this study contributed both academically and practically by showing how technology driven inventory management systems could improve operational efficiency, reduce human error, strengthen decision making, and support business growth in small automotive repair shops.

1.6 Definition of Terms

Automotive Repair Shop: An automotive repair shop was a business that provided maintenance and repair services for motor vehicles. Its services usually included routine tasks such as oil changes, tire rotations, and battery replacements, as well as bigger jobs like engine diagnostics, brake repairs, and transmission work. In this study, the automotive repair shop was the main setting for the proposed system. The shop depended on a steady supply of spare parts and consumables to do its daily work. Managing those supplies well was at the center of the problem this study addressed.

Inventory Tracking: Inventory tracking was the process of monitoring and recording the quantity, location, and movement of parts and supplies in the shop at any given time. It included updating stock counts whenever items were added, used during repairs, or found to be damaged or expired. Accurate tracking helped the shop avoid running out of needed parts, kept it from buying more than it needed, and supported better planning for money. In this system, inventory tracking was one of the core functions. It let staff see what was currently in stock, and it let managers decide what to buy based on real data and not on guesses.

Parts Forecasting: Parts forecasting was the practice of using past transaction data and usage patterns to predict which parts would be needed in the coming days, weeks, or months, and in what amounts. Instead of waiting for a part to run out and then reacting, the shop could plan ahead and make sure high demand items were on hand before they ran out. This reduced delays in service and helped the shop avoid both having too little and having too much stock. The forecasting function in this system used historical sales records to produce estimates that guided future purchasing decisions.

Real Time Monitoring: Real time monitoring was the ability of a system to show and update information as events happened, without any delay and without anyone entering data later. In this study, it meant that whenever a transaction was recorded, whether a part was sold, used in a repair, or restocked, the system changed the inventory count and other related records right away. This gave shop owners and managers an accurate, current picture of the shop at all times. It was especially useful when quick decisions had to be made about restocking or about repairs that were already underway.

Reorder Tracking: Reorder tracking was a feature of the system that helped the shop keep watch over parts flagged for reordering. When a part's stock level fell below its set minimum, the system recorded this and let the manager start a reorder. The manager entered the quantity needed by hand, and the system flagged the low stock parts. Restocking took a single click, which updated the inventory right away. This feature replaced the habit of relying on memory or informal notes. It gave the shop a clear and organized record of what had been ordered, from whom, and when it was expected to arrive.

Prototype: In software development, a prototype was an early working version of a system, built to show its main features and test its basic abilities before it was fully put into use in a live setting. At the prototype stage, developers and stakeholders could check whether the system worked as intended, find problems or things to improve, and collect feedback before committing to full development. In this study, the developed system was presented as a prototype. This meant it was working and had been tested, but it had not yet been formally deployed in a real automotive repair shop.

Stockout: A stockout happened when a part or supply needed to complete a repair was completely unavailable at the shop. This usually forced mechanics to pause or postpone the job until the item could be obtained. The result could be delays in service, unhappy customers, and lost income for the business. Stockouts were often caused by poor inventory monitoring or by failing to reorder items in time. One of the main goals of this system was to reduce how often stockouts happened by keeping stock levels visible and sending timely alerts when supplies were running low.

Minimum Stock Threshold: The minimum stock threshold was a preset value that marked the lowest acceptable quantity of a part before the system raised a restocking alert. Each part in the inventory could have its own threshold, based on how fast it sold, how long it took to reorder, and how important it was to daily work. When the available stock of a part dropped to or below its threshold, the system flagged it so the manager could act. Setting suitable thresholds was an important part of running the shop well, because it gave an early warning before a stockout actually happened.

Centralized System: A centralized system was a digital platform where all data about the shop's operations was stored, managed, and accessed from one place. Instead of keeping separate spreadsheets, paper logs, or files for different parts of the business, such as inventory counts and sales records, everything was kept together in one unified system. This made it easier for authorized users to find information quickly, lowered the risk of data that did not match, and made sure everyone was looking at the same current data. For a small business like an automotive repair shop, a centralized system could cut down a lot of the time spent on administrative work.

Access Level: Access level was the set of permissions given to a user in the system. It decided what information the user could see and what actions the user could take. Different roles in the shop needed different levels of access. For example, a regular staff member might be allowed to record sales and search for parts but not to change prices, delete records, or open financial summaries. The shop owner or manager, on the other hand, usually had full access to all features and data. Clear access levels protected sensitive business information and made sure each user only worked with the parts of the system that matched their role.

Transaction: A transaction, in this system, was any recorded event that involved the movement of parts or the exchange of payment. This included sales to customers, parts used during a repair job, and restocking recorded by the owner or manager. Each transaction was logged with details such as the date, the item involved, the quantity, and the amount paid or received. These records were the basis for inventory updates, sales reports, and financial tracking. Keeping transaction records accurate was necessary to understand how the shop was doing and to make decisions based on reliable data and not on estimates.

Sales Report: A sales report was a summary generated by the system that gave an overview of all transactions in a chosen period, such as a single day, a calendar month, or a whole year. The report usually showed total revenue earned, the individual parts or services sold, the quantities moved, and any breakdowns by payment method or category. Sales reports helped the owner and manager see how the business was performing over time and which products were selling well. They also showed periods of high or low activity, which could guide decisions about staffing, purchasing, or promotions.

Dashboard Metrics: The following terms described the specific measures and visual indicators on the system's main dashboard. They were meant to give the shop owner and manager a quick but useful view of the business's current status and recent performance.

Daily Revenue, Parts vs. Labor: This metric showed a breakdown of the shop's total income for each day, split between money earned from parts sales and money earned from labor or service fees. With both figures side by side, the manager could quickly see which days brought in the most income and whether that income came mostly from parts or from services. This helped with scheduling and pricing, and it showed which kinds of work were most profitable for the shop from day to day.

Monthly Revenue Trend: The monthly revenue trend showed a running view of the shop's total earnings for each month of the year. It let the owner or manager see whether the business was growing, going through a seasonal slowdown, or holding steady. By comparing one month with another, management could notice patterns, such as certain months that were always slower, and use that to plan staffing, promotions, or inventory orders. It gave a wider view of the business's financial health than what happened in the current week alone.

Top 10 Products Sold: This metric ranked the ten best selling parts or products in a chosen period and showed which items customers wanted most. Knowing which products moved fastest gave the shop practical guidance on where to focus restocking and how to use its storage space. It also helped the owner and manager plan

inventory better by showing which popular items needed enough stock at all times. This metric turned raw sales data into information the shop could act on, which made it easier to keep the right stock at the right time.

Revenue Split, Parts vs. Labor: Similar to the daily revenue breakdown but viewed as a whole, the revenue split showed the overall share of income that came from parts sales compared with labor and services. It was usually displayed as a chart or a percentage, so the owner could see at a glance where the shop's money was coming from. If the split changed unexpectedly, for example if parts revenue dropped sharply while labor stayed the same, it could point to a change in customer behavior or to a problem with stock availability that needed a closer look.

Payment Method Breakdown: This metric showed how customers paid for parts and services each month, and it separated cash payments from digital payments made through online payment and credit card. This information was useful for accounting, and it also showed what customers preferred and whether the shop was ready to handle both kinds of payment smoothly. If one method was used much more than the other, the shop could adjust its processes. For example, it could make sure the QR codes for digital payments were always visible and easy to reach, or keep enough cash on hand for change.

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CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Sánchez-Luna, Rodríguez-Campos, and Rico-Chagollán [1], in their study titled "Technology Transfer in Inventory and Order Management in Automotive Parts Stores," published in the Journal Applied Computing / Revista de Cómputo Aplicado, looked at how technology based inventory and order management systems were used in automotive parts stores. The study focused on how technology could improve inventory control, order processing, and operational efficiency in automotive businesses that handled large quantities of parts and supplies every day. These were businesses where a small mistake in a stock count could affect many customers in a single day.

The researchers explained that inventory and order management were critical to automotive operations because these businesses depended heavily on having parts available to keep their services running. Where transactions were frequent and inventory moved quickly, businesses needed systems that could manage stock information accurately and efficiently. The study pointed out that even businesses already using digital tools could still run into inefficiencies when their inventory systems were not fully built into daily workflows, or when inventory data was not updated consistently in real time. Having a digital tool was not the same as having a system that worked well in practice.

According to the study, common inventory problems in automotive parts stores included differences between recorded and actual stock levels, delays in updating inventory information, poor monitoring of orders, and trouble processing transactions accurately. These problems hurt performance by causing stock shortages, delayed customer service, inaccurate reports, and unnecessary financial losses. Each problem also made the others worse. A delayed update led to a wrong stock count, a wrong stock count led to a shortage, and a shortage led to a delayed customer. The researchers stressed that an inventory system had to do more than store data digitally. It also had to support real time monitoring and fit smoothly into the daily workflow to be fully effective.

The study also discussed the importance of better inventory visibility and smoother order management. When a system updated stock information right after a transaction, businesses could monitor their inventory more closely, spot low stock items earlier, and make better purchasing decisions. Real time visibility also let businesses respond faster to customer demand and reduced delays caused by unavailable parts. A business that could see its stock at any moment did not have to find out about a problem from an unhappy customer.

To address these challenges, the researchers proposed an improved web based inventory and order management system designed to raise inventory accuracy, improve monitoring, and support more efficient handling of orders and stock movement. The study emphasized that technology based systems could greatly reduce the

administrative burden of manual inventory management while making business records more reliable and easier to reach. Staff who spent less time on record keeping had more time for the work that actually served customers.

Another important point in the study was that adopting technology did not automatically guarantee better operations. The researchers explained that inventory systems had to match the actual workflow and needs of the business. Systems that were poorly fitted to daily operations could still produce inaccurate records and inefficient processes even when digital tools were in use. Because of this, the study highlighted the need to design systems that supported how automotive businesses naturally worked, instead of forcing users to adjust to overly complex software. A system that people found hard to use would be avoided or used wrongly, and the benefits would be lost.

The findings of Sánchez-Luna et al. [1] were highly relevant to the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop because the problems in the study closely matched those the shop had experienced. Like the automotive parts stores in the research, D Speedway had relied on manually maintained spreadsheets for inventory and sales tracking. This setup caused problems with stock accuracy, inventory visibility, delayed updates, and inefficient monitoring. The shop was dealing with the same kinds of problems the researchers described, even though it was a different business in a different place.

The study strongly supported the need for real time inventory tracking in the proposed system. Before development, D Speedway had no centralized platform that could automatically update stock records when parts were sold or used in repairs. As a result, staff sometimes found differences between recorded inventory and actual stock, which delayed repairs and disrupted customer service. By using real time inventory updates and centralized record management, the proposed system aimed to remove many of these inefficiencies. Once the record changed at the same moment as the stock, the gap between the two would no longer have a chance to grow.

The study also reinforced the need to build inventory systems directly into operational workflows. The proposed system was designed around the actual working environment of D Speedway. Instead of forcing staff to adapt to complicated processes, it was meant to simplify inventory management by automating stock updates, monitoring low stock items, generating reports, and supporting reorder planning in a way that fit naturally into the shop's daily operations. The staff would keep doing their work the way they already did, and the system would follow along in the background.

The study further supported improving inventory visibility and order management through automation. The features planned for the proposed system, including automated inventory monitoring, stock alerts, forecasting, and reorder recommendations, lined up directly with the technology improvements discussed in the research. These features were meant to reduce manual administrative work, improve inventory accuracy, and help the shop respond better to customer needs and changing demand. Each feature answered a problem the study had already described.

Overall, the study by Sánchez-Luna et al. [1] gave strong support for developing the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop. It showed that effective inventory and order management systems needed real time monitoring, accurate stock control, and a good fit with daily workflow to improve performance in automotive businesses. Its findings reinforced the value of a centralized, technology driven system that could improve inventory accuracy, efficiency, and service reliability in automotive repair settings.

According to Agboola, Malgwi, Mahmud, and Oguntoye [2], inventory management was one of the most important areas of operation in small and medium enterprises because it directly affected efficiency, cost control, and customer satisfaction. In their study titled "Development of a Web-Based Platform for Automating an Inventory Management of a Small and Medium Enterprise," published in the FUDMA Journal of Sciences (Vol. 6, No. 5, ISSN: 2384-5821), the researchers set out to create a computerized inventory management platform to replace the manual methods still used by Sangkemi Global Nigeria Limited, a small and medium enterprise in Adamawa State, Nigeria. The study started from a simple but important idea: the way most small businesses tracked their inventory was not working, and the longer they waited to change it, the worse the consequences became.

The study explained that many small businesses kept relying on spreadsheets, handwritten records, and disconnected files to monitor inventory and sales. On the surface these methods seemed like enough, since a spreadsheet could hold a lot of data and most people already knew how to use one. But Agboola et al. [2] pointed out that the problem was not the tool itself. It was what happened when the tool was asked to do more than it was made for. As a business grew and transactions increased, the spreadsheet approach became harder to sustain. What began as one manageable file turned into a group of files, each kept by different people at different times, each drifting a little further from the others until no single record could be fully trusted. The researchers were clear that these problems were not unique to one business or industry. They were the expected result of using manual systems in places that had outgrown them.

Agboola et al. [2] also pointed to the financial side of poor inventory management, which was often overlooked when talking about small businesses. The researchers cited that inventory costs for businesses still using manual processes could range from four percent to ninety percent of total business expenses, depending on the industry and how many errors were involved. That was an enormous range, and even at the low end it was a large share of spending that was being affected, often for the worse, by inaccurate and outdated records. For a small business like D Speedway, where parts inventory was both the main cost and the main driver of service, sitting anywhere on that range without knowing it was a real financial risk. The study presented this not as a scare tactic but as a fact that explained why manual inventory management was not a neutral choice. It carried measurable financial consequences that grew over time.

The researchers also discussed how inaccurate inventory records affected the business beyond the numbers. When stock information could not be trusted, businesses lost the ability to make informed purchasing decisions. Owners ended up guessing, ordering more of something because they were unsure how much was left, or waiting too long to reorder because the records showed more stock than really existed. Both outcomes had costs.

Agboola et al. [2] noted that overstocking raised storage costs and tied up money that could have been used elsewhere, while shortages delayed operations, disrupted the workflow, and pushed customers toward frustration and eventually toward competitors. A business did not have to be doing anything wrong for these problems to appear. They developed naturally when the way inventory was tracked did not keep up with the pace of the business. This was exactly what was happening at D Speedway, where the research team observed that the staff and owner were not careless people. They were capable people working in a system that was not built to handle what they were putting it through.

One of the more specific findings documented by Agboola et al. [2] was what happened to staff time when inventory records were unreliable. Because the records could not be trusted, employees stopped using them as their main reference and started doing by hand what the records were supposed to handle. Instead of checking a system, someone walked to the storage area. Instead of pulling a report, the owner sat down with a calculator. Instead of trusting the numbers in a file, a staff member compared two or three files to work out the real situation. None of this was productive time. It was time spent making up for a broken system, and it happened every day among every staff member who touched inventory data. The researchers made the point that this kind of hidden time loss was one of the biggest costs of manual inventory management. It was significant because it was real, and hidden because it never showed up as a line item on any report.

To address these challenges, Agboola et al. [2] developed a web based inventory management platform that could automate inventory monitoring and record keeping. The system was built at Sangkemi Global Nigeria Limited using the Iterative Waterfall Model, a development method that moved through structured phases but allowed corrections at each stage before moving forward, which lowered the risk of building something that did not actually solve the problem. The researchers chose a web based design because it could be reached from any device with an internet connection, did not need special software installed on each machine, and could support several users at the same time without the data conflicts that happened when many people worked in the same spreadsheet. The system stored all inventory data in a centralized digital database, updated stock records automatically whenever a transaction was recorded, and gave users fast access to current information without making them dig through files or rely on memory.

The system was built around three core modules. The first was a sales management module, which let staff record transactions directly in the system so that every sale was logged right away and the matching inventory count was adjusted automatically. This removed the manual step of updating a stock record separately after each sale, a step that Agboola et al. [2] identified as one of the main points where errors entered the system. The second was a product tracking module that showed real time stock levels for every item, so staff could check availability instantly from wherever they were working instead of walking to the storage area or opening a file that might be out of date. The third was a user management module that controlled what each type of user could access and do, with administrators having full access and sales staff seeing only the functions that applied to their role. This structure protected sensitive business data while keeping the interface clean and manageable for employees who only needed to do basic tasks.

Another important contribution of the study was its focus on real time visibility as a practical tool and not just a technical feature. Agboola et al. [2] explained that real time monitoring changed how a business responded to its own inventory situation. When stock levels updated instantly with each transaction, the owner did not need to schedule a stock check or wait for a report to know what was going on. The information was there whenever it was needed. This meant low stock situations could be caught while there was still time to reorder through normal channels at standard prices, instead of being discovered only when a mechanic was already in the middle of a repair and the needed part had already run out. For D Speedway, where the research team documented cases of exactly that kind of mid repair discovery, this difference was not small. It was the difference between a planned, organized purchase and a panicked rush order paid at a premium.

The study also highlighted the importance of centralized data management in a way that went beyond just staying organized. Agboola et al. [2] argued that when data was scattered across separate files kept by different people, the real problem was not only that the files did not match. It was that the business no longer had one reliable source of truth. When two records showed different numbers for the same item and there was no way to tell which one was correct, every decision based on either record carried a hidden margin of error. The owner might not even know the margin was there. They were reading a number and treating it as fact when it was really a guess in disguise. Putting all the information into one connected system removed that doubt. There was one number. It updated automatically. Everyone in the business was looking at the same data at the same time. Decisions made from that data stood on a foundation that could be trusted.

The findings of the study confirmed what the researchers set out to show: the automated web based platform greatly improved efficiency, reduced the workload on staff, and raised the accuracy of inventory records at Sangkemi Global Nigeria Limited. After the system was deployed, the owner could monitor all business processes through a single platform without relying on physical book entries or manual calculations. The record keeping burden that used to fall on individual employees to maintain correctly and on time was handled by the system automatically. Reports that once needed manual compiling were generated instantly from stored data. The study concluded that digital inventory management systems were no longer optional for growing businesses. They had become necessary tools for keeping operations stable where transaction volumes were too high and too complex for manual methods to handle reliably.

The study of Agboola et al. [2] was strongly relevant to the proposed Smart Inventory and Parts Planning System for D Speedway because the two situations were comparable in ways that went beyond surface similarity. Both involved small businesses that sold or used physical parts as the core of their operations. Both were growing faster than their manual systems could keep up with. Both faced the same set of consequences: inaccurate records, unreliable stock data, time consuming manual reporting, and purchasing decisions made without a clear view of what was on hand. Agboola et al. [2] documented these problems in a Nigerian small and medium enterprise. The research team documented them at D Speedway through site visits and staff interviews. The problems were the same because the cause was the same: manual systems being asked to do more than they were built to handle.

The solution that Agboola et al. [2] developed was a centralized web based platform that automated stock updates, gave real time visibility, brought data into one system, and generated reports without manual effort. It was the same kind of solution built for D Speedway. The proposed system addressed exactly the problems identified in this study. Disconnected records were replaced by a centralized database. Manual stock updates were replaced by automatic deductions after every recorded transaction. Delayed report generation was replaced by real time sales summaries. Reactive stockout discoveries were replaced by proactive low stock alerts that gave the shop time to reorder before a shortage disrupted a repair. The thinking behind both systems was the same: reduce the manual work, automate the record keeping, and give the owner the visibility needed to run the business well.

The emphasis on real time inventory visibility in the Agboola study also strongly supported the forecasting and reorder recommendation features planned for the proposed system. Real time visibility was not only about knowing what was on the shelf at the moment. It was the foundation that made forward planning possible. If stock data was accurate and current, the system could analyze how fast certain parts were used and estimate when they would need restocking. If inventory and sales history were recorded correctly, the system could suggest how much stock to replenish and which items needed attention first. None of these features would work properly if the underlying data was unreliable. Since D Speedway handled many vehicle repairs every day, accurate and updated inventory data was essential to make sure mechanics always had the parts they needed. Through automation, the proposed system reduced manual workload, minimized stock related disruptions, improved efficiency, and helped the business keep growing without the administrative problems that came from spreadsheet based inventory management.

The study also supported adopting technology early, while the business was still at a manageable size. D Speedway had been operating for only one year, but its inventory and tracking problems were already getting hard to manage. Agboola et al. [2] made the same observation about the businesses they studied: relying on manual inventory systems became harder to sustain as transaction volume and inventory complexity grew, and the longer a business waited to deal with the problem, the more its broken habits became part of daily routine and the harder the change became. Putting an automated inventory and parts planning system in place at this stage of D Speedway's growth was not too early. It was the right moment, while the scale was still manageable, while the data to be migrated was still fairly clean, and while staff could be trained on a new system without the disruption that came from changing things in the middle of a fully grown operation. The findings of Agboola et al. [2] confirmed that this kind of early action was not only practically smart but also strategically necessary for a business that planned to keep growing.

Ivan and Oetama [3], in their study titled "Inventory Management System Using Economic Order Quantity and Reorder Point," published in G-Tech: Jurnal Teknologi Terapan, looked at how an automotive workshop managed its inventory when it relied entirely on Microsoft Excel to track parts and stock levels. The researchers found that spreadsheets were manageable at first for recording inventory data, but they became less efficient and less reliable as the workshop grew and handled a wider range of automotive parts. Since every inventory calculation depended on manual updates, the workshop struggled to keep accurate stock records and to follow

inventory movement consistently. The more parts the workshop carried, the more there was to update by hand, and the harder it became to keep up.

The study explained that the workshop had no automated way to tell when inventory levels were getting critically low. Stock monitoring depended heavily on staff checking the records by hand and noticing shortages before they disrupted operations. Safety stock limits were applied to only a small number of fast moving items, while most of the inventory had no organized monitoring at all. Because of this, stockouts often happened without warning, which led to delayed repairs, interrupted work, and trouble keeping daily operations smooth. The researchers stressed that poor inventory planning and weak stock monitoring could affect the whole supply chain of a workshop, especially when a needed part was unavailable at the exact moment a repair was being done. A single missing part could hold up a job that was otherwise ready to be finished.

The researchers also discussed how manual inventory systems caused long term inefficiencies. Since the workshop depended on Excel files that needed constant manual updates, inventory accuracy rested entirely on how consistent and attentive the staff were. In busy automotive settings where employees focused mainly on servicing vehicles and helping customers, updating inventory files accurately and right away often became a secondary task. Repairs and customers came first, and the spreadsheet was updated whenever there was time. As a result, the gap between recorded stock and actual physical inventory slowly widened. The study noted that once staff lost confidence in the inventory records, they began relying more on physically checking stock than on the system itself, which defeated the purpose of keeping records in the first place.

To address these challenges, Ivan and Oetama [3] developed a computerized inventory management application that combined two established inventory control methods: Economic Order Quantity (EOQ) and Reorder Point (ROP). The Economic Order Quantity method was meant to find the most cost efficient amount of parts to order in each restocking cycle. EOQ balanced the cost of ordering too often against the cost of storing too much stock, which helped businesses lower unnecessary expenses while keeping enough inventory on hand. In simple terms, it helped answer how much to buy each time so that the business was neither ordering constantly nor filling its shelves with parts it did not need.

The Reorder Point (ROP) method focused on finding the stock level at which a part should be flagged for restocking. The system calculated reorder points from the average usage rate of a part and its past inventory movement. In other words, it answered when to buy, while EOQ answered how much to buy. By combining EOQ and ROP, the system turned inventory management from a reactive process into a proactive one. Instead of waiting until parts ran out completely before acting, the workshop could spot low stock early enough to replenish inventory before operations were affected.

Another important part of the study was its use of Rapid Application Development (RAD). The researchers explained that RAD let the system be built in smaller stages with steady feedback from actual workshop users. Instead of designing the system only from assumptions, the developers regularly consulted the workshop staff throughout the process to make sure the final product matched the real needs of the business. This improved

usability because the people who would eventually use the system helped shape how it worked. Problems with the design could be raised early, while they were still easy to change.

The results of the study showed that the developed system was highly effective and well accepted by its users. User Acceptance Testing with actual workshop personnel gave an average satisfaction rating of 98.44 percent among administrative users and 97 percent among general users. These results mattered because they showed the system was not only technically functional but also practical and easy to use in a real automotive workshop. The researchers stressed that user friendly design was essential in automotive businesses, where employees were not software specialists and needed systems that supported their work without adding unnecessary complexity. A system that was hard to use would only have added to the workload it was meant to reduce.

The study of Ivan and Oetama [3] was highly relevant to the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop because both studies dealt with nearly the same operational challenges. Like the workshop in the study, D Speedway had relied on manually maintained Excel spreadsheets to manage inventory and record sales. This setup caused many of the same problems the researchers identified, including inaccurate stock records, disconnected files, no real time inventory visibility, delayed reporting, and no automatic stock alerts. Even though the two businesses were in different places, their daily struggles were very similar.

The reactive approach to inventory described in the study closely matched the earlier situation at D Speedway. Before the proposed system, the shop often found out about shortages only after a needed part was already unavailable, which forced mechanics to pause repairs while the parts were replenished or restocked. This delayed service, added operational costs, and hurt customer satisfaction. The automated reorder alerts and inventory forecasting in the proposed system directly addressed these problems by helping the shop spot low stock items before shortages disrupted daily operations.

The EOQ and ROP concepts in the study also gave useful guidance for the forecasting and reorder recommendation features planned for D Speedway's system. By analyzing historical sales data and watching inventory movement, the system could recommend suitable reorder quantities and decide when new stock should be ordered. This let the shop keep enough inventory without overstocking parts that were not needed right away. The owner would no longer have to rely on a rough guess when deciding how much to buy.

The study also strongly supported the importance of usability in system design. Like the workshop employees in Ivan and Oetama's research, the staff at D Speedway were mainly automotive service workers and not technology specialists. Because of this, the proposed system had to stay simple, clear, and easy to operate in a busy repair shop. The high User Acceptance Testing results in the study showed that an inventory system could be both technically capable and user friendly when it was designed around the real needs of workshop staff.

Overall, the study gave both a technical foundation and a practical real world example in support of the Smart Inventory and Parts Planning System for D Speedway. It showed that automated inventory management,

proactive stock monitoring, reorder planning, and user centered design could greatly improve operational efficiency, inventory accuracy, and service reliability in automotive workshops.

Surbakti and Selly [4] conducted a study published in the International Journal of Scientific Research and Management (Vol. 11, No. 05, e-ISSN: 2321-3418) on designing a web based information system for a motorcycle spare parts store in Depok City, Indonesia, called Bintang Multi Sparepart Store. The store had been operating since 2017 and ran with only three full time employees, who handled everything from serving customers to managing stock to closing the books at the end of each day. With so few people covering so many jobs, there was little room for careful record keeping. Its main customers were owners of nearby repair workshops and regular buyers who came in looking for specific motorcycle parts. That customer profile was worth noting because it was not very different from D Speedway, where mechanics pulled parts from inventory every day and the business depended on having those parts available and accounted for at all times. The study was one of the more relevant pieces of literature for this project because it dealt with an actual automotive parts business and not a made up scenario, and the problems it found at Bintang Multi were the same ones this research team found at D Speedway when they visited the shop and talked to the staff.

Before looking at what the study built, it helped to understand what the store was dealing with before any system existed. Bintang Multi handled its daily operations entirely by hand. When a customer came in to buy a part, the employee first had to walk to the storage area to check whether the item was available. The study described the storage area as being quite far from the customer service area, so every stock check wasted time, both the employee's and the customer's. If the part was available, the employee went back and wrote a receipt for the customer, then recorded the transaction a second time in a physical sales book, noting the quantity, item type, item code, and price. This double entry happened for every sale throughout the day. At closing time, the owner sat down with all the sales book records and added up the day's total with a calculator. The same thing happened at the end of each month for the monthly summary. It sounded manageable, but Surbakti and Selly [4] pointed out that errors happened constantly under this setup. This was not because the employees were careless, but because manual record keeping almost guaranteed mistakes over time. The book got stained or torn, handwriting became hard to read, numbers were swapped, and entries were missed entirely. These small errors added up quietly until the records and the actual cash no longer matched, and the owner had to recount everything from scratch to find where it went wrong.

The stock fulfillment process had its own problems, separate from the sales recording. There was no tracking system that showed current stock levels, and no alert went out when a product was running low. The only way anyone knew a part had run out was when an employee went to the storage area for an order and found the shelf empty, and then reported it to the owner. The owner then prepared a list of items to restock by hand. Because this whole process depended on someone noticing and reporting a shortage at the right time, parts sometimes stayed out of stock for long periods simply because nobody caught it before it became a problem. The researchers specifically noted that stock was left unfulfilled for long stretches because the owner forgot to replenish inventory. This was not out of negligence, but because nothing in the system reminded the owner it needed to be done. This was exactly the kind of situation D Speedway dealt with. Mechanics started a repair and

found out partway through that the part they needed was already gone. That kind of disruption happened because there was nothing in place to catch a low stock situation before it became a zero stock situation.

The financial reporting problem at Bintang Multi was perhaps the most telling detail in the study. The profit and loss calculation depended on matching the sales book against the actual cash on hand, and that match was often off because of recording errors. Because of this, the owner had stopped trying to work out actual profit from the records. Instead, the owner used a rough estimate of ten percent of total sales as the assumed profit for the period. That was not a business decision. It was a workaround for a broken system. The owner of a business with daily transactions and a storage room full of parts had no reliable way to know what the store was really earning, because the records could not be trusted to give the right answer. This connected directly to D Speedway, where the owner expressed the same frustration: getting a clear read on how the business was doing took hours of manual work sorting through spreadsheet files, and even after all that effort, the figures were not always reliable because the files were updated by several people and did not always stay consistent with each other.

Surbakti and Selly [4] developed their solution using the System Development Life Cycle methodology, which guided the project through four structured phases: planning, analysis, design, and implementation. What made their approach worth highlighting was that they did not start with the system. They started with the business. Before any design work began, the researchers visited the store in person, watched the operations directly, and interviewed the owner to understand not only what the problems were but also how the day to day work actually flowed. That observation and those interviews let them design a system that fit the store's actual operations, instead of something the owner and staff would have had to adjust to. This was the same approach taken in this project. The research team visited D Speedway, watched how the shop operated, and talked to the owner and the staff before any feature decisions were made. The literature consistently supported this kind of needs first approach, and the study of Surbakti and Selly was one of the clearest examples of why it mattered. The system they built worked because the people who would use it shaped it, and not just technical assumptions about what an inventory system should include.

The system itself was built with PHP as the programming language and MySQL for the database, delivered through a web based interface. The choice to go web based was a practical one. Web applications did not need special software installed on each computer, could be reached from any device with an internet connection, and let several users work in the system at the same time without conflicts. All product data, including codes, names, categories, sizes, brands, stock quantities, prices, and descriptions, was stored in a centralized database connected directly to the transaction process. This meant that every time a sale was recorded, the system automatically updated the stock count for whatever was sold, without anyone doing it separately. That automatic update was the core of what made a system like this different from a spreadsheet. With a spreadsheet, the transaction and the inventory update were two separate manual steps, and any time someone skipped or delayed the second step, the records began to drift away from reality. With the system Surbakti and Selly [4] built, the two things happened together, and one action handled both. Daily and monthly sales reports were also generated automatically from the stored data, so the owner no longer had to sit down at the end of each day to compile and match figures from a record book. The information was already organized and available whenever it was needed.

The researchers also removed several manual steps from the stock fulfillment process. Because the system showed current stock levels at all times, employees no longer needed to walk to the storage area every time a customer placed an order. They could check availability through the system from wherever they were standing. This saved time for the employee and kept the customer from waiting while someone went to look. Reordering also became more organized, because the owner could see at a glance which items were running low without waiting for an employee to report a shortage. This shift from a reactive approach, where a stockout was found after the fact, to a more visible and proactive one was exactly what D Speedway needed. Before, the shop only knew a part was gone when someone went looking for it and came back empty handed. A system that showed stock levels in real time removed that blind spot entirely.

The testing phase covered the system from several angles. Unit testing, integration testing, and system testing were all done using a black box approach, where users were given scenario cases to run through the system so that any errors or weaknesses could be caught before the system was finalized. Acceptance testing was done through usability testing that measured five dimensions. Learnability assessed how quickly users could figure out the system on their first use. Efficiency measured how fast they could complete tasks once they were familiar with it. Memorability tested whether users could return after a break and still use the system without relearning everything. Error rate tracked how many mistakes users made while operating it. Satisfaction was measured through the System Usability Scale questionnaire, a widely used standard tool for judging whether a system is genuinely easy to use in real conditions. The system passed all of these tests, and the study concluded that the designed information system fully met the users' needs in both functionality and usability.

After the system was implemented, the changes at Bintang Multi Sparepart Store were concrete. The owner could monitor all business processes, including product availability, transaction records, and sales summaries, through one platform instead of piecing information together from a book, a calculator, and a storage room. Real time sales data was available at any point during the day without the owner having to stop and calculate by hand. The double entry problem was gone because the system handled the record and the stock update together. Reports that used to take hours to compile were generated by the system instantly. The study concluded that the web based information system successfully replaced the manual approach and gave the business the data reliability and visibility it had been missing.

Looking at this study alongside the situation at D Speedway, the connection was easy to see. Bintang Multi and D Speedway were different businesses in different countries, but the problems they faced were described in almost the same terms: manual records that could not keep up with the pace of daily operations, stock levels nobody could verify without checking physically, reports that took hours of manual effort and still were not fully reliable, and an owner trying to make business decisions without a clear and trustworthy picture of what was happening. Surbakti and Selly [4] showed that a web based system built around the real workflow of a spare parts business solved all of these things at once. The features that fixed Bintang Multi, namely centralized product data, automatic stock updates tied to sales, real time availability checking, and automated report generation, were the same features built into the Smart Inventory and Parts Planning System for D Speedway. This study did not just support the argument that a system like this was a good idea. It was evidence that it worked, in a business very much like the one this project was designed for.

2.2 Local Studies

Barcelona, Gungon, Laylo, Panganiban, Sultan Amanollah, and Altarejos [5] conducted a study that looked at how auto supply businesses in Candelaria, Quezon, Philippines managed their inventory. The study involved nine business owners and managers as respondents and used a descriptive correlational research design to understand how inventory practices affected the overall performance of auto supply businesses in the area.

The study found that many auto supply businesses in the locality still relied on outdated ways of managing inventory, particularly spreadsheets. These traditional methods took a lot of time and were prone to human error. They often led to inaccurate data, overlooked stocks, and internal losses that hurt daily business operations. This made it clear that there was a strong need for a more reliable and organized system, such as an automotive inventory system, that could handle stock management more accurately and efficiently. The researchers also identified two major problems that kept these businesses from managing their inventory well.

The first was the cost of compliance, which included securing the permits, licenses, and certifications required by local government units. The second was the unpredictable movement of stock levels, which made it hard to keep inventory stable and organized. Together, these issues led to unnecessary expenses, excessive stock buildup, and frequent shortages that directly affected the profitability and smooth operation of auto supply businesses. An automotive inventory system could help ease these burdens by giving real time tracking of stock levels, automated alerts for low inventory, and organized records that reduced the chance of human error. To help address these concerns, the study recommended adopting a Just in Time (JIT) inventory approach, where businesses ordered stock only when needed instead of keeping large amounts on hand.

This strategy helped lower storage costs, reduced the risk of overstocking, and kept automotive parts from becoming outdated or obsolete before they were sold or used. An automotive inventory system supported this kind of approach by giving business owners accurate and current information on their stock, which made it easier to decide when and what to reorder. Based on the findings of this study, it was clear that local auto supply businesses genuinely needed an automotive inventory system. The challenges the study pointed out were keeping track of stock by hand, dealing with incorrect data, managing unpredictable inventory levels, and having no access to real time updates. An automotive inventory system could help with every one of these. A system that watched stock automatically, produced up to date inventory reports, and kept the ordering process organized could make a big difference in how auto supply businesses operated. It could help them become more efficient, more accurate, and better prepared for the demands of running a business.

The findings of Barcelona et al. [5] were highly relevant to the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop because the problems in the study closely matched those the shop had experienced. Like the auto supply businesses in Candelaria, Quezon, D Speedway had relied on manually maintained Excel spreadsheets to record sales and track inventory. The study showed that this kind of setup was common among local automotive businesses, and that it brought the same results: time consuming work, human

error, inaccurate data, and stock that was overlooked until it was needed. This was what the research team saw at D Speedway during the shop visits and staff interviews.

The study's point about unpredictable stock movement also reflected the situation at D Speedway. The shop had seen demand for certain parts rise without warning, and because nobody noticed in time to reorder, the part ran out in the middle of a repair. The customer had to wait, and the shop often paid more to get the part quickly. These were the kinds of shortages and unplanned expenses the study described. The proposed system answered this by updating stock automatically after every transaction and sending low stock alerts, so that the shop could see a shortage coming and reorder at a normal price instead of in a rush.

The study's recommendation of a Just in Time approach also supported the forecasting and reorder recommendation features planned for the proposed system. Ordering only what was needed required accurate and current information about what was on hand and how fast it was being used. The proposed system gave the shop this information by keeping stock counts current, studying past sales to show which parts moved quickly, and suggesting reorder quantities for parts that were running low. This helped D Speedway avoid both overstocking, which tied up money, and understocking, which delayed repairs.

Overall, the study by Barcelona et al. [5] gave strong local support for developing the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop. It showed that spreadsheet based inventory management was a real and common problem among auto businesses in the Philippines, and that a system offering real time tracking, low stock alerts, and organized records was something they genuinely needed. These findings confirmed that the problems at D Speedway were not unusual, and that a centralized and automated system was a practical answer to them.

Bergonia [6], in the study titled "The Inventory Management Practices of the Retail Automotive Industry in Pangasinan" published in the Ascendens Asia Journal of Multidisciplinary Research Conference Proceedings, examined the inventory management practices being implemented by automotive businesses in Pangasinan, Philippines. The study focused specifically on how businesses in the automotive retail industry managed procurement, storage, stock monitoring, inventory information systems, and inventory-related decision-making processes. The research highlighted the growing importance of inventory management in automotive businesses, where daily operations heavily depended on the availability of vehicle parts, tools, supplies, and other materials necessary for repair and maintenance services.

The study explained that inventory management was one of the most important operational functions in automotive businesses because it directly affected workflow efficiency, customer satisfaction, and financial performance. Automotive businesses constantly dealt with different types of parts that varied in demand, availability, and pricing. Because of this, maintaining accurate inventory records and consistently monitoring stock levels were necessary to avoid disruptions in service operations. According to the study, businesses that failed to manage inventory effectively often experienced operational inefficiencies such as stock shortages, overstocking, delayed repairs, inaccurate reporting, and unnecessary expenses

Bergonia [6] found that while many automotive businesses had established procurement and storage procedures, their inventory information systems were only considered "fairly effective." This finding indicated that many businesses still relied heavily on traditional or partially manual methods of managing inventory data. Inventory records were often maintained through handwritten logs, spreadsheets, or disconnected files that required constant manual updating by employees. Although these methods allowed businesses to maintain basic inventory operations, they lacked the reliability, speed, and accuracy needed to support growing automotive businesses effectively.

The study emphasized that manual inventory systems created several long-term operational challenges. Since inventory updates depended entirely on employees manually recording every transaction, businesses became highly vulnerable to human error. Mistakes such as forgotten updates, duplicate entries, incorrect stock counts, and delayed recording of inventory movement became increasingly common, especially in busy automotive environments where staff members prioritized repairs and customer service over administrative tasks. Over time, these small inaccuracies accumulated and caused the inventory records to no longer match the actual physical stock available in storage.

Another major issue identified in the study was the difficulty businesses faced in maintaining real-time inventory visibility. Because inventory information was not updated automatically, business owners and employees often had no immediate way of knowing the exact quantity of parts available at any given moment. In many cases, staff members had to physically inspect storage areas or manually cross-check several files just to confirm whether a particular part was still in stock. The study explained that this process consumed unnecessary time, reduced productivity, and created delays in customer service operations.

The lack of real-time visibility also negatively affected decision-making processes within automotive businesses. Without accurate and updated inventory data, business owners struggled to determine which parts needed restocking, which products were moving quickly, and which items remained unsold for long periods. As a result, businesses became more vulnerable to both understocking and overstocking problems. Understocking created service delays because mechanics could not complete repairs immediately when parts were unavailable, while overstocking tied up financial resources in inventory that was not moving efficiently.

Bergonia [6], in the study titled “The Inventory Management Practices of the Retail Automotive Industry in Pangasinan,” published in the Ascendens Asia Journal of Multidisciplinary Research Conference Proceedings, looked at the inventory management practices of automotive businesses in Pangasinan, Philippines. The study focused on how businesses in the automotive retail industry handled procurement, storage, stock monitoring, inventory information systems, and inventory related decision making. The research highlighted the growing importance of inventory management in automotive businesses, where daily operations depended heavily on the availability of vehicle parts, tools, supplies, and other materials needed for repair and maintenance services.

The study explained that inventory management was one of the most important operations in automotive businesses because it directly affected workflow efficiency, customer satisfaction, and financial performance. Automotive businesses constantly dealt with different types of parts that varied in demand, availability, and price. Because of this, keeping accurate inventory records and checking stock levels consistently was necessary

to avoid disruptions in service. According to the study, businesses that failed to manage inventory well often ran into stock shortages, overstocking, delayed repairs, inaccurate reporting, and unnecessary expenses.

Bergonia [6] found that although many automotive businesses had established procurement and storage procedures, their inventory information systems were only considered “fairly effective.” This showed that many businesses still relied heavily on traditional or partly manual ways of managing inventory data. Inventory records were often kept in handwritten logs, spreadsheets, or disconnected files that employees had to update constantly by hand. These methods let businesses keep up basic inventory operations, but they lacked the reliability, speed, and accuracy needed to support growing automotive businesses.

The study emphasized that manual inventory systems created several long term challenges. Since inventory updates depended entirely on employees recording every transaction by hand, businesses were very open to human error. Mistakes such as forgotten updates, duplicate entries, wrong stock counts, and late recording of inventory movement became more common, especially in busy automotive settings where staff put repairs and customer service ahead of paperwork. Over time, these small inaccuracies piled up, and the inventory records no longer matched the physical stock in storage.

Another major issue in the study was how hard it was for businesses to keep real time visibility of their inventory. Because inventory information was not updated automatically, business owners and employees often had no quick way to know the exact quantity of parts on hand. In many cases, staff had to inspect storage areas by hand or compare several files just to confirm whether a part was still in stock. The study explained that this process wasted time, lowered productivity, and caused delays in serving customers.

The lack of real time visibility also hurt decision making in automotive businesses. Without accurate and current inventory data, business owners struggled to decide which parts needed restocking, which products were moving quickly, and which items stayed unsold for long periods. As a result, businesses were more likely to understock or overstock. Understocking caused service delays because mechanics could not finish repairs right away when parts were missing, while overstocking tied up money in inventory that was not moving.

Bergonia [6] further explained that inventory problems had a direct effect on customer satisfaction and business reputation. In automotive businesses, customers expected repairs and maintenance to be finished within a reasonable time. When needed parts were unavailable because of poor inventory monitoring, repairs were delayed and customers had to wait longer than expected. Repeated delays from stock shortages could damage customer trust and make repeat business less likely. The study stressed that good inventory management was not only a technical or administrative concern. It was also a key part of giving customers reliable service.

The study also highlighted the importance of centralized inventory information systems. According to the researcher, businesses that kept separate files for inventory, procurement, and sales records often found inconsistencies between them, because different employees updated each file independently at different times. This fragmented setup made reporting harder and raised the chance of conflicting data. Bergonia noted that centralized systems solved these issues by putting all inventory related information into one platform that could be accessed and updated consistently.

Another important point in the study was the growing need for automotive businesses to adopt computerized inventory management systems as their operations expanded. Bergonia explained that manual systems might seem manageable for small businesses at first, but they became inefficient as inventory volume, transaction frequency, and customer demand grew. Businesses that kept relying on spreadsheets and manual records often struggled to scale, because their administrative systems could not keep up with the growing complexity of inventory management.

The study recommended that automotive businesses adopt more advanced inventory information systems that could automate stock monitoring, improve record accuracy, produce faster reports, and support better decision making. The researcher emphasized that digital inventory systems could reduce administrative workload, improve operational efficiency, and give business owners more reliable information for planning and purchasing. Through automation, businesses could reduce inventory related disruptions and respond more quickly to customer needs.

The findings of Bergonia [6] were highly relevant to the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop because the challenges in the study closely matched the problems the shop had experienced. Like the businesses in the research, D Speedway had relied on manually maintained Excel spreadsheets to record inventory and sales transactions. This caused many of the same difficulties Bergonia identified, including inaccurate inventory records, disconnected files, delayed updates, inefficient reporting, and no real time stock monitoring.

The problem of fragmented and disconnected inventory records in the study was especially relevant to D Speedway's earlier operations. Before the proposed system, the shop managed sales records and inventory information through separate spreadsheets that staff updated by hand. Since these files were not connected, inventory counts could easily become inconsistent whenever updates were delayed or forgotten. This often forced staff to verify stock levels by physical inspection instead of trusting the records, which wasted time and lowered efficiency.

The study's discussion of real time inventory visibility also strongly supported the objectives of the proposed system. D Speedway previously had no automated way to update inventory records right away when a part was sold or used in a repair. Because of this, the shop sometimes found shortages only after mechanics had already started a repair job. These situations delayed operations, disrupted workflow, and hurt the customer experience. The proposed Smart Inventory and Parts Planning System addressed this directly through automatic inventory updates and low stock alerts that notified staff before shortages happened.

The recommendation to adopt centralized computerized inventory systems also lined up directly with the goals of this project. The proposed system for D Speedway aimed to replace the shop's fragmented spreadsheet process with one platform that handled sales recording, inventory monitoring, stock alerts, forecasting, and report generation. By centralizing inventory and sales data, the proposed system improved data consistency, reduced administrative workload, and gave the shop owner more accurate information for decision making.

The study also supported the importance of inventory forecasting and proactive stock planning. Bergonia emphasized that businesses needed reliable inventory data to make good purchasing decisions and avoid stock related disruptions. This related directly to the forecasting and reorder recommendation features planned for the

proposed system, which aimed to help D Speedway anticipate future inventory demand from historical sales patterns. By identifying which parts were used most often and monitoring inventory in real time, the system helped the shop reorder parts before shortages happened and lowered the risk of delayed repairs.

The study further reinforced the idea that inventory management systems needed to support the actual workflow of automotive businesses, instead of forcing them to adapt to complicated software. Automotive repair shops worked at a fast pace, and mechanics and staff focused mainly on servicing vehicles and helping customers. Because of this, inventory systems had to stay practical, user friendly, and easy to operate even for users with limited technical knowledge. This principle was reflected in the design goals of the proposed Smart Inventory and Parts Planning System, which aimed to simplify inventory management while keeping extra workload for staff to a minimum.

Overall, the study by Bergonia [6] gave strong evidence in support of developing the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop. The research showed that ineffective inventory information systems could seriously affect inventory accuracy, operational efficiency, reporting reliability, and customer satisfaction in automotive businesses. It also showed how centralized, computerized inventory systems could improve operations by automating stock monitoring, reducing manual errors, improving inventory visibility, and supporting better decision making. These findings reinforced the importance of a reliable inventory and parts planning system designed for the operational needs of D Speedway as it continued to grow.

Roque et al. [7] conducted a study titled “Development of an Inventory and Service Management System for an Auto Repair Shop Using Agile Methodology,” which was published in Philippine EJournals. The study focused on improving operational efficiency in a small automotive repair shop by addressing the limits of manual and semi digital inventory and service management practices.

The researchers began by identifying the common challenges in automotive workshops, particularly those that still relied on traditional tools such as Microsoft Excel, handwritten logs, or separate standalone files to manage operations. These fragmented systems often led to disorganized data handling, inconsistent updates, and difficulty in tracking the real time status of inventory and ongoing service jobs.

One of the key issues the study highlighted was the lack of integration between different operational processes. In many repair shops, inventory tracking, customer records, and service transactions were handled separately, so information did not flow automatically from one process to another. For example, when a part was used in a repair job, someone had to deduct it from the inventory record by hand. This dependence on manual work raised the risk of human error, such as forgotten updates, wrong quantity entries, or duplicate records.

The study further emphasized that these manual processes became even more troublesome in busy workshops, where staff were often switching between customer service, diagnostics, and repair work. Because of this workload, administrative tasks like updating inventory records were sometimes delayed or skipped entirely. Over time, these small inconsistencies added up and led to inaccurate stock levels and unreliable data.

Roque et al. [7] also found that the lack of real time data access greatly affected decision making in automotive repair shops. Without updated inventory information, shop managers could not quickly tell whether a required part was available or needed to be restocked. This often delayed repair services, especially when parts that were

assumed to be in stock were actually missing. In some cases, it led to urgent replenishment, which raised operating costs.

To address these challenges, the researchers developed a web based Inventory and Service Management System using Agile methodology. The system was designed to bring all key operational processes into one platform, including inventory management, service tracking, customer records, and transaction processing. This integration meant that all data in the system was connected and updated in real time.

A major feature of the system was automated inventory updating. Every time a transaction was recorded, whether a sale, part usage, or service completion, the system adjusted stock levels automatically without manual input. This greatly reduced the risk of human error and kept inventory data accurate and current at all times.

Another important feature was multi user accessibility, which let different staff members use and update the system at the same time without data conflicts. This mattered in workshops where several employees might be working on different tasks at once. The centralized database made sure that all users saw and worked with the same real time information.

The study also applied Agile development principles, which meant getting continuous feedback from actual users throughout development. This let the researchers refine the system based on real operational needs and not on assumptions. As a result, the final system was more practical, user friendly, and suited to the daily workflow of automotive repair staff.

Testing and evaluation results showed that the system greatly improved operational efficiency. Users reported easier access to inventory data, faster transaction processing, and better accuracy in stock monitoring. The reduction in manual workload also let staff focus more on core repair and customer service work instead of administrative tasks.

In relation to D Speedway, this study was highly relevant because it reflected the same working environment and challenges the shop had faced. Like the workshop in the study of Roque et al. [7], D Speedway had relied on Excel based inventory tracking, which created risks of inconsistent records, delayed updates, and manual encoding errors.

The idea of centralization in the study was especially important for D Speedway's proposed Smart Inventory and Parts Planning System. By combining inventory tracking and sales recording in one platform, the system removed the need for several disconnected spreadsheets. This made sure that all data was consistent, current, and accessible in real time.

The automated inventory updating developed in the study also directly supported one of the core objectives of the proposed system for D Speedway. Every time a part was used or sold, the system automatically updated the stock count, which reduced reliance on human memory and manual entry. This greatly reduced errors and made sure inventory records always matched actual stock levels.

Overall, Roque et al. [7] gave strong evidence that a centralized, automated inventory and service management system greatly improved efficiency, accuracy, and workflow coordination in automotive repair settings. The findings strongly supported developing a Smart Inventory and Parts Planning System for D Speedway, since

both systems aimed to solve the same basic problems of manual tracking, inconsistent data, and no real time inventory visibility.

Lirios et al. [8], in their study published in Innovatus titled “A Problem-Oriented Approach to Implementing an Auto Sales and Inventory System Through Agile Methodology,” looked at the operational challenges of an auto supply company that depended heavily on manual and semi digital inventory practices. The study focused on finding the root causes of inefficiency in inventory tracking and sales recording, particularly where transactions happened often and needed timely updates.

The researchers found that the company’s reliance on spreadsheet based systems caused ongoing problems with data accuracy and consistency. Since encoding was done by hand, inventory records were often updated late or wrongly, especially during peak hours when many transactions happened at once. This led to differences between actual stock levels and recorded data, which made it hard for the business to keep a reliable view of its available inventory.

The study further emphasized that the absence of a centralized system added greatly to the inefficiency. Sales records, inventory updates, and product tracking were stored in separate files that were not automatically synchronized. Because of this, even small transactions needed several manual updates across different documents. Over time, this fragmented setup raised the chance of missing entries, duplicate records, and mismatched stock information.

Another key issue was the difficulty of producing accurate reports for decision making. Management had to combine data from several sources by hand, which took time and invited human error. As a result, decisions about restocking, procurement, and sales forecasting were often based on incomplete or outdated information. This limited the company’s ability to respond quickly to changes in demand and supply.

To address these challenges, Lirios et al. [8] developed an automated inventory and sales management system using Agile methodology. The system was designed to bring all operational data into one platform, so that every transaction automatically updated both sales and inventory records in real time. This removed the need to cross check several files by hand and greatly reduced the risk of inconsistent data entry.

The system also introduced structured workflows for recording transactions, monitoring stock levels, and accessing business reports. These features let employees do daily tasks more efficiently, while management had immediate access to accurate and current information. Automation also reduced the dependence on individual memory and manual discipline, which were common sources of error in traditional systems.

The Agile approach also played a big part in the success of the system. By involving end users throughout development, the researchers could refine the system based on actual operational needs and not on assumptions. This step by step process made sure the final system matched how the business really worked, which made it more practical and easier to adopt.

The findings showed that the automated inventory system greatly improved data accuracy, reduced processing time, and improved overall workflow efficiency. It also made inventory records more reliable, which in turn improved the quality of decision making in the company. The reduced manual workload let employees focus more on operational tasks and less on correcting administrative errors.

This study was highly relevant to the proposed Smart Inventory and Parts Planning System for D Speedway Automotive Repair Shop. Like the company in Lirios et al. [8], D Speedway had relied on Excel based spreadsheets to manage sales and inventory data. This created the same weaknesses the study identified, including delayed updates, inconsistent records, and reliance on manual encoding for every transaction.

Both settings also shared an operating environment in the automotive industry, where inventory accuracy was critical to keeping repair work running smoothly. Inaccurate or outdated stock records could lead to service delays, unfinished repairs, and unhappy customers. The problems in the study closely matched the challenges D Speedway faced, particularly in making sure parts availability was always shown accurately.

By applying the principles highlighted in the study, namely centralized data management, automated transactions, and real time inventory synchronization, the proposed system aimed to remove the inefficiencies caused by manual spreadsheet tracking. This included reducing human error, improving report accuracy, and making sure stock levels were updated right after each transaction.

The study also gave strong reason to build a system that was both automated and user centered. Just as Lirios et al. showed the importance of matching system design to actual user workflows, the proposed system for D Speedway was designed to fit naturally into existing shop operations without needing major changes in staff behavior.

Overall, Lirios et al. [8] established that manual and fragmented inventory systems were no longer enough for businesses with frequent transactions and fast moving stock. Their findings strongly supported the move toward automated, centralized inventory systems, which gave a solid theoretical and practical foundation for developing a Smart Inventory and Parts Planning System for D Speedway Automotive Repair Shop.

Sison, Oreiro, Camalit, and Ng [9] conducted a study published in Innovatus titled “A Problem-Oriented Approach to Implementing an Inventory and Point-of-Sale System for Company KCP.” The study focused on the operational challenges of a business that relied heavily on manual and semi digital methods for handling sales transactions and inventory records. The researchers looked at how the company’s existing practices affected the efficiency of its operations, particularly transaction processing, stock monitoring, data organization, and reporting accuracy. Their work centered on understanding the problems people actually faced in the workplace and building a system that addressed those weaknesses directly, instead of adding technology only for the sake of modernizing.

The researchers found that the company’s Point-of-Sale and inventory process depended too much on manual data entry and disconnected record keeping. Employees had to encode transactions separately, and stock updates were done by hand after sales were completed. Since the sales process and inventory monitoring were not synchronized, the system was very open to human error. During busy hours, when many transactions happened at once, employees often struggled to keep records updated consistently. This led to delayed updates, missing entries, and inaccurate inventory counts that no longer matched the stock actually available in the business.

The study emphasized that the problem was not simply the use of spreadsheets or digital files. It was the lack of connection between the different records being kept. Sales records, inventory lists, and refund records existed separately, and employees had to compare files by hand whenever something needed to be verified. This

fragmented setup caused confusion and slowed down normal work, because employees could not immediately tell which records were accurate and current. The researchers explained that once inconsistencies began to appear across files, staff gradually lost confidence in the records and had to rely more on physical checking and manual confirmation than on the system itself.

Another major finding was the problem of data redundancy and weak data integrity caused by repeated and disconnected encoding. Since employees had to enter the same information into several files or sections, duplicate entries and conflicting data became very common. For example, a transaction might have been recorded correctly in the sales report but never reflected in the inventory file, or an inventory adjustment might have been entered more than once during reconciliation. Over time, these small inconsistencies built up and made reports less and less reliable for management.

The study pointed out that unreliable data caused more than administrative inconvenience. It also affected the quality of business decisions. Managers depended on accurate reports to decide which products were selling quickly, which items to reorder, and how much inventory to keep. When reports came from inconsistent records, purchasing decisions became less accurate and less efficient. In some cases, businesses reordered products they did not need, while in other cases they failed to restock important items on time because the system wrongly showed that inventory was still available.

The researchers also highlighted how inefficient inventory monitoring led to delays in the company's operations. Since stock levels were not updated in real time, employees often found shortages only while processing customer transactions. This reactive approach disrupted the workflow because customer requests could not always be filled right away. Employees then had to spend extra time confirming stock availability or checking storage areas by hand to see whether items were actually there.

The effects of these delays went beyond internal operations and reached customers as well. Customers who expected quick service had to wait longer because employees first had to settle inventory problems before completing transactions. The study explained that repeated delays and stock related interruptions could hurt customer satisfaction and lower confidence in the business. The company also overstocked slow moving items because it had no accurate way to track consumption or forecast demand. Without reliable records of how products actually moved, inventory decisions were based mostly on estimates and not on verified data.

To solve these problems, Sison et al. developed an automated Inventory and Point-of-Sale system that brought the major business operations into one platform. The system connected sales processing and inventory monitoring, so every completed transaction automatically updated the matching stock level in real time. This removed the need for separate manual inventory adjustments and greatly reduced the chance of inconsistent records across different files.

The system also introduced structured database management features that improved how information was organized and how reliable it was. Instead of depending on scattered spreadsheets and independently kept documents, the system stored all records in one centralized database where transactions, inventory data, and reports were connected. This kept all information synchronized and consistent throughout the system.

Another important contribution of the study was its focus on improving transaction speed and operational efficiency. Because employees no longer had to switch between spreadsheets or check records by hand during transactions, the time needed to process customer purchases dropped significantly. Routine administrative tasks that used to take large amounts of time became simpler through automation, which let employees focus more on customer service and their operational duties and less on repetitive paperwork and correcting data.

The researchers further stressed the importance of usability and practicality when building information systems for real business settings. They recognized that not all employees had advanced technical skills, especially in small and medium sized businesses where workers were more experienced in operations than in information technology. Because of this, the system was designed to be simple and accessible. Menus, transaction forms, and inventory functions were arranged so that employees could understand and use them with minimal training.

This user centered approach became one of the strengths of the study because it addressed a common problem in system adoption, which was resistance to change. The researchers explained that even technically advanced systems often failed when users found them difficult, confusing, or disruptive to their normal workflow. By designing the system around the actual needs and routines of employees, the study made it more likely that the technology would be accepted and used consistently in the organization.

The results of the study showed that the centralized Inventory and Point-of-Sale system greatly improved operational efficiency in the company. Data accuracy improved because transactions and inventory updates were synchronized automatically. Processing time dropped because employees no longer had to compare several files by hand or enter the same information again and again. Inventory monitoring became more reliable, which let management keep better control of stock levels and reduce both shortages and unnecessary overstocking.

The study also showed that automation could strengthen decision making in a business. Because reports came from organized and continuously updated records, management got faster access to accurate information about sales performance, inventory movement, and purchasing needs. This let the company make better informed operational decisions based on actual data and not on assumptions or delayed reports.

This study was highly relevant to the proposed Smart Inventory and Parts Planning System for D Speedway Automotive Repair Shop. Similar to the company examined by Sison et al. [9], D Speedway had relied on manual and spreadsheet based methods to handle sales and inventory management. The shop experienced many of the same difficulties the study identified, including inconsistent inventory updates, disconnected records, delayed reporting, and no real time stock visibility.

Both settings also involved transaction heavy operations where products and inventory moved constantly throughout the day. In an automotive repair shop like D Speedway, inventory management was especially important because repairs depended on having specific automotive parts available right away. Even small inaccuracies in stock monitoring could interrupt repair work, delay customer service, and create unnecessary stress for mechanics and staff.

The findings of the study clearly showed that the challenges at D Speedway were not isolated problems. They were common among businesses that kept relying on manual record management. The issues Sison et al.

identified, namely fragmented records, delayed inventory updates, inconsistent data, and inefficient reporting, closely matched the operational situation seen at D Speedway during the researchers' interviews and site visits.

By applying the principles in the study, particularly centralized data management, automated transactions, real time inventory synchronization, and simplified reporting, the proposed Smart Inventory and Parts Planning System aimed to remove many of the inefficiencies that had affected the shop. The proposed system let inventory levels update automatically after every transaction, reduced the need for repeated manual encoding, and gave staff a more reliable way to monitor stock availability.

The study also supported the importance of designing systems that fit naturally into the workflow of the business, instead of forcing employees to completely change how they worked. This principle mattered greatly for D Speedway because the proposed system was meant not only to improve technical operations but also to stay simple, understandable, and practical for the daily use of shop personnel.

Overall, the study conducted by Sison et al. [9] gave strong support for developing an integrated Smart Inventory and Parts Planning System. It showed how centralized and automated inventory management could improve accuracy, strengthen operational efficiency, reduce administrative workload, and support better business decisions. The study therefore served as both a practical reference and a strong conceptual foundation for the system developed for D Speedway Automotive Repair Shop.

Llabres [10], in the study titled "Auto Parts Inventory Management System," published in the International Journal of Scientific and Management Research (Vol. 6, No. 9), looked at how a computerized inventory management system could be used to manage the stock of an auto parts business. The study focused on the difficulty of tracking, recording, and monitoring a large and varied collection of automotive parts, where each item differed in price, supplier, demand, and rate of turnover. These were businesses where a single inaccurate stock count could mean a lost sale or a customer who had to be turned away.

The researcher explained that inventory management was central to the daily operations of auto parts businesses because they depended on having the right parts available at the right time. When inventory was recorded manually or through disconnected files, the records easily became outdated, incomplete, or inconsistent. Staff then had to spend extra time searching for items, confirming quantities, and checking records against the physical stock. The study pointed out that these problems grew as the number of parts and transactions increased, which made manual methods harder to sustain over time.

To address these challenges, Llabres [10] developed an auto parts inventory management system that placed product records, stock quantities, and transactions in one computerized platform. Instead of relying on handwritten logs or separate spreadsheets, the system stored inventory information in a central database where it could be searched, updated, and reviewed more easily. This reduced the amount of manual recording needed and made current stock information easier to reach for the people who needed it.

The study also emphasized that an inventory system was more than a place to store data. A well-designed system improved accountability and efficiency by keeping organized records of what was received, what was sold, and what remained on hand. With accurate and accessible records, owners and staff could make purchasing

decisions based on actual stock information instead of memory or estimation. Time that used to go into record keeping and stock checking could be given to serving customers.

The findings of Llabres [10] were relevant to the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop because the problems in the study closely matched those the shop had experienced. Like the auto parts business in the research, D Speedway handled many kinds of parts, such as oils, filters, spark plugs, and other replacement items, each with its own price, usage rate, and stock level. The shop also managed this inventory through manually maintained Excel spreadsheets, which led to delayed updates, inconsistent records, and difficulty in knowing which parts were actually available.

The study supported the need for a centralized database in the proposed system. Before development, D Speedway kept inventory, sales, and purchase records in separate files that were updated by hand. By storing parts, stock levels, and transactions in one connected database, the proposed system removed the need to cross-check several files and gave staff a single, current source of inventory information. This reflected the approach of Llabres, where centralized records made inventory easier to monitor and maintain.

The study also reinforced the value of accurate stock records for decision-making. The features planned for the proposed system, including automatic inventory updates after every transaction, low-stock alerts, forecasting, and reorder recommendations, built on this idea. By keeping stock counts current, the system could warn the owner before a part ran out and suggest what to reorder, which helped the shop avoid both stockouts in the middle of a repair and overstocking of parts that were not needed.

Overall, the study by Llabres [10] gave additional support for developing the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop. It showed that a computerized inventory system could improve record accuracy, reduce manual workload, and give businesses more reliable information for managing automotive parts. These findings reinforced the value of a centralized and automated system for the inventory needs of D Speedway.

2.3 Related Literature

Several foundational concepts from business and software development literature directly shaped the design and objectives of the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop. These were not abstract theories — they were practical principles that explained why the problems previously experienced at D Speedway existed, why manual systems eventually failed to keep up with the demands of a growing business, and why the features built into the proposed system were designed the way they were.

Inventory Management. Inventory management was one of the most critical operational functions in any business that depended on physical stock to deliver its services. The core principle was straightforward: businesses had to avoid two failure modes at the same time. The first was overstocking, where too many parts

were ordered and stored, tying up money that could have been used elsewhere in the business and creating the risk that slow-moving items would become obsolete before they were ever used. The second was understocking, where too few parts were on hand to meet demand, leading to delays, incomplete repairs, and customers who were forced to wait or take their business elsewhere. Both outcomes carried real costs, and in automotive repair environments, where customers expected their vehicles to be serviced within a predictable timeframe, the consequences of either extreme showed up quickly in the form of lost revenue and damaged reputation.

What made inventory management genuinely difficult was that the right balance was not a fixed number—it shifted constantly based on how quickly parts were being used, what new vehicles were coming into the shop, and what seasonal patterns existed in demand. A business relying on manual records and personal memory to track all of this was essentially asking one or two people to maintain a running calculation across dozens or hundreds of individual parts, update it after every transaction, and catch any developing shortage before it disrupted operations. This was not a sustainable approach. It worked at a very small scale and for a short period of time, but as the volume of transactions increased and the variety of parts grew, manual tracking became a source of error rather than a source of control. The proposed system directly addressed this by automating the tracking process, removing the dependency on individual memory, and making the current state of inventory visible at any moment without requiring anyone to calculate or compile it manually.

Real-Time Inventory Monitoring. One of the most consistent findings across inventory management literature was that the timing of information mattered as much as its accuracy. A record that accurately reflected stock levels from three days ago was not the same as a record that reflected stock levels right now, and in a busy automotive repair shop where multiple repairs happened throughout the day and parts were pulled from storage constantly, that gap between what the record said and what was actually on the shelf could grow very quickly. When inventory records were updated manually at the end of the day, or only when someone remembered to do it, there was always a window of time during which the numbers in the system did not match the physical reality of the storage area. During that window, a mechanic might have been sent to retrieve a part that the records showed as available but that had actually been used two hours earlier and never logged. The repair stopped, time was wasted, and the customer waited.

Real-time inventory monitoring eliminated that window by connecting every transaction directly to the inventory count. When a part was recorded as used or sold, the system immediately adjusted the corresponding stock level without any separate manual step. This meant the records were always current, and anyone checking the system at any point during the day saw an accurate picture of what was actually available. For D Speedway, where the research team observed that inventory shortages were sometimes discovered only after a repair was already underway, real-time monitoring was not a luxury feature — it was the core of what made the proposed system more reliable than the spreadsheet-based approach previously in use. The difference between finding out a part was gone before the repair started versus finding out in the middle of it was the difference between a managed situation and a disrupted one.

Sales Reporting and Business Decision-Making. Accurate and accessible sales data was one of the foundations of sound business decision-making, and the speed at which that data could be retrieved was directly connected to how useful it actually was. When a business owner had to manually pull figures from multiple files,

reconcile inconsistencies between records, and compile everything into a usable format before any analysis could begin, the reporting process itself became a barrier. It took time that could have been spent on other things, it introduced the possibility of calculation errors during compilation, and it produced a result that reflected the state of the business several hours or days ago rather than right now. For a small automotive repair shop where purchasing decisions, staffing adjustments, and service priorities all depended on knowing which parts were moving and how fast, a reporting process that was this slow and this labor-intensive was a genuine operational problem.

The proposed system addressed this by generating sales summaries and inventory reports automatically from the data that was already being recorded through normal transactions. No separate compilation step was needed. The owner of D Speedway could check how a particular part had been selling over the past week, or pull the total value of parts used during the current month, in a matter of seconds rather than hours. This changed the relationship between the business owner and the data. When reports were easy to access, they got consulted more often. When they got consulted more often, purchasing decisions were grounded in actual patterns rather than estimates. When purchasing decisions were grounded in actual patterns, the business was better positioned to maintain the right stock levels, avoid unnecessary expenses, and respond quickly when demand shifted.

Reorder Point and Demand Forecasting. A reorder point was the specific stock level at which a new order for a particular part should be placed, calculated based on how quickly that part was being consumed and how long it took for a new delivery to arrive. The logic behind it was simple but powerful: rather than waiting until a part completely ran out before placing an order — by which point it was already too late to avoid a disruption — the business set a threshold in advance and placed the order while there was still enough stock to keep operating through the lead time. Done correctly, this meant the shop never actually ran out of a part during normal operations. The new stock arrived before the existing stock was fully depleted, and the workflow continued without interruption.

Applying this in practice required two things: accurate data about how fast each part was being used, and a system that monitored stock levels continuously and raised an alert when the reorder threshold was reached. Neither of these was possible without a properly functioning inventory management system. If the usage data was unreliable because records were updated inconsistently, the reorder point calculation would be off, and the alert would either fire too early — prompting unnecessary orders — or too late — after the shortage had already begun. If the stock levels were not monitored in real time, the alert might not fire at all until the next time someone manually checked the records, by which point the part might already be gone. The proposed system for D Speedway combined accurate real-time stock tracking with automated low-stock alerts and usage-based forecasting, giving the shop the tools to stay ahead of shortages rather than reacting to them after the fact. Over time, as the system accumulated more transaction history, the forecasting became more precise because it drew on a longer and more complete record of actual demand patterns.

System Design and Usability. A system that was technically well-built but difficult to use in practice would not deliver the results it was designed to produce. This was one of the most consistent findings in software development and information systems literature, and it was particularly relevant in environments like automotive repair shops, where the people using the system every day were not technology specialists. They

were mechanics, service staff, and shop managers whose primary focus was on vehicles and customers, not on navigating software interfaces. If a system required complicated data entry, a large number of steps to complete a basic task, or a significant adjustment in how staff members went about their normal routine, the likelihood that it would be used consistently and correctly dropped considerably. Systems that were resisted or used incorrectly did not improve the operational problems they were built to solve — they just added a new layer of complexity on top of the existing ones.

Good system design in this context meant building software around the actual workflow of the business rather than asking the business to adapt to the software. It meant making the most common tasks — recording a transaction, checking stock, reviewing a sales summary — as simple and fast as possible. It meant reducing the number of steps required to complete each action, presenting information in a format that was immediately understandable without interpretation, and minimizing the chance that a user would make an error during normal use. The decision to observe D Speedway's daily operations before any design decisions were made, and to involve staff members in testing before the system was finalized, reflected this principle directly. The proposed system was shaped by how the shop actually worked, not by assumptions about how an inventory system should function in theory.

Web-Based Systems and Multi-User Accessibility. Building the proposed system as a web-based platform rather than a desktop application was a practical decision grounded in how small automotive businesses actually operated. In a repair shop environment, multiple people interacted with inventory and sales data throughout the day — mechanics pulling parts, counter staff recording sales, and the owner reviewing stock levels and reports. If the system had lived on a single desktop computer, access would have become a bottleneck. Only one person could have used it at a time, and anyone who was not physically at that machine could not have checked or updated the records without walking over to it. This would have created delays and increased the likelihood that updates would be deferred until someone was at the right machine, which was exactly the kind of inconsistency that led to inaccurate records.

A web-based system removed this constraint entirely. Because it ran through a browser rather than a locally installed program, it could be accessed from any device on the network — a desktop at the front counter, a tablet in the storage area, or a phone in the workshop — without any additional setup. All users interacted with the same centralized database at the same time, which meant that when one person recorded a transaction, everyone else immediately saw the updated stock level. There was no version conflict, no duplicate file, and no risk of one person's update overwriting another's. For D Speedway, where the owner and staff were often in different parts of the shop and needed current information at different points in the day, this kind of shared real-time access was not a minor convenience — it was a fundamental requirement for the system to work as intended. A centralized web-based platform ensured that the inventory records the owner looked at were the same records the mechanics worked from, at all times.

2.4 Synthesis of the Reviewed Literature

Looking at everything reviewed in this chapter together, one thing was hard to ignore: the problems D Speedway dealt with were not unique to the shop. Every study covered in this chapter, regardless of where it

was conducted or what kind of automotive business it examined, kept running into the same story. A small business started out managing its inventory manually — spreadsheets, handwritten logs, or a combination of both. It worked well enough at first, when the number of transactions was low and the variety of parts was still manageable. But as the business grew and the volume of daily activity increased, the manual approach started to crack. Records fell behind. Stock counts became unreliable. The owner made purchasing decisions based on memory rather than data. Parts ran out without warning. Reports took hours to put together and still did not tell the full story. These were not the kinds of problems that happened because of poor management or careless employees — they happened because manual systems had a ceiling, and once a business grew past that ceiling, the system started working against it instead of for it.

What the reviewed literature established very clearly was that this pattern repeated itself across different types of automotive businesses and different scales of operation. Whether the business was a spare parts store, a motorcycle shop, or a general repair shop, the moment it started handling a volume of transactions that a spreadsheet could not comfortably keep up with, the same set of problems appeared. Inventory records drifted away from reality because updates depended on people remembering to do them consistently, and in a busy shop environment, inventory updates were almost always the first thing that got skipped when things got hectic. Stock shortages were discovered only after they had already caused a problem — a mechanic was already mid-repair when someone realized the needed part was gone, or a customer was already waiting when staff discovered the part they ordered was actually out of stock. Reports that should have informed daily decisions instead got compiled at the end of the week or the end of the month, by which point whatever information they contained was too late to act on. The businesses in the reviewed literature were not struggling because they were doing something wrong. They were struggling because the tools they were using were not built for the kind of work being asked of them.

The foreign studies covered in this chapter were particularly useful because they went beyond identifying the problem and actually documented what happened when a proper digital system was put in place. Across every case where an automated inventory management system was developed and implemented, the results followed the same pattern: inventory accuracy improved, stock-related disruptions decreased, the time staff spent on manual administrative tasks dropped significantly, and the owner gained access to reliable business information without having to compile it by hand. These were not small improvements — they represented a fundamental change in how the business operated on a daily basis. When inventory records updated automatically after every transaction, the owner stopped guessing and started knowing. When low-stock alerts fired before a part ran out rather than after, the shop stopped reacting to shortages and started preventing them. When sales reports were generated automatically from stored data, business decisions were made from facts rather than estimates. The literature showed, consistently and without exception, that these outcomes were achievable for small businesses of exactly the kind and scale that D Speedway represented.

The local studies added something important to this picture that the foreign research alone could not provide. The challenges documented in automotive businesses closer to home confirmed that the same operational problems experienced at D Speedway were not just a general business phenomenon — they were present specifically in small automotive shops operating under conditions that closely mirrored D Speedway's own situation. Small staff, growing transaction volume, inventory managed through spreadsheets, no centralized

system, purchasing decisions driven by personal judgment rather than tracked data. The businesses described in the local studies dealt with exactly this, and the findings made clear that the problems were structural rather than circumstantial. They did not go away on their own as the business matured. If anything, they compounded. The longer a business ran on a manual system past the point where that system could handle the load, the more embedded the habits became and the harder the eventual transition got. The local studies did not just confirm that the problem existed — they confirmed that it existed in the same context as D Speedway, which made their findings directly applicable to this project.

Taken together, the reviewed literature pointed to four things that any effective inventory system for an automotive repair shop needed to be able to do. First, it needed to update inventory records automatically every time a transaction occurred, without requiring any separate manual step. The moment human memory or human consistency became a dependency in the update process, errors became inevitable. Second, it needed to make current stock levels visible in real time so that the gap between what the records said and what was physically available was eliminated rather than just reduced. Third, it needed to be capable of tracking usage patterns over time and using that data to anticipate when parts would need to be reordered, so the shop could act on that information before a shortage occurred rather than after. Fourth, it needed to be simple enough to use reliably in a busy shop environment where the people operating it were not technology specialists and could not afford to spend significant time learning complicated software. These were not wish-list features — they were the minimum set of capabilities that the literature consistently showed were necessary for an inventory system to actually solve the problems it was supposed to solve.

One more thing that came through clearly across the reviewed studies was how much the design process mattered, not just the design itself. The systems that worked well — the ones that got adopted by staff and continued to be used consistently after implementation — were all built through a process that started with understanding the actual business before making any technical decisions. The developers visited the businesses in person, observed how daily operations worked, talked to the people who would be using the system, and let those observations shape the design. The systems that were built purely on technical assumptions without this kind of grounding tended to produce tools that staff found confusing or that did not fit naturally into how the work actually got done. For a shop like D Speedway, where the goal was a system that mechanics and counter staff could use confidently in the middle of a busy workday without it becoming another thing to worry about, this principle was not just worth noting — it was central to how this project was approached.

What did not yet exist in the literature was a system that brought all of these elements together in a single integrated tool built specifically for a small automotive repair shop of D Speedway's profile. The reviewed studies showed pieces of the solution working in various settings — real-time tracking here, demand forecasting there, automated reporting somewhere else, usability-centered design in another. Each of those contributions was valuable and each one informed a part of what was built for this project. But none of them was built for this shop, and none of them addressed the specific combination of needs that D Speedway had: a young and growing repair business, operating with limited staff, handling a wide variety of vehicle types, managing a parts inventory that was already becoming too complex for spreadsheets, and needing a system that was both capable enough to handle that complexity and simple enough to be used by people whose expertise was in fixing vehicles, not managing software. That was the gap this project was designed to fill. The proposed Smart

Inventory and Parts Planning System for D Speedway was not adapted from something built for a different context. It was developed from the ground up, shaped by the consistent findings of the reviewed literature and by direct knowledge of how this specific shop operated day to day. The literature told what the system needed to accomplish. The actual conditions at D Speedway told how it needed to accomplish them.

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CHAPTER III

METHODOLOGY

3.1 Research Design

This study employed a descriptive-developmental research design to guide the development of the proposed Smart Inventory and Parts Planning System for D Speedway Auto Repair Shop. This research design was appropriate because the study sought not only to examine the existing inventory management practices of the business but also to develop a system that addressed the operational issues identified during the investigation. By combining description and development within a single framework, the study established a direct connection between the problems observed in the existing process and the solutions incorporated into the proposed system.

The descriptive phase of the study focused on examining the current inventory and sales management procedures of D Speedway Auto Repair Shop. Before the development of the system began, the researchers established a thorough understanding of the shop's existing operations. This phase provided a clear picture of how inventory was monitored, how sales transactions were recorded, how stock levels were tracked, and how purchasing decisions were made. Since every business had its own workflow, practices, and operational challenges, understanding the specific environment of D Speedway was essential in ensuring that the proposed system met its actual needs.

Information was gathered through direct observation of daily activities within the shop and through interviews with the owner and staff members who were directly involved in inventory and sales-related tasks. Observing actual work processes provided firsthand information on how operations were carried out instead of relying only on existing records or assumptions. Particular attention was given to the methods used for recording transactions, updating inventory records, monitoring available stock, and preparing reports. Interviews with personnel also helped identify the common difficulties encountered during daily operations and the information needed to perform their responsibilities more efficiently.

The descriptive phase also included a review of the documents and records used by the business, such as inventory lists, sales records, purchase records, receipts, and manually prepared reports. Examining these documents helped verify the information gathered through observation and interviews while providing additional details about the existing record-keeping process. This review made it possible to identify inconsistencies, incomplete records, and other issues that affected the accuracy of inventory management.

Through these activities, the study identified operational challenges that affected the efficiency and accuracy of inventory management. The existing practices revealed issues such as delayed inventory updates, inaccurate stock records, difficulties in monitoring inventory levels, and the time required to prepare reports manually. The study also examined how these challenges affected decision-making, customer service, and overall business operations. Understanding these concerns was important because they served as the basis for determining the features and functions that were included in the proposed system.

The descriptive phase played a significant role in the success of the project because it established a clear foundation for system development. Rather than adopting features commonly found in generic inventory systems, the proposed solution was designed according to the actual requirements of D Speedway Auto Repair Shop. This approach helped ensure that every feature included in the system had a specific purpose and directly addressed the problems identified during data gathering.

Following the completion of the descriptive phase, the developmental phase began. During this stage, the Smart Inventory and Parts Planning System was designed and developed based on the information gathered from the business. The development process focused on creating a system capable of improving inventory monitoring, simplifying record management, and providing accurate information to support daily operations and decision-making.

The proposed system included features intended to address the operational gaps identified during the descriptive phase. These features consisted of automatic inventory updates whenever a transaction was recorded, real-time monitoring of stock quantities, low-stock notifications, demand forecasting, reorder recommendations, and automated report generation. Each component was developed to reduce manual work, minimize human error, and improve the accessibility of inventory information.

An iterative approach was followed throughout the developmental phase. Instead of completing the entire system in a single development cycle, individual components and functions were continuously reviewed and refined as development progressed. This approach allowed improvements to be made whenever issues related to usability, functionality, or workflow compatibility were identified. Necessary revisions were based on observations and feedback gathered during system demonstrations and testing.

As different features became available, the owner and staff members were given opportunities to interact with the system and evaluate its functionality. Their observations and suggestions contributed to the refinement of the system, ensuring that its interface and workflow remained consistent with the actual practices of the shop. This process helped ensure that the completed system supported existing operations without creating unnecessary complications for its users.

After the development process was completed, the system underwent testing and evaluation within the actual operating environment of D Speedway Auto Repair Shop. Conducting the evaluation in the workplace made it possible to assess the performance of the system under normal operating conditions. Actual inventory transactions, stock monitoring activities, and reporting tasks were performed using the system to determine whether it functioned effectively in supporting daily operations.

The evaluation focused on determining whether the system improved inventory accuracy, enhanced stock monitoring, reduced the time required to manage records, and provided useful information for inventory planning and purchasing decisions. The usability of the system was also assessed to determine whether users could perform their tasks efficiently with minimal difficulty. Any limitations identified during testing were documented and considered in evaluating the overall effectiveness of the proposed system.

The descriptive-developmental research design provided a structured process for achieving the objectives of the study. The descriptive phase established a detailed understanding of the existing operational environment, while the developmental phase transformed those findings into a practical technological solution. By maintaining a clear relationship between the identification of operational problems and the development of system features, the study ensured that the proposed Smart Inventory and Parts Planning System addressed the actual needs of D Speedway Auto Repair Shop. Through this approach, the developed system contributed to more efficient inventory management, improved planning of automotive parts, more accurate record-keeping, and better overall operational performance.

3.2 System Development Methodology

The team followed the Agile methodology throughout the development process. The reason for choosing Agile over something like Waterfall was simple: when building for a real business, requirements tend to shift once people actually see what the system looks like. Agile accommodates that by building in short cycles. Each cycle produced a working piece of the system that could be tested and adjusted before moving forward. This iterative approach allowed the development team to respond effectively to feedback received during each stage of development instead of waiting until the project was nearly complete. Since the proposed Smart Inventory and Parts Planning System was intended to support the day-to-day operations of D Speedway Auto Repair Shop, it was important that the development process remained flexible enough to accommodate changes in user requirements and workflow improvements. Agile encourages continuous communication between developers and users, allowing issues to be identified early and improvements to be incorporated immediately into succeeding iterations. Rather than treating system development as a fixed sequence of activities, Agile views development as an evolving process where continuous refinement contributes to a more reliable and user-centered solution. This methodology also promoted collaboration among the researchers and the shop personnel, ensuring that the resulting system remained aligned with the actual operational needs of the business throughout the entire development process. [7], [8]

The Waterfall model, by comparison, works by completing one phase entirely before moving on to the next. Planning is done first, then design, then development, then testing, in that strict order. The problem with applying that to a project like this one was that D Speedway's staff might not fully know what they needed from the system until they actually saw a version of it running. Something that sounds good in a requirements document might turn out to be confusing in practice, or a feature that nobody thought to mention during planning might turn out to be essential once the team started using the system day to day. Waterfall does not leave much room to go back and adjust once a phase is closed. Agile does, and that flexibility was exactly what this kind of project called for. In projects involving small and medium-sized businesses, operational requirements often become clearer only after users begin interacting with an early version of the system. Because of this, following a rigid development sequence could increase the likelihood of producing a system that technically satisfies the documented requirements but fails to address practical workplace needs. Any necessary modifications identified late in the development process would require significant time and effort if a traditional Waterfall approach were followed. Agile minimized this risk by encouraging regular evaluation, continuous refinement, and incremental improvements throughout development. As feedback was gathered after

each development cycle, adjustments could be implemented without disrupting the overall progress of the project, resulting in a system that evolved together with the users' understanding of their own operational requirements.

Another reason Agile fit here was that it kept the development grounded. Instead of building everything and then testing it all at once at the very end, Agile delivered working pieces of the system regularly throughout the process. That meant problems got caught early, when they were still easy to fix, rather than at the end when the cost of going back would have been much higher. It also meant D Speedway could start seeing what the system looked like and how it worked before it was fully finished, which gave the team a chance to weigh in while there was still time to make changes. That kind of ongoing involvement from the people who would actually use the system tended to produce a much better end result than handing something over at the finish line and hoping it matched what was needed. Continuous user participation also increased confidence in the development process because the owner and staff members were able to observe how the system gradually evolved from one iteration to the next. Their participation allowed the researchers to verify whether implemented features accurately reflected actual business processes and whether additional adjustments were needed before development continued. This collaborative approach strengthened communication between developers and users while reducing misunderstandings that commonly occur when users are only involved during the beginning and end of a project. Furthermore, identifying issues during the early stages of development significantly reduced the possibility of costly revisions later in the project. As a result, the Agile methodology contributed not only to technical improvements but also to the overall quality, usability, and practicality of the Smart Inventory and Parts Planning System developed for D Speedway Auto Repair Shop.

The development moved through the following phases:

Planning: This phase was about getting a clear picture of where the shop stood before anything got built. That meant sitting down with the owner and walking through how things were currently being handled: which spreadsheets were being used, who updated them, how often, and what kind of information was being tracked in each one. It also meant looking at where the process broke down: where entries got missed, where files conflicted, where the owner was spending time doing things manually that the system would eventually handle automatically. The goal of this phase was not just to understand the problem in general terms but to map it out specifically enough that the team knew exactly what the system needed to do and in what order those things needed to be built. Without a clear plan going in, even a well-designed development process tends to drift, and time gets spent on things that are not actually the priority. Planning kept the project focused from the start. In addition to identifying operational issues, this phase established the overall direction of the project by defining its objectives, scope, expected outputs, and development priorities. The researchers carefully reviewed the existing workflow of D Speedway Auto Repair Shop to determine which business processes required immediate improvement and which could be incorporated into future enhancements. This helped ensure that development efforts remained concentrated on solving the most significant operational problems first. Planning also included identifying the resources required for development, estimating the time needed for each activity, and organizing development tasks into manageable iterations consistent with the Agile methodology. Potential risks, such as incomplete requirements, data inconsistencies, or implementation delays, were also considered during this stage

so that appropriate mitigation strategies could be prepared. By investing sufficient time in planning before coding began, the development team established a clear roadmap that guided every succeeding phase of the project and minimized unnecessary revisions during implementation.

Requirements Gathering: If the planning phase was about understanding the problem from a high level, requirements gathering was about getting into the details. This was where the team spent time at the shop, watching how work actually flowed throughout the day, sitting with staff as they handled transactions and checked inventory, and asking the kinds of questions that only come up when seeing the process in person rather than just hearing it described. Interviews were conducted with the owner, the mechanics, and whoever was responsible for managing the spreadsheets, because each of those people had a different relationship with the data and a different set of needs from the system. What the owner needed to see in a report was not the same as what a mechanic needed to know when checking stock, and what the staff member updating inventory needed was different again. All of those perspectives had to be captured and accounted for before the design work began, because trying to add them in later almost always means redoing work that has already been done. The output of this phase was a clear, specific list of requirements: what the system had to do, what it should do, and what was out of scope for this version. This phase served as the foundation for all succeeding development activities because the quality of the finished system depended heavily on the accuracy and completeness of the requirements gathered. During this stage, the researchers documented both functional and non-functional requirements to ensure that the system would satisfy not only operational needs but also expectations regarding usability, reliability, efficiency, and performance. Functional requirements described the specific activities that the system should perform, such as recording sales transactions, updating inventory records automatically, generating reports, issuing low-stock notifications, and supporting demand forecasting. Non-functional requirements included considerations related to system responsiveness, ease of navigation, data accuracy, security, and maintainability. Throughout the requirements gathering process, continuous communication with the owner and staff helped clarify uncertainties and validate collected information. Whenever inconsistencies or differing opinions arose, additional discussions were conducted to ensure that the documented requirements accurately reflected the actual needs of the business. The resulting requirements specification became the primary reference throughout the design and development phases, reducing ambiguity and ensuring that every implemented feature could be traced back to a clearly identified business requirement.

System Design: With the requirements in hand, this phase focused on translating what the system needed to do into a concrete plan for how it would be built. The database structure had to be designed to hold all the relevant information (parts, stock levels, transactions, and sales history) in a way that was organized, efficient, and easy to query when reports were needed. The interface layout had to be thought through from the perspective of the people who would be using it every day, most of whom were not technical users. Buttons needed to be where people expected them. Information needed to be displayed clearly without requiring any kind of interpretation. The fewer steps it took to complete a common task, the better. System logic, meaning the rules that govern how data flows and how the system responds to different actions, was also defined here. When does a low-stock alert trigger? How does a sale automatically update the inventory count? What happens when a part is restocked and added back to inventory? These were the kinds of questions that had to be answered at the design stage so that

the development work that followed had a solid blueprint to build from rather than decisions being made on the fly.

During this phase, the researchers prepared the overall architecture of the Smart Inventory and Parts Planning System to ensure that every component functioned together as a complete and integrated solution. Relationships among different data entities were carefully defined to maintain consistency, prevent duplication of records, and support efficient retrieval of information whenever reports or inventory inquiries were generated. The user interface was designed with simplicity and accessibility in mind so that employees with limited technical knowledge could easily learn and operate the system with minimal training. Navigation menus, forms, buttons, and report layouts were arranged logically to support the natural workflow already practiced within D Speedway Auto Repair Shop. Consideration was also given to error prevention by incorporating input validation, confirmation messages, and automated calculations that reduced the likelihood of incorrect data entry. In addition, the researchers prepared process flow diagrams and database relationships that served as references during programming, ensuring that the implemented system remained consistent with the approved design specifications. By thoroughly planning both the technical structure and user experience before coding began, the design phase minimized development errors and established a strong foundation for successful system implementation.

Development: This was the longest phase of the project and the one where the actual system started to take shape. The decision to build feature by feature rather than all at once was a deliberate one. Starting with sales recording and inventory tracking made sense because those were the most fundamental parts of the system, and everything else depended on that data being captured correctly. Once those core features were working and tested, the team moved on to the next layer: generating reports the owner could actually use, building in the reorder alerts that flagged low stock before it became a problem, and adding the forecasting tools that helped the shop plan ahead based on historical patterns. Each feature went through its own round of testing before development moved forward. That way, if something was not working the way it should, the problem got caught immediately while the code was still fresh and the fix was straightforward. Building everything first and testing everything at the end is how a system ends up full of interconnected problems that are hard to untangle. Testing as the team went kept the build clean and the progress steady. Throughout the development phase, the Agile methodology allowed the researchers to implement system components incrementally while continuously evaluating whether each completed feature satisfied the documented requirements. Every development iteration produced a functional portion of the system that could immediately be reviewed and validated before additional features were introduced. This incremental approach reduced development risks because errors and inconsistencies could be corrected before they affected other components of the application. During programming, particular attention was given to ensuring the accuracy of inventory computations, synchronization of sales transactions with inventory records, proper generation of reports, and reliable execution of forecasting and reorder recommendation functions. Security measures, including user authentication and controlled access to system functions, were also incorporated to protect sensitive inventory and business information. As development progressed, revisions were made whenever improvements in functionality, interface design, or workflow efficiency were identified through continuous testing and stakeholder feedback. This flexible development process ensured that the completed Smart Inventory and Parts Planning System

remained aligned with the operational practices of D Speedway Auto Repair Shop while achieving the project's objectives of improving inventory management, reducing manual work, increasing record accuracy, and supporting better inventory planning and business decision-making.

Testing and Evaluation: Testing happened in layers, and each layer served a different purpose. Unit testing checked that each individual component of the system worked correctly on its own: that a sale got recorded properly, that the inventory count updated to the right number, that a report pulled the correct figures. Integration testing then checked that all of those individual components worked correctly together, because something that functions fine on its own can still break when it interacts with another part of the system. These two rounds of testing were done by the development team and focused on catching technical issues before the system was handed over to anyone outside the team. Conducting testing in multiple stages ensured that errors could be identified systematically and corrected before they affected the overall functionality of the application. During unit testing, each module was executed independently to verify that its inputs, processes, and outputs behaved according to the intended design specifications. This included validating inventory computations, confirming the accuracy of sales recording functions, verifying report generation, and ensuring that automatic stock updates were performed correctly after every transaction. Once individual modules demonstrated satisfactory performance, integration testing was conducted to verify that data flowed correctly between interconnected components. This stage examined whether transactions initiated in one module were accurately reflected in other related modules, ensuring consistency throughout the system. The researchers also evaluated the interaction between the database, application logic, and user interface to confirm that information remained synchronized and that system responses were accurate under normal operating conditions. Any inconsistencies, software defects, or performance issues discovered during testing were documented, analyzed, and corrected before proceeding to the next stage of evaluation. This systematic testing process contributed significantly to the overall reliability, stability, and quality of the Smart Inventory and Parts Planning System by ensuring that each component functioned correctly both independently and as part of the complete application.

User acceptance testing was different. This was where real staff at D Speedway, the people who would actually be using the system day to day, sat down with it and tried to use it the way they would in a normal workday. They recorded sales, checked stock levels, ran reports, and navigated through the system the way they would once it was deployed. The goal here was not just to find bugs, though that was part of it. It was to find out whether the system actually worked for the people it was built for: whether the interface made sense to them, whether the flow of tasks felt natural, and whether there was anything they expected the system to do that it did not do. Feedback from this round of testing was taken seriously and folded back into the system before the final version was delivered. A system that the development team thinks is finished but that the actual users find confusing or incomplete is not a finished system. User acceptance testing was what closed that gap. During this phase, the owner and employees evaluated the system within the context of their actual daily responsibilities, allowing the researchers to observe how effectively the application supported real operational activities. Users were encouraged to complete routine inventory management tasks without extensive guidance so that their natural interaction with the system could be observed. This helped determine whether menus, forms, reports, notifications, and navigation controls were intuitive enough for regular business use. Participants also assessed the responsiveness of the system, the clarity of displayed information, and the overall ease of completing

common inventory and sales transactions. Any observations regarding confusing workflows, unnecessary steps, missing features, or interface improvements were carefully documented and reviewed by the development team. Whenever feasible, suggested improvements were incorporated into subsequent revisions before the final deployment of the system. The feedback obtained during user acceptance testing also served as an important measure of the system's usability, practicality, and readiness for implementation because it reflected the experiences of the individuals who would rely on the application during everyday business operations. Successfully completing this phase demonstrated that the Smart Inventory and Parts Planning System was not only technically functional but also capable of supporting the operational requirements of D Speedway Auto Repair Shop in a manner that was efficient, user-friendly, and aligned with established business practices. By incorporating user feedback into the final version of the application, the researchers ensured that the completed system addressed both the technical objectives of the project and the practical expectations of its intended users, resulting in a solution that effectively improved inventory management, inventory planning, and overall operational efficiency.

3.3 System Architecture

The Smart Inventory and Parts Planning System for D Speedway was built on a three-layer architecture. This kind of structure is a standard and well-established approach for web-based information systems, and the reason it is used so widely is not because it looks good on a diagram; it is because it actually makes systems easier to build correctly, easier to fix when something goes wrong, and easier to update later without breaking everything else. When the visual design, the processing logic, and the data storage are kept separate from each other, a change in one layer does not automatically create problems in the other two. That separation mattered a lot in a real working environment because needs change over time. A new report format, a new field in the inventory form, or a new alert condition: any of these could be addressed without touching the parts of the system that were already working. For a small business like D Speedway that might want to expand or adjust the system as it grows, that flexibility was not a minor technical detail. It was what made the system sustainable beyond just the first month of use. The adoption of a three-layer architecture also contributed to the overall quality, maintainability, and scalability of the proposed system. By organizing the application into clearly defined layers, the researchers were able to assign specific responsibilities to each component, making the overall system easier to understand, develop, and maintain. This architectural approach reduced unnecessary dependencies between system components and promoted a modular design in which each layer could perform its designated function independently while still working together as part of a unified application. Such an arrangement simplified debugging and troubleshooting because problems could be isolated within a specific layer instead of requiring extensive investigation across the entire system. Furthermore, separating responsibilities made future enhancements more manageable, allowing developers to introduce additional features, modify business rules, or improve the user interface without significantly affecting the remaining components of the application. This architectural design also improved code organization, encouraged software reusability, and reduced the likelihood of introducing errors during future system updates. From a long-term perspective, implementing a layered architecture increased the reliability of the Smart Inventory and Parts Planning System by ensuring that

changes could be implemented systematically while preserving the stability of existing functions. As the operational requirements of D Speedway Auto Repair Shop continue to evolve, this architectural structure provides the flexibility needed to accommodate future improvements, additional modules, and increased transaction volumes without requiring a complete redesign of the application.

Each of the three layers had a distinct role, and together they handled every interaction that happened when the owner or manager used the system, from the moment they logged in to the moment a sales report was generated and printed. The following sections describe each layer in detail, explain what it was responsible for, and trace how data moved through the entire system during a typical transaction. The interaction among these layers followed a structured sequence in which every user action initiated a controlled flow of information throughout the application. Whenever a user performed an activity, such as recording a sale, updating inventory information, adding new automotive parts, or generating reports, the request first passed through the presentation layer, where user input was collected and validated. The request was then forwarded to the application layer, where the system processed the information according to predefined business rules and operational logic. Finally, the processed information was stored, retrieved, or updated within the database through the data layer before the results were returned to the user interface for display. This organized flow ensured that each layer performed only the responsibilities assigned to it, reducing unnecessary complexity while improving overall system efficiency. The clear separation of responsibilities also enhanced data consistency because all validation, processing, and database transactions followed standardized procedures before information became available to users. As a result, inventory records, sales transactions, forecasting data, reorder recommendations, and generated reports remained synchronized throughout the system. This structured communication among the presentation layer, application layer, and data layer enabled the Smart Inventory and Parts Planning System to provide accurate, reliable, and responsive services that supported the daily inventory management activities of D Speedway Auto Repair Shop. In addition, the layered architecture established a strong technical foundation for future system maintenance and expansion, ensuring that the application could continue to support the growing operational needs of the business while maintaining consistent performance, security, and reliability.

User Interface Layer

The User Interface Layer was what the owner and manager actually saw when they opened the system on their browser. It was the part of the system that the user interacted with directly: every button they clicked, every form they filled out, every number they read on a report came from this layer. Because D Speedway's staff were automotive professionals and not IT specialists, this layer was designed with one guiding principle: if someone with no technology background could not figure out how to use it in a single session, the design needed to be fixed. This emphasis on simplicity and accessibility guided every interface design decision throughout the development process. Rather than focusing on complex layouts or advanced visual effects, the researchers prioritized clarity, consistency, and ease of use. The interface was designed to minimize the amount of training required before users could operate the system confidently. Common tasks such as recording sales, checking inventory availability, viewing reports, and updating stock information were made straightforward by presenting only the information necessary for each activity while avoiding unnecessary complexity. The researchers also considered the daily working environment of the auto repair shop, where employees often needed to perform

transactions quickly while serving customers or managing repair operations. Because of this, the interface emphasized efficiency, allowing users to complete routine tasks with as few steps as possible. Every element displayed on the screen was intentionally positioned to support the natural workflow of the business, helping users become familiar with the system through regular use rather than extensive technical training.

The first screen a user encountered was the login page. It required a username and password, and the system used these credentials to determine what that user was allowed to see and do. The owner had full access to every part of the system, including inventory records, sales history, reorder management, expense tracking, reports, and all dashboard data. Staff members who were assigned limited roles saw only the parts of the system relevant to their responsibilities, such as recording sales transactions and checking stock availability. This role-based access structure existed for a practical reason: it prevented someone from accidentally editing or deleting records they had no business touching in the first place, which is a real risk in any shared digital system. In addition to improving security, the login process helped establish accountability by ensuring that all activities performed within the system were associated with authenticated users. This allowed the business to maintain better control over inventory records and operational transactions while reducing the possibility of unauthorized access to confidential business information. User authentication also protected sensitive data, including financial records, inventory quantities, and sales history, from unauthorized modification. By assigning different levels of system access according to user responsibilities, the application ensured that employees could only perform tasks appropriate to their assigned roles. This not only improved system security but also simplified the interface for staff members because unnecessary menus and administrative functions remained hidden from users who did not require them. Consequently, the role-based access mechanism contributed to both operational efficiency and data integrity while supporting secure day-to-day system usage.

After logging in, the user was taken directly to the main dashboard. The dashboard was the operational center of the entire system, and it was designed to give the owner a clear, current picture of the business without requiring any manual data gathering. It displayed a set of key performance indicators that updated automatically as transactions were recorded throughout the day. These included total daily revenue broken down into parts sales and labor fees, the monthly revenue trend showing how income moved over the current year, the top ten best-selling parts within a selected period, a revenue split showing what proportion of income was coming from parts versus services, a color-coded inventory alert panel showing which parts had dropped to or below their minimum stock threshold, and a payment method breakdown showing the split between cash and GCash transactions for the month. Presenting these indicators on a single dashboard enabled the owner to monitor multiple aspects of business performance simultaneously without opening separate modules or manually compiling information from different records. Since the displayed information was generated directly from the system database, all figures reflected the most recent transactions recorded within the application. This real-time presentation of business information improved decision-making by allowing the owner to identify sales trends, monitor inventory status, and evaluate operational performance immediately as business activities occurred. The dashboard also served as a centralized monitoring tool that reduced the time required to review daily operations while improving overall awareness of business performance. Through visual summaries, charts, and automatically updated statistics, users could quickly identify areas requiring attention, such as declining stock

levels, increasing sales of specific automotive parts, or changes in revenue patterns, allowing management decisions to be based on current and accurate information rather than manually prepared reports.

These dashboard metrics were not decorative. Each one was chosen because it answered a specific question that the owner at D Speedway used to have to ask repeatedly under the manual system. Before, knowing how much revenue came in on a given day meant opening a spreadsheet and manually adding up rows of data. Knowing which parts were running low meant physically checking the storage area or hoping someone had remembered to update the inventory file. Knowing the monthly revenue trend meant pulling multiple files and cross-referencing them. The dashboard replaced all of that with a single screen that had already done the work. The owner could open the browser, look at the dashboard for two minutes, and know more about the current state of the business than they could previously learn from an hour of manual file review. This transformation significantly reduced the amount of time spent gathering operational information while improving the accuracy and reliability of business monitoring. Instead of relying on manual calculations that were susceptible to errors or omissions, the system performed all necessary computations automatically using the most recent data stored in the database. The immediate availability of summarized information allowed management to respond more quickly to changing inventory conditions, identify high-performing products, monitor revenue growth, and make informed purchasing decisions. Furthermore, presenting operational data visually through charts, summary cards, and graphical indicators improved information interpretation by allowing important trends and exceptions to be recognized at a glance. As a result, the dashboard became not merely a display of information but an effective decision-support tool that assisted the owner in managing inventory, evaluating sales performance, and planning future business activities more efficiently.

Beyond the dashboard, the User Interface Layer included several other sections that the owner and manager used for day-to-day operations. The sales recording form was where transactions were entered: the user selected the parts used or sold, entered the quantities, recorded the payment method, and the system took it from there. The inventory section displayed the current stock level of every item in the catalog, searchable by name or category, with visual indicators showing which items were in a healthy stock range, which were approaching the minimum threshold, and which had already fallen below it. The reorder management section allowed the manager to record pending restocking activities, track their status, and update inventory records once the stock was replenished. The expense tracking section captured operational costs outside of parts purchases, which fed into the profit and loss calculations available in the reports section. The reports section allowed the owner to generate daily, monthly, and annual summaries of sales performance, inventory movement, and revenue, and to export or print them as needed. Each of these interface modules was designed to support a specific operational process while maintaining a consistent appearance and navigation structure throughout the application. Organizing system functions into clearly separated modules reduced user confusion by allowing employees to focus only on the tasks relevant to their current activity. The integration among these modules also eliminated redundant data entry because information entered into one section automatically became available wherever it was needed throughout the system. For example, recording a sales transaction immediately updated inventory quantities, influenced dashboard statistics, contributed to revenue reports, and provided historical information for forecasting without requiring additional manual input. Likewise, inventory updates and reorder activities automatically affected stock monitoring functions and future purchasing recommendations. This high level of

integration ensured that operational data remained accurate, synchronized, and readily available across all areas of the application, significantly improving efficiency compared to the previous spreadsheet-based process.

All of these interface elements were built using a design that kept visual noise to a minimum. The color scheme used red to indicate critical low-stock situations, yellow for approaching-threshold warnings, and green for healthy stock levels. This color coding allowed a staff member to scan the inventory page and immediately understand which items needed attention without reading every row in detail. Text labels were written in plain language rather than technical jargon, and navigation between sections was done through a consistent sidebar menu that behaved the same way throughout the system. The goal throughout the design process was to make sure that the system felt familiar enough by the second time someone used it that they would not need to think about how to operate it. They could just use it. In addition to supporting ease of use, these interface design principles contributed to reducing operational errors and improving overall user productivity. Consistent layouts, predictable navigation, and clearly labeled controls minimized the learning curve for new users while allowing experienced users to complete routine tasks more efficiently. The use of universally recognized visual indicators and straightforward language reduced the possibility of misunderstanding important inventory information or selecting incorrect system functions. Attention was also given to maintaining a clean and organized screen layout by grouping related information together and avoiding unnecessary visual distractions. This enabled users to locate information quickly and perform tasks with greater confidence, particularly during busy business hours when rapid access to inventory information was essential. Overall, the User Interface Layer served as the communication bridge between users and the underlying system, translating complex inventory management processes into simple, understandable interactions that supported efficient daily operations while providing an intuitive, user-friendly experience for the owner and staff of D Speedway Auto Repair Shop.

Application Layer

The Application Layer was the part of the system that the user never saw but depended on entirely. It was the engine running underneath the interface: the set of processes and logic that received input from the user, acted on it, updated the database, and sent back the right information to display on the screen. Every time something happened in the User Interface Layer, the Application Layer was what made it actually do something meaningful. It served as the central processing component of the Smart Inventory and Parts Planning System by implementing the business rules that governed how information was validated, processed, and managed throughout the application. Rather than simply transferring data between the interface and the database, this layer interpreted every user request, determined the appropriate action based on predefined operational rules, and ensured that all system processes were executed accurately and consistently. By centralizing business logic within a single layer, the system maintained uniform behavior regardless of where a transaction originated within the application. This architectural approach reduced code duplication, simplified maintenance, and ensured that inventory updates, report generation, forecasting, and other automated functions followed standardized procedures. As a result, the Application Layer became the primary component responsible for maintaining the accuracy, consistency, and reliability of the system’s operations.

When a sale was recorded, the Application Layer did not just save a record of the transaction and stop there. It immediately executed a sequence of operations in the background. First, it saved the transaction details to the database: the date, the parts involved, the quantities, the prices, the payment method, and the user who recorded it. Second, it subtracted the correct quantities from the inventory records for every part included in that transaction. Third, it checked each of those parts against its minimum stock threshold. If a part's new stock level had fallen at or below the threshold, the Application Layer flagged that item as low stock and queued a visual alert to appear on the dashboard and the inventory page. Fourth, it updated the revenue totals and KPI calculations that fed the dashboard metrics so that the next time the dashboard was viewed, it reflected the transaction that was just recorded. All of this happened within the same moment the user clicked the save button. The system did not wait to be told to check stock levels or update the dashboard. It did that automatically, every single time. Executing these operations as a single automated process significantly improved both operational efficiency and data accuracy compared to the manual procedures previously used by the business. Because every related activity occurred automatically after a transaction was recorded, the possibility of forgetting to update inventory records or financial summaries was greatly reduced. This automation ensured that sales information, inventory balances, dashboard statistics, and reporting data remained synchronized throughout the system at all times. Furthermore, processing these operations immediately after user input eliminated delays that could otherwise result in outdated inventory information or inaccurate business reports. The coordinated execution of multiple background processes also demonstrated the importance of the Application Layer in integrating different system modules into a unified workflow, allowing users to complete a single task while the application simultaneously performed all associated calculations, validations, and record updates without requiring additional user intervention.

The Application Layer also handled the forecasting calculations that supported the reorder recommendation feature. At defined intervals, it analyzed the transaction history stored in the database to calculate the average consumption rate for each part over a rolling period. It then used that rate to estimate how many units of each part the shop was likely to need in the coming weeks based on the pace at which those parts had been used in recent history. These forecasting outputs were stored and made available through the inventory and reorder sections of the interface, where the manager could see not just current stock levels but a projected depletion timeline for each item. This did not require anyone to calculate anything manually. The Application Layer ran the analysis using the data that was already in the database from every recorded sale. Automating these forecasting activities allowed the system to transform historical transaction records into practical information that supported inventory planning and purchasing decisions. Rather than relying solely on current stock quantities, the system considered actual inventory consumption patterns to estimate future requirements more accurately. This enabled management to anticipate shortages before they occurred, reducing the likelihood of stockouts that could delay repair services or affect customer satisfaction. Because the calculations were performed automatically using continuously updated transaction data, forecasting results remained current without requiring additional administrative effort. The forecasting module therefore functioned as an important decision-support component within the Application Layer, providing management with timely information that contributed to more efficient inventory control and better allocation of purchasing resources.

For the reorder recommendation feature specifically, the Application Layer cross-referenced the forecasted depletion timeline with the existing inventory records to determine which items did not yet have sufficient stock despite having a projected shortage. It also analyzed historical inventory movement and sales data associated with each part and presented relevant information alongside the reorder recommendation, allowing the manager to make informed restocking decisions without manually searching through records. The manager could view the part that needed restocking, the suggested quantity based on historical inventory and sales patterns, and the current stock level, all in one place.

They then made the decision and created the reorder record manually, which the Application Layer tracked through a status system from pending to completed. This feature simplified the inventory replenishment process by consolidating all relevant inventory information into a single interface, allowing management to make informed decisions more efficiently. Instead of reviewing multiple records individually to determine inventory needs, users received organized recommendations generated through automated analysis of inventory levels and historical consumption patterns. Although the final restocking decision remained under the control of the manager, the system significantly reduced the amount of manual analysis required before initiating a reorder. Tracking the status of reorder activities also improved inventory management transparency by allowing management to monitor each request from initial recommendation through inventory replenishment. This organized workflow minimized the risk of overlooked restocking requests while supporting more systematic inventory management practices.

The login and session management logic also lived in the Application Layer. When a user submitted their credentials on the login page, the Application Layer checked them against the database, confirmed the role associated with that account, and established a session that controlled what the user could access throughout their time in the system. When the session ended, either because the user logged out or because the session timed out after a period of inactivity, the Application Layer cleared it so that the next person to open the browser could not access the previous user's session. This was a basic but essential security measure for a system that contained business-sensitive financial and inventory data. In addition to protecting confidential business information, session management ensured that user activities remained secure throughout every interaction with the system. By continuously verifying active sessions and restricting access according to authenticated user roles, the application reduced the possibility of unauthorized access to sensitive operational data. Automatic session expiration also protected the business in situations where users unintentionally left the system open on a shared computer, preventing unauthorized individuals from viewing or modifying inventory records and financial information. The implementation of authentication and session management further supported accountability because all recorded transactions could be associated with the authenticated user responsible for performing them. These security mechanisms strengthened the overall reliability of the application while ensuring that inventory management and financial records remained protected from unauthorized access or unintended modification.

The Application Layer was built using PHP, which handled all of the server-side logic described above. PHP was chosen because it is well-suited for web-based systems of this scale, it has extensive support and documentation available, and it works seamlessly with the MySQL database used in the Database Layer. The

communication between the Application Layer and the User Interface Layer was handled through standard HTTP requests, which meant the system ran in a normal web browser without requiring any special software to be installed on the user's device. The selection of PHP as the primary server-side programming language also contributed to the maintainability and scalability of the proposed system. Its compatibility with widely used web technologies simplified the integration of database operations, user authentication, report generation, and automated inventory management processes within a single application environment. Because PHP is supported by numerous development frameworks, libraries, and community resources, future maintenance and enhancement of the system can be performed more efficiently if additional functionality becomes necessary. Using standard web technologies likewise improved system accessibility by allowing users to access the application through commonly available web browsers across different devices without requiring specialized hardware or proprietary software. This implementation approach reduced deployment costs while increasing convenience for users, making the Smart Inventory and Parts Planning System a practical and sustainable solution for the operational requirements of D Speedway Auto Repair Shop.

Database Layer

The Database Layer was where all of the shop's data lived. Every part in the catalog, every recorded transaction, every user account, every reorder record, every expense entry, and every stock level was stored here. It was the foundation that both other layers depended on. The User Interface Layer read from it to display information, and the Application Layer wrote to it whenever something changed. Without an accurate and well-structured database, neither of the other layers would have been able to do their jobs properly. As the central repository of information, the Database Layer ensured that all business data remained organized, consistent, and readily available whenever requested by the application. Every operation performed within the system ultimately relied on the information stored in this layer, making it one of the most critical components of the overall architecture. By maintaining a centralized database, the system eliminated the need for multiple independent files and spreadsheets that could easily become outdated or inconsistent. Instead, every transaction, inventory update, and operational record was maintained within a single integrated database, ensuring that all users accessed the same accurate information. This centralized approach improved data integrity by reducing duplication, minimizing inconsistencies, and ensuring that all modules within the system referenced a common and reliable source of information. Consequently, the Database Layer played a fundamental role in supporting the efficiency, accuracy, and reliability of the Smart Inventory and Parts Planning System.

The database was built in MySQL and organized into a set of related tables, each one designed to hold a specific type of information. The parts catalog table stored the details of every item the shop carried: the part name, category, unit, selling price, cost price, minimum stock threshold, and current quantity on hand. The transactions table recorded every sale entered into the system, with each row linked to the specific parts involved, the quantities sold, the prices at the time of sale, the payment method, and the date and time the transaction was recorded. The transaction items table handled the relationship between a single transaction and the multiple parts it might include, since a single repair job often involved more than one part being used at the same time. Organizing information into separate but interconnected tables followed the principles of relational database design, allowing related data to be stored efficiently while avoiding unnecessary duplication. Relationships

established through primary and foreign keys ensured that data remained consistent across the entire database and that records could be accurately connected whenever information was retrieved. This relational structure also simplified report generation because information from multiple tables could be combined through structured database queries to produce meaningful summaries of inventory movement, sales performance, and financial activity. Furthermore, separating transactional information from inventory records improved database maintainability by allowing updates to occur independently while preserving the integrity of related data. The use of MySQL also provided reliable data storage, efficient query processing, and compatibility with the PHP-based Application Layer, making it an appropriate database management system for a web-based inventory management application.

The reorder records table tracked every restocking activity initiated by the manager through the system. Each record contained the part being restocked, the quantity recommended or recorded, the date the reorder was created, and the current status of the activity. When a reorder was marked as completed, the Application Layer updated the corresponding inventory quantity in the parts catalog table, closing the loop from low-stock alert to replenished inventory within the same system.

Maintaining a dedicated reorder records table enabled the system to support a more organized inventory replenishment process while preserving complete historical records of restocking activities. Every reorder record remained available for future reference, allowing management to review previous inventory decisions, monitor stock movement, and analyze replenishment trends over time. Recording the status of each reorder activity also improved operational visibility by allowing users to monitor whether inventory had already been scheduled for replenishment or had already been added back into stock. This integration between inventory records and reorder transactions created a continuous inventory management cycle that supported timely restocking decisions and improved stock availability.

The expenses table recorded operational costs that were separate from parts purchasing: utilities, labor, and other overhead expenses that factored into the shop's overall financial picture. These records were used by the Application Layer when generating the profit and loss summaries in the reports section, giving the owner a more complete view of financial performance than sales revenue alone would provide. The users table stored the login credentials and role assignments for every account in the system, and it was what the Application Layer checked against during authentication. Each user record contained a hashed password rather than the password itself, meaning even if someone gained access to the database directly, they would not be able to read the passwords stored there. Including expense information within the database enabled the system to evaluate business performance more comprehensively by considering both income and operational expenditures. This allowed financial reports to present a more realistic assessment of profitability instead of relying solely on sales revenue. Maintaining user account information within a dedicated table also strengthened system administration by allowing user roles, permissions, and authentication data to be managed efficiently from a centralized location. The use of password hashing represented an important security practice because sensitive authentication information was protected even if unauthorized individuals obtained direct access to the database files. Additional database constraints and validation mechanisms further protected the integrity of stored information by ensuring that records satisfied predefined rules before being accepted into the database. Together, these

measures enhanced both the security and reliability of the system while safeguarding sensitive business information against unauthorized access and accidental data corruption.

The structure of the database was designed with two priorities in mind: speed and reliability. Fast lookups mattered because the dashboard needed to load quickly every time the owner opened it. If the database queries took too long to run, the system would have become frustrating to use in a busy shop environment. The tables were indexed on the columns that were queried most often, such as part names in the catalog, dates in the transaction history, and status values in the reorder records, which kept query times low even as the volume of stored data grew over time. Reliability mattered because the database was the single source of truth for every business decision the owner made using the system. Data that is stored incorrectly or that can be lost due to a system failure is worse than no data at all, because it creates false confidence. The database structure reflected in the Entity Relationship Diagram in the Supporting Diagrams section was designed to prevent those kinds of structural errors from the ground up. Beyond performance optimization, the database was also designed to support long-term maintainability and future expansion of the application. Proper indexing strategies reduced the amount of time required to retrieve frequently accessed information, ensuring that reports, dashboard statistics, inventory searches, and forecasting calculations remained responsive even as the database accumulated larger volumes of historical transaction data. Data integrity was further strengthened through carefully defined relationships, validation rules, and referential constraints that prevented inconsistent or orphaned records from being created during routine system operations. These design practices ensured that every inventory transaction, reorder activity, financial record, and user account remained logically connected and accurately represented throughout the database. The Entity Relationship Diagram served as the blueprint for organizing these relationships, providing a clear representation of how each table interacted with the others while supporting efficient data management. Overall, the Database Layer provided a stable, secure, and high-performance foundation that enabled the Smart Inventory and Parts Planning System to process information accurately, support informed business decisions, and maintain reliable operational records for D Speedway Auto Repair Shop over the long term.

Flow of a Typical Transaction Through the System

To make the three-layer architecture concrete rather than abstract, it helps to trace what actually happened from the moment the owner or manager opened the system to the moment a completed transaction had been recorded and the inventory had been updated. This flow represented the most common daily interaction with the system and the one the entire architecture was designed to support efficiently. Examining this sequence of activities provides a practical illustration of how the User Interface Layer, Application Layer, and Database Layer worked together as an integrated system rather than as independent components. Although each layer had a separate responsibility, they continuously communicated with one another to process user requests, manage inventory information, update business records, and present accurate information back to the user. Understanding the complete transaction flow also demonstrates how automation replaced the manual procedures previously performed using spreadsheets and handwritten records. Instead of requiring users to perform several independent tasks after every transaction, the system coordinated these activities automatically through its

layered architecture. This streamlined workflow reduced processing time, minimized the likelihood of human error, and ensured that all inventory and financial records remained synchronized throughout the application.

The manager would open a web browser and navigate to the system's login page. They entered their username and password and clicked the login button. The User Interface Layer sent those credentials to the Application Layer, which checked them against the users table in the database. If the credentials matched and the account was active, the Application Layer established a session, retrieved the user's role, and told the User Interface Layer to load the dashboard. The dashboard page requested the current KPI data (the day's revenue, current low-stock alerts, recent transactions, monthly trend figures, and top-selling parts) from the Application Layer, which pulled all of that from the database and returned it. The dashboard loaded and the manager could see the current state of the shop at a glance before doing anything else. This login sequence represented the initial communication among the three architectural layers and demonstrated how authentication, authorization, and information retrieval occurred before any operational activity could be performed. By validating user credentials prior to granting access, the system protected sensitive business information while ensuring that users viewed only the features appropriate to their assigned responsibilities. Once authentication was completed successfully, the dashboard immediately presented updated operational information retrieved directly from the database, allowing the manager to begin the workday with an accurate overview of business performance. Presenting current revenue figures, inventory alerts, transaction summaries, and performance indicators immediately after login enabled management to identify operational priorities without conducting manual record reviews. This efficient retrieval and presentation of information highlighted the responsiveness of the system and demonstrated how the three-layer architecture supported rapid access to critical business data.

The manager would then navigate to the sales recording section. A form loaded with fields for selecting the parts involved in the transaction, entering the quantities used, specifying the labor fee if applicable, and choosing the payment method. The manager filled in the form based on the repair that had just been completed. When they clicked save, the User Interface Layer sent all of that form data to the Application Layer. The Application Layer validated the data first, checking that all required fields were filled and that the quantities entered were not greater than the current stock levels. If validation passed, it began the write sequence: it saved the transaction header record to the transactions table, saved each individual part entry to the transaction items table, and then ran an update query on the parts catalog table to subtract the correct quantity from the stock level of each part that was used. Performing these validation procedures before saving information protected the integrity of the database by preventing incomplete, inaccurate, or inconsistent records from being stored. Automatic validation also minimized the possibility of recording transactions that exceeded available inventory or omitted essential information required for financial reporting. Once validation was completed successfully, all related database updates were executed as part of a coordinated transaction, ensuring that inventory records, sales information, and transaction histories remained synchronized. This integrated processing eliminated the repetitive manual updates previously required after every sale and significantly improved the speed and reliability of inventory management. Because all updates occurred automatically within a single workflow, users could complete sales transactions confidently, knowing that inventory balances and financial records had already been updated correctly.

Immediately after updating the stock levels, the Application Layer ran a threshold check. It compared the new stock level of each affected part against its minimum stock threshold. If any part's new level was at or below its threshold, the Application Layer marked that part as low stock in the database. The next time the dashboard or inventory page was loaded, which could be within seconds if another tab was already open, the color-coded alert for that part appeared automatically. The manager did not have to remember to check. The system flagged it and put it in front of them. This automated monitoring process represented one of the most important improvements introduced by the Smart Inventory and Parts Planning System because it replaced manual inventory checking with continuous real-time monitoring. Rather than depending on employees to periodically review stock quantities, the application evaluated inventory levels immediately after every transaction and responded automatically whenever predefined conditions were met. This proactive approach enabled the business to identify potential shortages before they affected daily operations or customer service. Automatic low-stock alerts also reduced the likelihood of stockouts caused by overlooked inventory records or delayed manual inspections. By presenting visual notifications directly on the dashboard and inventory pages, the system ensured that inventory concerns remained visible to management until appropriate corrective actions had been taken.

If the manager noticed a low-stock alert and decided to act on it, they navigated to the reorder management section. The system displayed the flagged parts alongside their current stock levels, minimum stock thresholds, and suggested reorder quantities calculated from historical inventory and sales data. The manager reviewed the recommendations, adjusted the quantities if necessary based on their judgment, and created a reorder record. The Application Layer saved this record to the reorder records table with a status of pending. When the inventory was replenished and the manager updated the reorder status as completed, the Application Layer added the replenished quantity to the part's stock level in the parts catalog table and updated the reorder record accordingly. The alert was then cleared from the dashboard. This workflow demonstrated how inventory monitoring, forecasting, reorder recommendations, and inventory replenishment were integrated into a continuous operational cycle within a single system. Although the application generated recommendations based on historical inventory usage, the final restocking decision remained with the manager, allowing professional judgment to be applied whenever exceptional circumstances affected inventory requirements. Recording reorder transactions and tracking their progress through predefined status levels also improved transparency by allowing management to monitor every stage of the replenishment process. Once inventory was received and recorded, the automatic update of stock quantities ensured that all inventory records accurately reflected current availability without requiring additional manual adjustments. The immediate removal of resolved alerts further contributed to a clean and organized dashboard by ensuring that users focused only on inventory issues requiring attention.

At any point during or after this process, the manager could navigate to the reports section and generate a sales summary for the current day, the current month, or any selected period. The Application Layer queried the transaction history from the database, aggregated the data by date, part, category, or payment method depending on what the report required, and returned the formatted results to the User Interface Layer for display. If the manager wanted to review the profit and loss for the month, the Application Layer combined the sales revenue from the transactions table with the expense records from the expenses table and returned the net figure. All of

this was available within seconds and without any manual compilation. The reporting capability transformed raw operational data into meaningful business information that supported monitoring, evaluation, and strategic decision-making. Because reports were generated dynamically using the most recent database records, management no longer needed to manually consolidate information from multiple spreadsheets before analyzing business performance. Automated report generation improved both efficiency and accuracy by eliminating repetitive calculations while ensuring that financial summaries, inventory reports, and sales analyses reflected current business conditions. The flexibility to generate reports for different time periods also allowed management to evaluate short-term operational performance as well as long-term business trends, supporting more informed planning, budgeting, and inventory purchasing decisions.

What this flow demonstrated was that the three-layer architecture was not just a structural choice made for technical reasons; it was the reason the system could do all of these things together, reliably, in real time, for a business that used to have to do every single one of these steps by hand. The User Interface Layer made the system accessible to the owner and staff without requiring technical knowledge. The Application Layer handled all of the logic that would otherwise fall to human memory and discipline. The Database Layer stored everything accurately and made it retrievable instantly. Each layer had a specific job, and together they produced a system that fit into D Speedway's daily operations without asking anyone to work harder, only smarter. The interaction among these three architectural layers illustrated the effectiveness of the system's overall design in supporting the operational requirements of D Speedway Auto Repair Shop. Each layer fulfilled its designated responsibility while communicating seamlessly with the others, creating a coordinated environment where user requests, business processes, and data management operated efficiently as a single integrated application. This separation of responsibilities simplified maintenance, improved system reliability, and allowed future enhancements to be implemented with minimal impact on existing functionality. More importantly, the architecture enabled the automation of numerous manual inventory management activities, reducing administrative workload while improving the accuracy, consistency, and availability of business information. Through this integrated transaction flow, the Smart Inventory and Parts Planning System successfully demonstrated how an appropriately designed information system could modernize inventory management practices, strengthen decision-making, and support the long-term operational efficiency of D Speedway Auto Repair Shop.

3.4 Tools and Technologies Used

The following tools were used to build the system:

- Frontend: HTML, CSS, and JavaScript for building the interface that users interacted with.
- Backend: PHP, given its compatibility with MySQL and the team's familiarity with it.
- Forecasting: CHART.JS for the parts demand forecasting component.
- Database: MySQL, which was already reflected in the ERD design.
- Development Environment: Visual Studio Code & XAMPP

3.5 Data Gathering Techniques

In order to build a system that actually addressed the real needs of D Speedway, it was important to first develop a thorough understanding of how the shop operated. This meant going beyond assumptions and making sure that every design decision made throughout the development process was grounded in actual information gathered directly from the source. Two data gathering methods were used for this purpose: observation and interviews. Each method served a different function, and together they provided a well-rounded picture of the shop's processes at the time, its pain points, and what the people working there genuinely needed from a new system.

The reason these two methods were chosen over others, such as surveys or questionnaires, was that D Speedway is a small, single-branch automotive repair shop with a small team, and the kind of information needed to design this system well could not be captured through a checklist. It required being present in the environment and talking directly to the people who dealt with the problems every day. Surveys are useful for collecting data from large numbers of respondents, but for a study like this one where the goal was to understand the specific, day-to-day operational reality of one particular shop, direct observation and face-to-face interviews produced far more useful and accurate information. The combination of both methods also served as a form of cross-verification: what was observed in practice could be compared against what the owner and staff reported in interviews, and where those two sources agreed, there was strong confidence that the finding was accurate and worth designing around.

Observation

The first method used was observation. Time was spent visiting D Speedway and watching how daily operations were actually carried out on the ground. Rather than relying on second-hand descriptions or asking staff to explain their workflow verbally, things were seen firsthand, which tended to reveal details that people would not necessarily think to mention if they were just answering questions. People who had been doing the same tasks the same way for a long time often stopped noticing the small workarounds they had built into their routines. Direct observation was one of the most reliable ways to bring those kinds of details to the surface because it captured what was actually happening in practice, which is not always the same as what people believe is happening or what they would describe out loud.

The observation sessions took place during actual working hours so that the shop could be seen under real operational conditions, not during a quiet period set aside specifically for the visit, but during the kind of busy, unscripted day that D Speedway's staff dealt with regularly. This mattered because the problems being investigated were problems that appeared under pressure: when multiple jobs were running at the same time, when a customer was waiting at the counter, when someone needed to check a part but also needed to answer a phone. Those were the conditions under which manual systems showed their weaknesses most clearly, and those were the conditions under which the new system needed to perform.

During the observation sessions, attention was given to several specific aspects of how the shop was running at the time. The following were the key areas that were closely monitored and documented:

● **Transaction Recording Process:** How sales were being recorded was examined in detail, including whether the method used was paper-based, spreadsheet-based, or something else entirely. The time it took to complete a single transaction entry was noted, along with any moments where the process slowed down, became complicated, or got skipped entirely because something else demanded the staff member's attention at the same time.

● **Stock Level Checking:** How staff members determined whether a part was available was observed. This included whether they physically walked to the storage area to count items, consulted a spreadsheet file, relied on memory, or used some combination of all three. The accuracy and consistency of each approach were noted, as well as how much time these checks took away from other tasks.

● **Inventory Update Workflow:** The process by which inventory records were updated after a part was used or sold was tracked. Attention was paid to how quickly updates happened after a transaction occurred, whether updates were sometimes delayed until the end of a shift or forgotten entirely, and whether different staff members handled this task in consistent or inconsistent ways.

● **Sales Report Generation:** How the owner or manager compiled sales reports was examined, including which files were opened, how data was extracted and combined, and how long the process took from start to finish. Any patterns of inconsistency or confusion between different files were noted.

● **Reordering and Stock Replenishment:** The process by which the shop decided to reorder parts was observed. This included how and when the decision was made, whether it was triggered by a physical discovery that something had run out or by a proactive check of records, who made the decision, and how the order was placed and tracked afterward.

● **Overall Shop Pace and Workflow Patterns:** The general rhythm of the shop throughout the day was observed, including how busy different periods were, how staff moved between their primary automotive work and administrative tasks, and what kinds of interruptions or unexpected problems tended to disrupt normal operations. This broader picture was important for understanding the environment in which the new system needed to operate.

All observations were written down carefully in real time so that the details stayed accurate and could be referred back to during the design and development stages. The notes taken during these sessions did not just describe what was seen but also captured the sequence of events, the approximate time each task took, and any points where something clearly went wrong or required an extra step to correct. These records served as a direct reference for deciding which system features were most urgently needed and how those features needed to be designed to fit into the shop's actual workflow rather than an idealized version of it.

Interviews

The second method used was interviews. After the observation phase was completed, the shop owner and selected staff members were sat down with directly and asked specific questions about their experience with how things were being handled and what they wanted to see in something new. While observation gave a clear picture of how things were done from the outside, interviews gave the people involved a chance to speak for themselves, to explain what frustrated them, what they found difficult to deal with on a daily basis, and what they believed could genuinely be improved about the way the shop managed its records and inventory.

The decision to conduct interviews after observation rather than before it was intentional. Going into the interview sessions with firsthand knowledge of how the shop actually operated allowed for more focused and meaningful questions. Instead of asking vague, open-ended questions about inventory management in general, the questions could refer directly to what was observed (specific moments, specific tasks, specific situations) and ask the owner and staff to elaborate on those. This made the interviews more productive because the conversation started from a shared, concrete understanding of the operational reality rather than from scratch.

The interview participants were selected based on their role and their level of direct involvement with the processes the system was being built to address. The shop owner was interviewed because they were the primary decision-maker and the person who would ultimately be responsible for using the system's reporting, forecasting, and reorder features on a regular basis. Staff members who were involved in day-to-day transaction recording and inventory checking were also included because they were the ones who experienced the friction of the old system most directly during working hours. Their input was especially important for ensuring that the system's interface and workflow were designed in a way that actually fit how they did their jobs.

The questions prepared for these interviews were focused and purposeful. The following were the main areas the interview questions covered:

● **Daily Pain Points and Time-Consuming Processes:** Staff were asked to describe which parts of their workflow at the time took the most time and caused the most frustration. This included questions about how long it took to record a transaction, how often they had to stop what they were doing to check on stock, and how much of their day was spent on administrative tasks rather than on the automotive work they were hired to do.
● **Common Errors and Recurring Mistakes:** The owner and staff were asked to describe the kinds of mistakes that happened most often in the old system. This included miscounted stock, missing transaction entries, incorrect figures in reports, data entered in the wrong file, or situations where two people's records contradicted each other. Understanding where errors tended to occur directly informed how the system was designed to prevent those same mistakes from happening again.
● **Stock Shortage Experiences:** Both the owner and staff were asked about specific instances where a part ran out unexpectedly during a repair job, and what happened as a result. This included questions about how often these situations occurred, what the usual response was when they happened, and what the cost of that response tended to be, both in terms of money paid for rush orders and time lost while the job was on hold.
● **Reporting Challenges:** The owner was asked in detail about how they generated sales and inventory reports, how long the process took, and how confident they were in the accuracy of the figures they ended up with. Questions were also asked about what kinds of business decisions depended on those reports and whether the reporting process at the time gave them the information they actually needed to make those decisions well.
● **Feature Priorities and Desired Capabilities:** The owner and staff were asked what they would most want a new system to be capable of doing. This was where a clearer sense was gained of which features mattered most to the people who would be working with the system on a daily basis. It would not have made sense to spend development time building something the staff did not find useful or practical, so gathering this kind of direct input early on was a step that could not be skipped.
● **Concerns About Adoption and Usability:** The owner and staff were also asked about their comfort level with technology and whether they had any concerns about learning or using a new system. This information was used to guide the design of the interface, particularly in terms of how much complexity was appropriate and where simplicity needed to take priority over additional features.

The interviews were conducted in a semi-structured format. This meant that a prepared set of questions was brought into each session, but the conversation was not rigidly limited to those questions alone. If an answer opened up an important direction that was not anticipated, that thread was followed. The goal of the interview was not to fill out a checklist; it was to understand the actual experience of the people running D Speedway, and that required a format that allowed the conversation to go where it naturally needed to go.

The responses gathered during each interview session were written down carefully, and where the participant's permission was given, notes were taken in as much detail as possible so that nothing important was lost. After all interviews were completed, the responses were reviewed alongside the observation notes collected during the first phase. Where both sources pointed to the same problem or the same need, that was treated as a strong and reliable signal that the system needed to address it. Where the two sources contradicted each other (for example, where the observed reality did not match what someone described in the interview), those discrepancies were examined more closely to understand why the gap existed and which source was more representative of the actual situation.

Taken together, the observation and interview data served as the foundation for every design and development decision made throughout this project. The features built into the system, the way the interface was organized, the alerts that were triggered, and the reports that were generated all traced back to something specific that was seen or heard at D Speedway during the data gathering phase. That connection between the system's design and the shop's actual operational reality is what the research team believed gave the final product its practical value, not just as a technical output of the study but as something the owner and staff could genuinely use and benefit from every day.

3.6 Testing Procedures

Testing was not treated as a single activity that happened only after the entire system had been built. Rather, it was conducted continuously and in phases throughout the whole development process. This approach was chosen because catching problems early, while a feature was still being worked on, was far less complicated than trying to trace an error back to its source after several other features had already been added on top of it. When a system is tested only at the end, errors can be deeply buried and difficult to isolate, and fixing them often means revisiting code that has already been integrated with other parts of the system, which risks creating new problems in the process of solving old ones. Testing at each stage prevented that chain of complications from developing in the first place.

Three distinct testing phases were carried out over the course of the project: unit testing, integration testing, and User Acceptance Testing. Each phase served a different purpose and looked at the system from a different angle. Unit testing confirmed that each individual feature worked correctly on its own before being connected to the rest of the system. Integration testing confirmed that all of those individually working features behaved correctly when they were made to interact with each other. User Acceptance Testing confirmed that the system as a whole was practical, understandable, and genuinely useful for the people at D Speedway who used it every day. Together, these three phases covered the system from the level of a single function all the way through to the full experience of a real user doing real work.

Unit Testing

The first phase carried out was unit testing. As the name suggests, this type of testing focused on one unit or one feature at a time, completely separate from everything else in the system. The idea behind this was

straightforward: before anything was connected to anything else, each individual piece needed to work correctly on its own. No feature was considered ready to move forward until it had been tested and confirmed to be functioning the way it was supposed to. This approach kept development clean and organized because it meant that when something did not behave correctly, there was no need to search through an entire integrated system to find the cause. The problem was isolated by default because only one thing was being looked at at a time.

Unit testing was performed by the developers immediately after each feature was coded, before moving on to the next one. Each test involved feeding the feature a set of inputs (both valid ones and deliberately invalid or edge-case ones) and verifying that the output matched what was expected. Testing only the straightforward cases was not enough. A feature also needed to be tested against unusual inputs, missing data, extreme quantities, and boundary conditions to make sure it did not produce unexpected behavior under the kinds of circumstances that come up in a real working environment. A stock update function that worked correctly when one unit was sold but broke when twenty were sold at once was not ready, even if it passed the basic test.

The following were the specific features and functions that went through unit testing:

● Sales Transaction Recording: The function that saved a sale to the database was tested to confirm that all required fields were saved correctly, that the transaction was linked properly to the right parts and quantities, and that incomplete or incorrectly formatted inputs were rejected with an appropriate error message rather than saved as corrupted records.
● Automatic Inventory Deduction: The function that subtracted quantities from the parts catalog after a sale was recorded was tested to confirm that the correct quantity was deducted from the correct item, that the deduction applied to every part included in a multi-part transaction, and that the stock count did not go below zero even if the entered quantity exceeded what was currently available.
● Minimum Stock Threshold Alert: The function that checked whether an item's updated stock level had dropped to or below its defined minimum threshold was tested to confirm that the alert was triggered at the right level, that it was not triggered prematurely, and that it correctly identified the specific item that needed attention rather than flagging unrelated parts.
● Reorder Record Creation and Status Tracking: The function that created a reorder record and managed its status was tested to confirm that new reorder entries were saved with the correct part, quantity, and date, that the status could be updated correctly throughout the replenishment process without errors, and that marking a reorder as completed correctly added the restocked quantity back to the inventory count.
● Sales Report Generation: The reporting function was tested to confirm that it correctly pulled transaction data from the database for the selected time period, that the totals calculated were accurate, and that the report displayed the breakdown of parts revenue and labor revenue correctly without mixing or omitting any figures.
● Dashboard KPI Calculations: Each metric displayed on the dashboard (daily revenue, monthly trend, top-selling parts, stock alert panel, revenue split, and payment method breakdown) was tested individually to confirm that the figures were being pulled and calculated correctly and that they updated when new transactions were recorded.
● User Login and Authentication: The login function was tested to confirm that valid credentials granted access, that invalid credentials were rejected with an appropriate message, that access levels were correctly applied after

login so that each user only saw what their role permitted, and that session timeout worked as intended.

● Expense Recording: The function that saved an expense entry was tested to confirm that the correct amount, category, date, and description were saved, and that the expense figures were correctly reflected in the profit and loss calculations in the reports section.

After each function passed its unit tests, a record of the test cases used and the results observed was kept. This documentation served as a reference during integration testing and made it easier to re-run specific tests if a later change to the codebase needed to be verified against earlier behavior.

Integration Testing

Once a number of features had been individually built and confirmed to be working on their own, the next phase that took place was integration testing. At this point, the focus shifted away from individual features and toward how all of those features functioned when they were connected and running together as a single system. It was entirely possible for two features to work perfectly on their own but then behave unexpectedly when they were made to interact with each other, and integration testing was specifically designed to catch that kind of problem. The reason this happens is that individual features are developed with certain assumptions about what the other parts of the system will provide or expect, and sometimes those assumptions do not match up perfectly when everything is finally combined.

The main thing verified during this phase was whether the system's different components were properly communicating and responding to each other in the right sequence. The system was built around a chain of connected operations, and the reliability of that chain as a whole was what integration testing was designed to confirm. If any link in the chain broke, if one component failed to pass information correctly to the next, or if two components interfered with each other in an unexpected way, the system would not have behaved reliably under real working conditions. Finding and fixing those breaks during integration testing was far better than discovering them during actual use at the shop.

The following were the key integration scenarios tested during this phase:

● Sale-to-Inventory Update Chain: When a sale was recorded through the sales form, the system was supposed to automatically subtract the correct quantities from the parts catalog in the database. This chain was tested by recording transactions involving different combinations of parts and quantities and then verifying that the inventory counts in the database reflected exactly what was sold. Edge cases such as transactions involving the same part multiple times in a single sale were also tested.

● Inventory Update-to-Alert Trigger Chain: When an inventory update brought a part's stock level to or below its minimum threshold, the system was supposed to immediately flag that item and display a low-stock alert on the dashboard and inventory page. This chain was tested by deliberately recording transactions that brought specific parts to and below their thresholds and verifying that the alerts appeared correctly in both locations at the right moment.

● Reorder Received-to-Inventory Restoration Chain: When a reorder record was marked as received, the system

was supposed to add the restocked quantity back to the part's current inventory count and clear the low-stock alert if the new level was above the threshold. This entire sequence was tested from the creation of the reorder record through to the confirmation of receipt and the verification of the updated stock level.

- Transaction Data-to-Dashboard KPI Update Chain: When a new transaction was recorded, the dashboard figures, particularly the daily revenue total, the top-selling parts ranking, and the payment method breakdown, were supposed to update to reflect the new data. This was tested by recording multiple transactions in sequence and verifying that the dashboard reflected each one accurately and without delay.
- Expense Entry-to-Profit and Loss Report Chain: When an expense was recorded, it was supposed to be correctly factored into the profit and loss summary in the reports section. This was tested by recording expenses of various amounts and categories and then generating reports to verify that the net figures were accurate and that the breakdown between revenue and expenses was correctly displayed.
- Login-to-Access Level Enforcement Chain: When a user logged in with a specific role, the system was supposed to restrict their access to only the sections and functions permitted for that role. This was tested by logging in with accounts assigned to different roles and attempting to access sections that should have been restricted, verifying that the system correctly denied those attempts without errors.

Integration testing was repeated each time a new feature was added to the system, not just once at the end. This meant that every new addition was checked against everything that was already in place, and any conflict or unexpected behavior the new feature introduced was caught immediately rather than being discovered later when the problem had become more difficult to trace. A log of each integration test (what was tested, what inputs were used, what was expected, and what actually happened) was maintained throughout this phase so that the testing process was documented and repeatable.

User Acceptance Testing (UAT)

The final phase of testing was User Acceptance Testing, more commonly referred to as UAT. This stage was different from the previous two in that it was not focused on the technical side of the system. By the time UAT began, the assumption was that the features were working correctly and that the system was stable enough to be used without constant supervision. What UAT was really concerned with was whether the system was actually practical and usable from the perspective of the people it was built for. A system can pass every technical test and still fall short if the people using it find it confusing, slow, or misaligned with how they actually work. UAT was the safeguard against that outcome.

For this phase, selected participants from D Speedway were asked to use the system in an environment that closely mirrored their actual working conditions. The participants included both the shop owner and staff members who were directly involved in the day-to-day processes the system was designed to support. They were not asked to use the system in an artificial or simplified setting; they went through the same kinds of tasks they handled on a regular basis but did those things through the new system instead of however they used to manage them. This real-conditions approach mattered because it was the only way to find out whether the system held up in practice, not just in a controlled test environment.

The following were the specific tasks UAT participants were asked to carry out during the evaluation sessions:

● Logging in to the system using assigned credentials and navigating to the main dashboard, evaluating whether the login process was straightforward and whether the dashboard gave them the information they needed without requiring additional steps.
● Recording a sales transaction involving multiple parts, evaluating whether the sales form was easy to understand, whether part selection and quantity entry felt natural, and whether the confirmation after saving was clear enough that they felt confident the transaction was recorded correctly.
● Checking current stock levels for specific parts using the inventory section, evaluating whether finding a specific item was quick and whether the stock status indicators were clear enough to understand at a glance without needing an explanation.
● Responding to a low-stock alert by navigating to the reorder management section and creating a reorder record, evaluating whether the alert was noticeable enough to catch attention during a busy working day, whether the reorder form was simple to complete, and whether the reorder recommendation information was presented clearly enough to support timely restocking decisions.
● Generating a sales report for a specified time period, evaluating whether the report was generated quickly, whether the figures presented were easy to read and interpret, and whether the report gave the owner the information they actually needed to understand business performance.
● Updating a reorder record from pending to received and verifying that the inventory count had been restored, evaluating whether the status update process was intuitive and whether the inventory change was immediately visible after the update was saved.
● Navigating freely through the system to explore any section they were curious about, evaluating overall ease of navigation, consistency of the interface across sections, and whether any part of the system felt unexpectedly complicated or unclear.

Throughout each session, participant behavior was observed closely. Notes were taken on where they hesitated, where they made mistakes, where they looked confused, and where the process moved quickly and smoothly. After each task was completed, the participants were asked to share their honest reactions, not just whether they managed to complete the task but whether they found it easy, whether it felt natural, and whether there was anything about the process they thought could be better. These impressions were captured in as much detail as possible because the kind of feedback that comes from actually using a system is different from the kind that comes from looking at it on paper.

3.7 Evaluation Criteria

Once testing was complete, the system was evaluated on the following:

● Functionality: The evaluation determined whether every feature did what it was supposed to do, such as automatic stock updates, low-stock alerts, and report generation [7], [8].
● Usability: The evaluation determined whether D Speedway's owner and manager could use the system without needing technical help [3], [4].
● Accuracy: The evaluation determined whether the stock counts, sales totals, and forecasting outputs were

correct [2], [8].

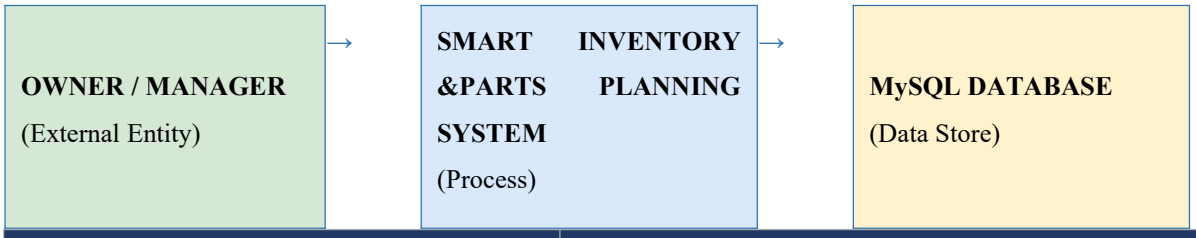
● Efficiency: The evaluation determined whether it took noticeably less time and effort to manage inventory and reporting compared to the old Excel-based process [2], [9].

● Reliability: The evaluation determined whether the system ran consistently without errors, crashes, or data loss [4].

SUPPORTING DIAGRAMS

(Included within Chapter 3 — these diagrams illustrate the structure, data flow, and database design of the Smart Inventory and Parts Planning System for D Speedway.)

Figure 1: Data Flow Diagram (DFD – Level 0)



INPUT FLOWS	OUTPUT FLOWS
User records a sale or parts used	Inventory levels updated automatically
User requests a sales or inventory report	Report displayed on screen; CSV export available
System reads historical transactions	Forecasting results returned to dashboard
Stock drops below reorder threshold	Visual alert displayed on dashboard and inventory page
Manager manually creates reorder record	Reorder saved with status: Pending → Confirmed → Received

Table 1: ERD Relationships

ENTITY / TABLE	RELATED TO	RELATIONSHIP DESCRIPTION
users	sales, inventory_transactions, expenses	A user (owner or manager) can record many sales, transactions, and expenses. All events are traced via created_by.
products	categories	Each product belongs to one category. A category can contain many products.
product_stock	products, inventory_transactions	product_stock is a computed view of current stock, combining product data with all inventory_transactions for that product.
sales	sale_items	Each sale has one or more line items (parts or labor) stored in sale_items.
sale_items	products	Each parts line item references a

		specific product from the catalog.
products	inventory_transactions	Every stock change (sale, restock, adjustment) is logged as a separate inventory_transaction record.
products	inventory_transactions	When the owner or manager records a restock, a new inventory_transactions entry with a positive quantity change is added, which immediately updates the current stock level for that product.

Figure 3: Use Case Diagram

The table below lists the main actors in the system and the actions each one can perform.

ACTOR	USE CASE (Action)	WHAT THIS MEANS
Manager	Log In	Authenticate using email and password (bcrypt verified).
	Record a Sale	Enter sale date, customer name, plate number, parts used, and labor charges.
	View Current Stock	Check real-time quantity on hand for any part from the product_stock view.
	Receive Visual Low-Stock Alert	View color-coded badges on dashboard and inventory page when stock is at or below threshold.
	Restock a Part	Manually enter quantity received; system logs it as a restock in inventory_transactions.
Owner	All Manager actions	Owner has full access to everything the manager can do.
	View Sales & Inventory Reports	Generate daily, monthly, or yearly summaries from sales and inventory data.
	View Forecasting Output	See demand analysis per product based on inventory_transactions history.
	Manage Reorder Records	Create, confirm, and mark reorder records as

		completed. Quantity entered manually by the manager.
	Record Expenses	Log operational costs in the expenses table; reflected in dashboard KPIs.
	Manage User Accounts	Add, edit, or delete manager accounts. Owner account cannot be deleted.
	Export Data to CSV	Export sales history and inventory records for offline use or record-keeping.
System	Auto-update Inventory	After every sale, write an inventory_transaction record and recalculate current_stock in product_stock.
	Display Low-Stock Alert	When current_stock ≤ reorder_threshold, flag item with yellow/red badge on dashboard and inventory page.
	Run Demand Forecast	Analyze inventory_transactions history to display demand trends per product in the Forecasting tab.

Figure 3: Use Case Diagram — actions for Owner, Manager, and the system.

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CHAPTER IV

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

The Smart Inventory and Parts Planning System was the finished product of everything covered in the previous chapter: the observations, the interviews, the planning sessions, and the several rounds of development and testing that followed. What came out of all that was a working, web-based platform that D Speedway could log into from any computer with a browser. The shop could use it to record sales, track inventory, manage reorders, keep an eye on expenses, and pull reports without touching a single Excel file. This section walks through the system as it was built, module by module, to give a clear picture of what was produced and how the different pieces fit together in practice [2], [4].

At its core, the system was built around a single idea that came up again and again during data gathering: everything the shop needed to know about its sales and inventory should live in one place, update itself automatically, and be available the moment someone needed it. That idea shaped every module described below. None of these modules existed in isolation. A sale recorded in one part of the system immediately rippled through inventory counts, dashboard figures, and reorder alerts elsewhere, without anyone having to connect the dots by hand.

The system ran on PHP and MySQL, with the interface built using HTML, CSS, and JavaScript, and Chart.js handling the visual side of the forecasting feature. All of this sat on top of a MySQL database made up of nine interconnected tables: users, products, categories, sales, sale_items, inventory_transactions, expenses, otp_codes, and audit_log. Together, these stored every piece of information the shop's day-to-day operations depended on.

General Structure of the System

The system was organized around a set of modules, each built to handle a specific part of the shop's operations. Some of these modules were things the owner and manager used directly through the interface, and some ran quietly in the background, doing work the user never had to think about. Together, they covered the full range of what came up during requirements gathering: recording sales, tracking stock, catching low inventory before it

became a problem, planning ahead for future demand, managing reorders, tracking expenses, and generating reports the owner could actually use [2], [4].

The main modules that made up the finished system were:

● User Authentication and Access Control: Handled logging in, verifying credentials, and controlling what each type of user was allowed to see and do.
● Dashboard: The main landing screen after login, showing a live snapshot of the shop's current performance and inventory status.
● Sales Recording Module: Where staff logged transactions involving parts and labor.
● Inventory Monitoring Module: Tracked current stock levels for every part in the catalog and flagged items that were running low.
● Reorder Management Module: Let the manager act on low-stock alerts by creating and tracking reorder records.
● Expense Tracking Module: Captured operational costs outside of parts purchasing, feeding into the shop's profit and loss picture.
● Reports Module: Generated sales and inventory summaries for any period the owner needed.
● Forecasting Module: Analyzed past sales data to estimate which parts were likely to be needed soon.

Each of these is described in more detail below, but it is worth saying upfront that none of them were built as standalone features bolted onto a generic template. Every one of them traced back to a specific problem documented during the descriptive phase of this study: the spreadsheet mismatches, the manual reconciliation, the mechanics discovering a part was gone mid-repair. The system existed to make those specific situations stop happening.

User Authentication and Access Control

Every interaction with the system started at the login screen, where a user entered their email and password. The system checked those credentials against the users table, and if they matched, it verified the account's role before letting the person in. This was not just a formality. It was the mechanism that determined what that person saw and did for the rest of their session. The owner logged in with full access to everything: sales, inventory, reports, forecasting, reorder management, expenses, and user account management. A manager account, by comparison, had access to the day-to-day operational functions, such as recording sales, checking stock, and managing reorders. It could not add or remove other user accounts or make certain administrative changes reserved for the owner [2].

Passwords were stored using bcrypt hashing rather than as plain text, so even someone with direct access to the database would not have been able to read anyone's actual password. This kind of basic security practice mattered given that the system held financial and business-sensitive information the shop would not have wanted exposed if something ever went wrong on the technical side.

Dashboard

Once logged in, the user landed directly on the dashboard, which functioned as the operational center of the whole system. This was the screen the owner was expected to check first thing in the morning and periodically throughout the day, and it was built specifically to answer the questions the owner used to have to dig through spreadsheets to figure out.

The dashboard showed daily revenue broken down into parts and labor, so the owner could see at a glance whether a given day's income came more from parts sold or services performed. It showed the monthly revenue trend, giving a longer view of how the business was doing over the current year rather than reacting to the natural swings of any single day. It listed the top-selling parts within a selected period, which was useful both for stocking decisions and for noticing when something suddenly started moving faster or slower than usual. It also showed a revenue split between parts and services, a breakdown of payments by cash, online payment, and credit card for the month, and, probably the most immediately useful piece, a color-coded low-stock alert panel that flagged any part sitting at or below its minimum threshold the moment the page loaded.

None of these figures required the owner to do anything to generate them. They pulled directly from the sales, sale_items, and product_stock tables in real time, so the numbers on screen reflected whatever had happened in the shop up to that exact moment, not whatever someone last remembered to update in a spreadsheet [4], [9].

Sales Recording Module

This was probably the single most frequently used part of the entire system, since it was involved in every transaction that happened at the shop. When a repair was finished, whoever was at the counter opened the sales form, entered the customer's name and plate number, and added line items for whatever parts were used and whatever labor fee applied. Parts and labor were entered separately. The sale_items table stored each line with a line_type marking it as either a part or a labor charge, which kept the revenue split on the dashboard accurate instead of lumping everything together into one number that did not tell the owner much.

A single transaction could include as many line items as the job actually needed. A repair involving a battery replacement and a set of spark plugs, for example, got logged as one sale with two separate part entries underneath it, both tied back to the same sale record. Once the form was saved, the system handled everything else automatically. It wrote the sale to the sales table, wrote each line item to sale_items, and logged a corresponding entry in inventory_transactions for every part involved, which was what actually drove the stock count down. Nobody had to remember to go update inventory separately. The moment the sale was saved, the inventory reflected it [4], [7]. Every transaction got an automatic, sequential reference number that kept counting up and never repeated.

To make part selection faster during a sale, the sales form also included a vehicle-brand filter that narrowed the parts list down to whatever brand the customer's vehicle actually belonged to. Instead of scrolling through the shop's entire catalog to find a compatible part, a staff member could select the vehicle brand (Toyota, Honda, Yamaha, and so on) from a dropdown, and the parts list on screen would immediately shrink to just the items tagged under that brand. This mattered more than it might have seemed on the surface, since several parts across

different brands shared similar or even identical names, and picking the wrong one during a busy counter interaction was one of the small but recurring mistakes the team noticed during the shop visits. The filter did not restrict what could ultimately be added to a sale. A staff member could still search or browse outside the selected brand if a job called for it. It simply made the common case faster by surfacing the most relevant parts first.

Inventory Monitoring Module

The inventory section was where staff could look up any part in the catalog and see its current stock level, searchable by name or category. What made this section meaningfully different from the spreadsheet it replaced was that the number on screen was always accurate to the moment [1], [2]. It was calculated directly from product_stock, which combined each product's base data with every inventory_transactions record tied to it: every sale, every restock, every adjustment.

Each part carried its own reorder_threshold, set individually rather than applied as one blanket number across the whole catalog, since a fast-moving item like a certain brake pad needed a different threshold than something that only got used occasionally. When a part's current stock dropped to or below its threshold, it got flagged with a color-coded badge, using the same color logic as the dashboard, so a staff member scanning the inventory page could immediately tell what needed attention without reading every row closely.

Restocking a part that was not tied to a formal reorder was also handled through this module. If a delivery came in, a staff member entered the quantity received, and the system logged it as a restock-type entry in inventory_transactions, which pushed the current_stock figure back up immediately.

Reorder Management Module

This module picked up where the low-stock alert left off. When the manager saw a part flagged on the dashboard or inventory page, they could open the reorder management section and see the same flagged items listed together, along with a recommended quantity based on the part's past movement [3], [5]. The manager reviewed this, adjusted the quantity if their own judgment indicated otherwise, and restocked the item in a single action. The system logged the reorder and updated stock immediately, with no separate pending or confirmation steps to manage. The low-stock flag cleared automatically once the part's count rose above its threshold. The system did not perform restocking actions on its own. The final restocking decision remained with the manager, allowing professional judgment to be applied whenever inventory requirements changed.

Expense Tracking Module

Not every cost the shop dealt with came from parts. Utilities, labor overhead, and other operational expenses got logged here, each with a category, description, amount, and date. These records lived in the expenses table, separate from the sales data, but they were pulled together with sales revenue whenever the owner generated a profit and loss report, giving a fuller financial picture than sales figures alone would have provided.

Reports Module

The reports section let the owner generate sales and inventory summaries for any period they needed: a single day, a full month, or a custom range. Because everything was already sitting in the database in an organized, connected form, generating one of these reports was just a matter of the system querying the sales, sale_items, and expenses tables and formatting the results, rather than someone manually opening several files and cross-checking them by hand. Reports could also be exported to CSV, which mattered for record-keeping purposes that went beyond just glancing at numbers on a screen [4], [9].

Forecasting Module

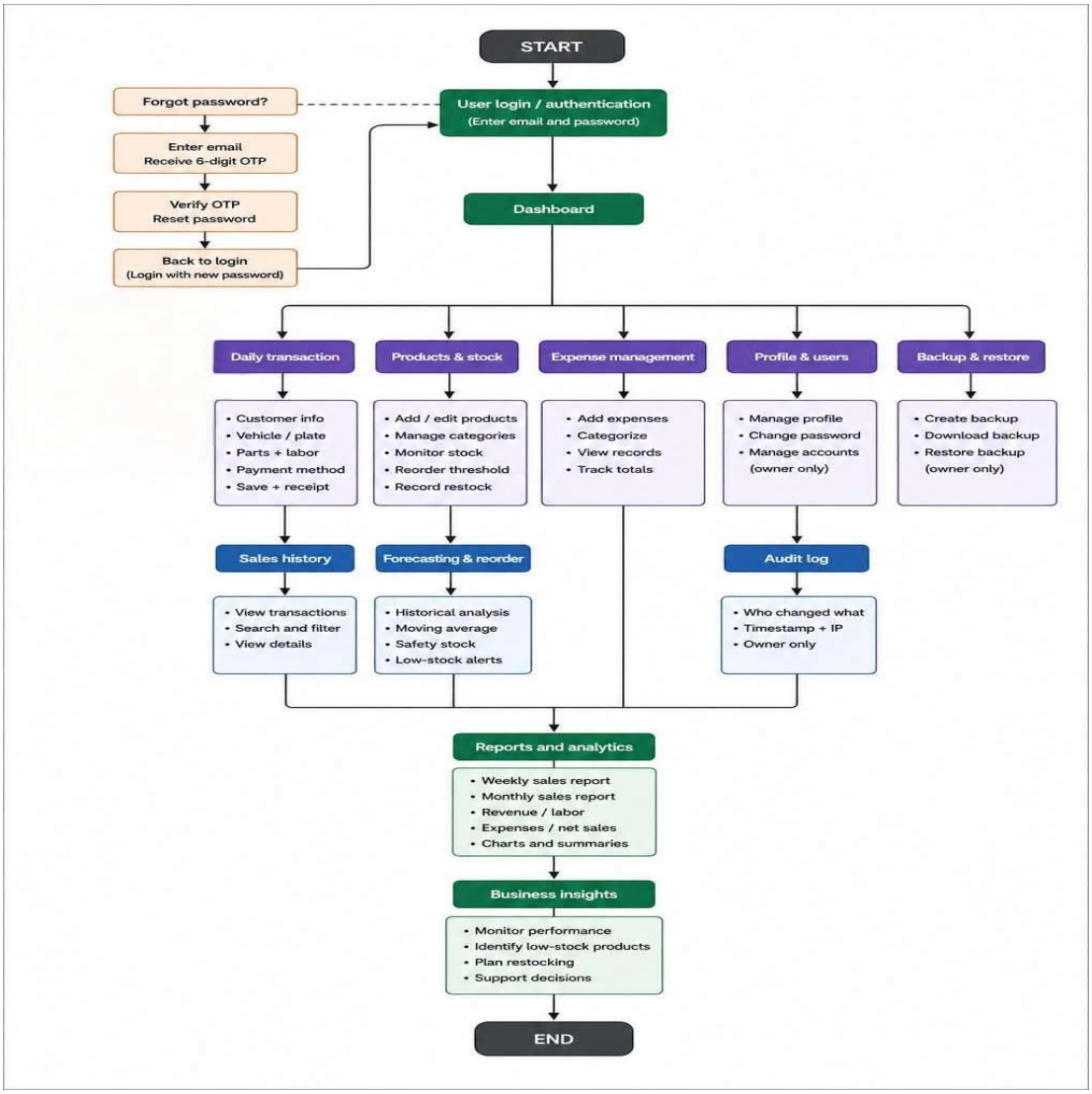
The forecasting tab looked back through a product's history in inventory_transactions and showed a simple demand trend for that item, indicating whether it had been moving faster, slower, or at roughly the same pace over recent weeks. This was not meant to replace the owner's own sense of the business. It was meant to give the owner a starting point grounded in actual numbers rather than memory alone. Chart.js turned this history into a visual line the owner could read at a glance, which mattered a lot more than it might sound, since a chart trending upward communicated the same information as a table of raw figures but in a form that took a couple of seconds to understand instead of a couple of minutes.

How the Modules Worked Together

Looking at these modules side by side, the thing that stood out was how much they depended on each other rather than functioning as separate, disconnected tools. A sale recorded through the sales module did not just sit in the sales table. It immediately affected inventory counts, which affected whether a low-stock badge showed up, which affected what appeared in the reorder management module, which eventually affected what showed up in the forecasting tab once enough history built up. None of that chain required a person to carry information from one place to another by hand. That was really the whole point of building the system this way instead of as a handful of separate spreadsheets: the connections between sales, stock, and planning were supposed to be automatic, not something a staff member had to remember to maintain every single day.

The sections that follow go into more detail on how the system was actually implemented, how it was tested against real conditions at D Speedway, and what the results of that testing looked like.

Figure 4.1 Illustrates the overall workflow of the developed system



System Workflow Diagram

4.3 System Development Methodology

The Smart Inventory and Parts Planning System was not simply coded without a clear process. If no defined process had been followed, the system risked being shaped by developer assumptions instead of actual business needs, which was something the project aimed to avoid from the start. Because of this, the Agile Software Development Methodology was used, an iterative approach built on constant collaboration, incremental development, and regular evaluation throughout the software development life cycle. This methodology was applied not only to the development of the system but also to its documentation, so that what was written reflected how the system actually worked rather than how it was originally planned. Agile was chosen mainly because the requirements kept changing as the project progressed, and it allowed for regular feedback from D Speedway. This was important since the project depended heavily on the shop's actual day-to-day operations rather than assumptions about how the business worked [7], [8].

Through Agile, the project was broken down into smaller parts called sprints. At the end of each sprint, a working part of the system was produced, tested, and reviewed before moving on to the next one. This made development less risky and allowed problems to be caught early, while they were still small and easy to fix, instead of being discovered later after more features had been built on an unstable base. Documentation was also reviewed and updated during each sprint so it would stay consistent with how the system actually functioned as development continued.

The methodology followed seven phases: Requirements Gathering, System Planning, System Design, Software Development, Testing, Deployment, and Maintenance. Each phase depended on the one before it, and the next sections explain what happened during each of these phases.

1. Requirements Gathering

This project started by figuring out what D Speedway actually needed, instead of assuming what a repair shop would need based on general knowledge or other systems that had been seen before. It would have been easy to skip straight to designing a system based on common features found in typical inventory software, but that approach would have risked building something that looked complete on paper while missing the specific problems D Speedway was dealing with. Because of this, the phase began with direct observation rather than assumption [3], [4].

This phase involved visiting the shop directly, observing how staff handled sales and inventory during a normal working day, and paying attention to the small, repetitive tasks that staff performed without necessarily thinking of them as problems, since those routine frustrations often revealed more about what the system needed to fix than a formal complaint would. Conversations were also held with the owner and staff about what frustrated them most about the current setup, including how stock levels were tracked, how sales were recorded, and how often mistakes or delays came up because of the fragmented way information was being stored.

No list of features was decided beforehand, and this was intentional. Instead, the visits and conversations were used to build that list from scratch, based on what was actually observed and said, rather than fitting the shop's situation into a template that had already been decided in advance. This was where problems like the scattered Excel files, unreliable stock counts, and the lack of any early warning before a part ran out were identified, along with smaller but still important issues, such as how much time was spent manually cross-checking spreadsheets just to confirm whether a part was actually in stock. Without this phase being done properly and thoroughly, everything built afterward would have been guesswork disguised as a solution, since a system designed around incorrect assumptions tends to solve problems that do not actually exist while leaving the real ones untouched. The output of this phase was a documented list of the shop's pain points and the features needed to address them. This list became the foundation for every decision made in the phases that followed, and it was referred to repeatedly whenever a design decision needed to be checked against what the shop had actually asked for.

2. System Planning

Once a clear picture of what D Speedway needed was established, the next step was turning that understanding into an actual project plan: what would be built, in what order, using what tools, and within what timeframe. This phase focused less on the shop itself and more on organizing the project internally so that development would not lose direction or drift away from the original goals once the actual building of the system started, which was a real risk given how easy it is for a project to expand in scope once development is underway.

Planning covered several practical decisions, each of which had consequences for how smoothly later phases would go. One of the first was choosing Agile as the overall development approach instead of a rigid, linear one, since requirements were expected to shift once real users interacted with early versions of the system, and a rigid approach would have made it difficult to adjust course without essentially restarting. Another major decision involved settling on the technology stack (PHP, MySQL, and a browser-based frontend) based on what could run reliably on the shop's existing computer without needing special installations, additional hardware, or ongoing technical support that the shop would not have easily been able to manage on its own. Sequencing the work was also decided during this phase, so that foundational features like sales recording and inventory tracking were built before features that depended on them, like forecasting, since attempting to build forecasting without reliable transaction data already being captured would have produced a feature with nothing meaningful to work from.

This phase also involved being clear and honest about what would not be included in this version of the system, since attempting to include everything the shop might eventually want would have stretched the project beyond what could realistically be delivered within the available time. Defining these boundaries early, rather than letting scope expand gradually as development went on, helped keep the project realistic and manageable, and gave the team a clear reference point whenever a new request came up later that fell outside what had originally been planned.

3. System Design

With the plan in place, the system's structure was designed before any real code was written, since attempting to build features without first working out how they would fit together tends to produce a system that is difficult to maintain and prone to inconsistencies. This phase turned the requirements gathered earlier into something concrete enough to actually build from: a database schema, an overall system architecture, and a rough layout for how the interface would look and function.

The system was designed around a three-layer architecture, separating the interface that users would interact with, the underlying logic that processed and handled data, and the database that stored everything permanently. That way, a change made in one part of the system would not risk unexpectedly breaking something unrelated in another part. This separation also made it easier to test and adjust individual pieces of the system later without having to touch everything else at the same time. The database design went through several drafts before settling into the nine interconnected tables described later in this study. Getting the relationships between parts, transactions, and inventory records right early on mattered significantly more than it might have seemed at first glance, because a mistake in the database structure tended to cause problems that were considerably harder to

fix once real data had started flowing through it, sometimes requiring extensive rework that could have been avoided with more careful planning upfront.

The interface was also sketched out during this phase, with particular emphasis on keeping things simple enough for staff who were not particularly comfortable with technology. This concern came up repeatedly during requirements gathering and shaped many of the smaller design decisions made throughout this phase, such as minimizing the number of steps needed to complete common tasks and avoiding unnecessary technical language in the interface itself.

4. Software Development

This was the phase where the system began to exist as actual working software instead of diagrams, schemas, and planning documents. Development followed the Agile approach established during planning, moving in short cycles that each produced a specific, functioning piece of the system. The team did not attempt to build the entire system at once and hope that all the individual pieces would come together smoothly at the end, an approach that tends to leave problems undiscovered until much later in the process.

Development started with the most foundational features, recording a sale and automatically updating inventory, since nearly everything else in the system depended on that data being captured accurately and consistently from the very beginning. From there, development moved outward to the dashboard, the low-stock alerts, the reorder management tools, and finally the forecasting feature. Forecasting required a meaningful amount of real transaction history in order to be useful, so it was deliberately built last rather than earlier in the process. Each feature was reviewed and adjusted as it was built, rather than being finalized immediately and left untouched. Feedback gathered from informal check-ins with the shop owner and staff occasionally required revisiting features that had technically already been completed. This was exactly the kind of flexibility Agile had been chosen to allow for in the first place, rather than treating an earlier decision as permanent simply because it had already been implemented.

5. Testing

A system that had been built but not properly tested was not truly a finished system in any meaningful sense. It was simply code that had not yet failed in front of anyone, and there is an important difference between the two. Testing was treated as a continuous part of development rather than something reserved entirely for the end of the project, and it occurred in layers, with each layer checking the system from a different angle and catching different kinds of problems.

Unit testing checked that each individual feature worked correctly on its own, using both normal, expected inputs and deliberately unusual or invalid ones, in order to observe how the system responded when something did not go exactly as planned. Integration testing checked that all of those individually working features behaved correctly once connected together as a complete system, since two features that each worked fine in isolation could still cause unexpected problems once they were made to interact with one another, particularly around shared data like inventory counts. Finally, User Acceptance Testing brought the system directly to the people

who would actually be using it day to day, the owner and staff at D Speedway, who tested real tasks under real, everyday conditions and gave honest feedback about what felt confusing, what felt slow, and what actually worked the way it was supposed to. Only after a feature had passed through all three of these layers was it considered genuinely finished, rather than simply appearing finished based on a quick initial check.

6. Deployment

Once the system had been built and tested, this phase covered the process of actually installing it and getting it running within the shop's real, day-to-day environment, rather than leaving it sitting on a development machine where nobody at D Speedway could use it. Deployment involved setting the system up on the shop's existing computer, migrating whatever historical data made sense to carry over from the old spreadsheets so the system did not have to start completely from scratch with no existing records, and walking the owner and staff through the system directly, so that the first time they used it for an actual transaction was not also the very first time they had ever seen it in action.

This phase also intentionally included a short adjustment period, during which staff used the system for real transactions while still keeping the old spreadsheet method available on the side, just in case something unexpected came up during those first few days of real use. This overlap was not included because of any doubt that the system would work, but because a business could not reasonably afford to have its records interrupted or lost even for a single day if something unforeseen went wrong. Once confidence had been established that the system was holding up reliably under real, everyday use, the shop was able to rely on it fully without needing the spreadsheet as a backup any longer.

7. Maintenance

Maintenance was less about a single event and more about an ongoing responsibility that continued well after deployment had technically been completed. This included keeping regular backups of the database so that the shop's records would not be placed at risk if something ever went wrong with the system itself, checking in periodically to see whether anything needed adjusting as the shop's own day-to-day operations naturally changed over time, and remaining available to fix bugs or make small improvements based on how the system actually performed once it was being used for real, day after day, rather than only during a controlled testing environment.

New parts would inevitably be added to the shop's catalog over time, inventory requirements might change, or the owner might, after spending more time using the system, request a small adjustment to how a particular report was presented. Building in a plan for this kind of ongoing support from the very beginning, rather than treating deployment as the finish line of the project, was what kept the system from becoming just another tool that worked well for a few weeks and then quietly fell out of use once something inevitably needed fixing. Because the system was still being presented as a prototype at the time of this study, full-scale, long-term maintenance fell partly outside the immediate scope of the project. The plan for it was still built into the overall methodology from the start, rather than being treated as an afterthought to be dealt with later.

Taken together, these seven phases gave the project a clear and connected path, starting from an early, uncertain understanding of D Speedway's problems and continuing all the way through to a working, tested system that the shop could actually rely on and continue relying on going forward. Each phase depended directly on the one before it, and skipping or rushing through any single phase would have weakened everything built afterward, regardless of how much effort went into the later stages. That connected, step-by-step structure was, in the end, what made it possible to build a system that was not just technically functional, but genuinely useful to the people it was built for.

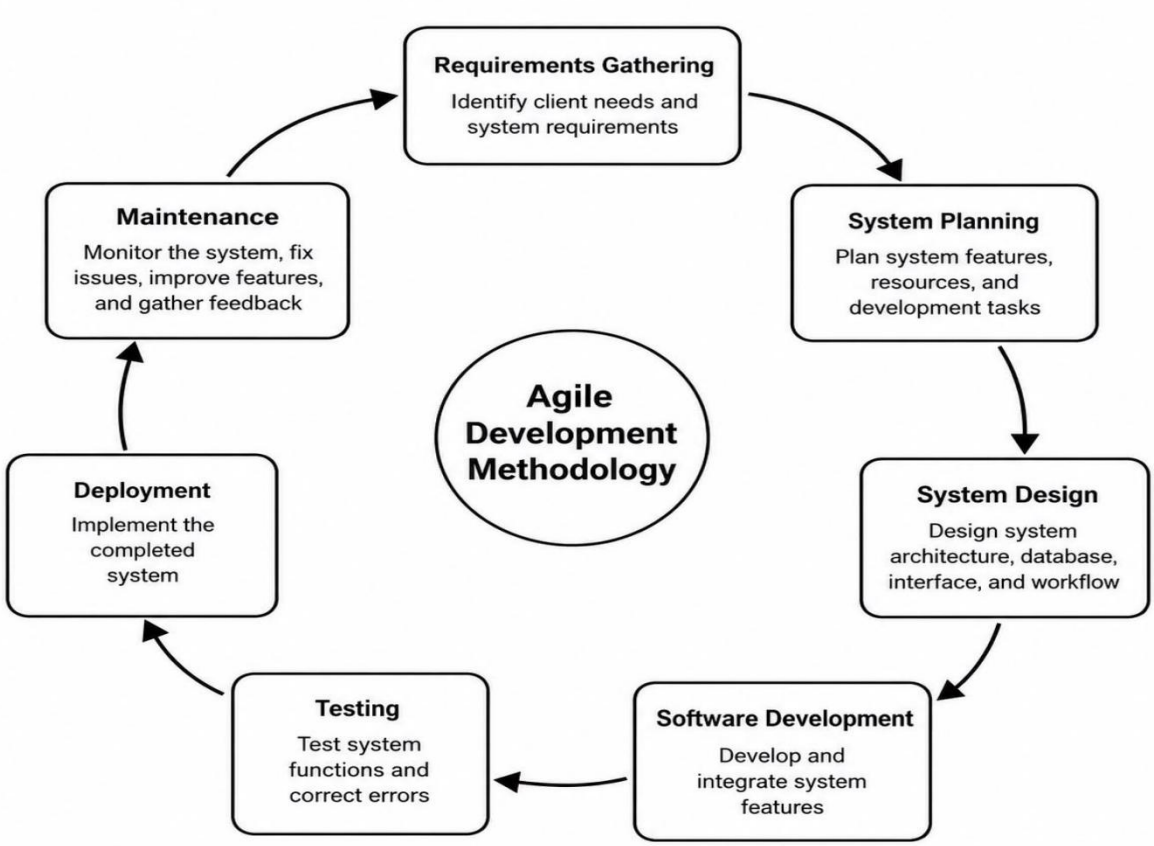


Figure 4.2 Agile Methodology Diagram

4.4 System Architecture

The Smart Inventory and Parts Planning System was built around a three-layer architecture, splitting the system into a presentation layer, an application layer, and a data layer. The team went with this structure for a simple reason: it kept the different parts of the system from being tangled up with each other. A change made to how

something looked on screen did not risk breaking the logic underneath it, and a change to the database did not force the whole interface to be rebuilt. For a system that was meant to keep running at D Speedway long after this study was finished, that kind of separation mattered more than it might have seemed at first.

This decision actually traced back to something the team noticed while still gathering requirements. The shop’s old spreadsheet setup did not really have layers at all. The file someone typed numbers into was the same file that stored the data and the same file that got read whenever someone needed a report. That was part of why one small mistake, like a wrong number entered in the wrong cell, could throw off everything downstream. Splitting the new system into three separate layers was the team’s way of making sure that kind of tangled, all-in-one setup would not happen again, even as more features got added over time.

It also made the system easier to work on as a team. Since the three layers did not depend on each other’s internal details, one member could be fixing something in the interface while another was adjusting how the database handled a certain table, without the two of them stepping on each other’s work. That turned out to matter more than expected once the project moved from planning into actual coding, since five people were touching the same codebase throughout development.

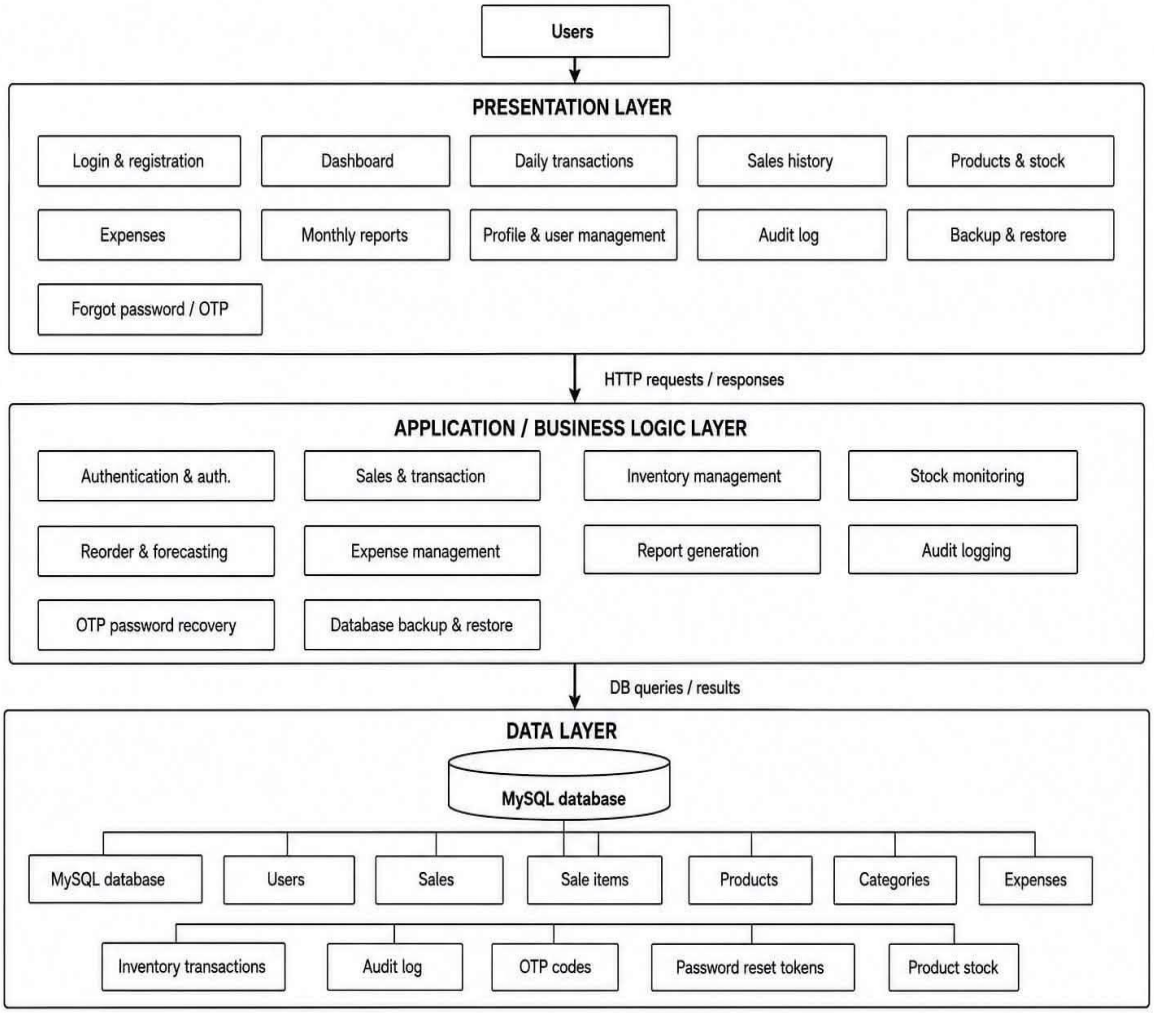


Figure 4.3 Architecture Diagram

Presentation Layer

This was the layer the owner and staff actually saw and clicked through. Nothing here touched the database directly. Every screen in this layer existed to collect input from a person or display information back to them in a way that made sense at a glance.

Login and Registration

This was the very first screen anyone saw when they opened the system. Staff typed in their email and password here before they were let into anything else. Registration existed mostly for the owner to bring new staff accounts into the system without having to touch the database by hand, something that used to require manually editing a spreadsheet and hoping nobody typed a name wrong. If the credentials did not match what was on file, the person simply did not get past this screen, which was the first and simplest layer of protection for the shop's data.

Dashboard

This was the landing page after login, and it was designed to answer the question "how's the shop doing right now?" without anyone having to dig through a report. It pulled together revenue for the day, which parts were selling the most, and which parts were running low, all in one glance. The idea came directly from watching how the owner used to start every morning, flipping through several spreadsheet tabs just to get a rough sense of where things stood. The dashboard collapsed that into one screen.

Daily Transactions

This was the form staff filled out every time a sale happened, whether it involved a part, labor, or both together. It was arguably the most-used screen in the whole system, since every single transaction in the shop started here. Keeping this form simple mattered a lot. It only asked for what was actually needed to complete a sale, because a staff member filling this out with a customer standing at the counter did not have time to click through unnecessary fields.

Sales History

Once a sale was recorded through Daily Transactions, it did not disappear. It landed here. This screen let staff or the owner look back through past transactions, search for a specific sale, or double check what was sold on a given day. It existed mainly to replace the shop's old habit of flipping through paper receipts or old spreadsheet rows to answer a simple question like "did we sell that part last week?"

Products and Stock

This was where the shop's full catalog of parts lived on screen: names, categories, prices, and how many of each part were currently on the shelf. Staff used this constantly, not just to check numbers but to search for a part before promising a customer it was in stock. This screen was one of the most requested features during the shop visits, since walking to the storage area just to check a number was one of the exact inefficiencies the team kept running into.

Expenses

This was where operational costs that were not tied to parts sales got logged, things like rent, utilities, or supplies. It was kept as its own separate screen because mixing expenses into the sales records was one of the messier habits from the old spreadsheet system, and separating the two made both easier to track accurately.

Monthly Reports

This was where sales and inventory summaries could be generated and exported, usually by the owner at the end of the month. Instead of manually adding up numbers from several files, this screen pulled everything together automatically. It existed specifically because the owner used to spend a noticeable chunk of time every month just compiling numbers that the system could now put together in seconds.

Profile and User Management

This screen let a user manage their own account details, and for the owner, it doubled as a place to manage staff accounts and roles. It was a smaller screen compared to the others, but it was what kept account-level control in the owner's hands instead of requiring a developer to step in every time a staff member's access needed to change.

Audit Log

This screen gave the owner a read-only view of every significant action taken inside the system: who added a product, who deleted a sale, who changed a price, and exactly when it happened. It existed so that if something ever looked wrong, the owner had an actual record to check instead of having to ask around or guess. Only the owner could access this screen.

Backup and Restore

This screen let the owner create a full backup of the system database at any time, download that backup file for safekeeping, or restore the system from a previous backup if something went wrong. It was included specifically because losing the shop's records to a crash or a corrupted file was a real risk for a business that had relied entirely on local spreadsheets before, and having a working restore option meant that risk was no longer a single point of failure.

Forgot Password / OTP

This was the screen a user landed on when they could not remember their password. Instead of a reset link that could be ignored or expire before being clicked, the system sent a 6-digit one-time code to the user's registered email address. The user typed that code into this screen, verified it, and was then allowed to set a new password. The code expired after ten minutes to prevent it from being misused if the email was left open.

Application / Business Logic Layer

This was where the actual thinking happened. Nothing the user did on the presentation layer ever reached the database directly. It all passed through here first, which was what made sure one action, like saving a sale, correctly turned into an updated stock count, a possible low-stock flag, and an updated dashboard number, all without the user doing anything extra.

Authentication and Authorization

This was the logic behind the Login screen. It checked whether the typed-in credentials actually matched an account, and if they did, it decided what that account was allowed to see and do next. The owner's account could reach everything, while a staff account was limited to the day-to-day screens they actually needed. This distinction mattered because not every piece of shop data was meant to be equally open to everyone using the system.

Sales and Transaction Processing

This was the logic that ran the moment a staff member hit save on the Daily Transactions form. It recorded the sale, broke it down into its individual line items, and made sure every part of that transaction tied back correctly to the same sale record. If someone tried to sell more of a part than the shop actually had, this layer also rejected the transaction outright instead of letting the stock count slide into a negative number.

Inventory Management

This handled everything related to keeping the product catalog accurate: adding new parts, editing existing ones, and organizing them into categories. It was kept separate from Stock Monitoring because managing what a part was (its name, price, category) was a different job from tracking how many of it were currently on hand.

Stock Monitoring

Every time a sale went through, this was the logic that immediately subtracted the right quantity from the right part and checked the updated number against that part's low-stock threshold. If a part dropped to or below its threshold, this was what triggered the alert that eventually showed up on the Dashboard. This entire sequence happened the instant a sale was saved, without the staff member doing anything beyond hitting the save button.

Smart Reorder and Forecasting

This layer looked at past sales activity to estimate which parts were likely to run out soon and suggested reorder quantities based on that history. It was the piece of the system that tried to get ahead of a stockout instead of just reacting to one after it happened, which was something the old spreadsheet setup had no way of doing at all.

Expense Management

This was the logic behind the Expenses screen. It validated entries, categorized them, and kept them organized separately from sales data so the two never got mixed together the way they sometimes did in the shop's old files.

Report Generation

This was what actually built the Monthly Reports the owner saw on screen. It pulled sales, inventory, and expense data together, organized it, and formatted it so nothing had to be compiled by hand. It was essentially the engine behind a screen the owner used but never had to think about the mechanics of.

Audit Logging

This quietly kept a record of significant actions taken in the system: who changed what, and when. It did not show up on the staff-facing screens directly, but it existed so that if something ever needed to be traced back (a

stock number that looked off, a record that got edited), there was an actual trail to follow instead of everyone just guessing.

OTP Password Recovery

This was the logic that ran whenever a user requested a password reset. It generated a secure 6-digit one-time code, stored it with a ten-minute expiry, and sent it to the user's registered email address through Gmail SMTP using PHPMailer. Once the user submitted the correct code on the Forgot Password screen, this layer verified it, marked it as used, and allowed the password update to proceed. If the code had already expired or been used, the request was rejected outright.

Database Backup and Restore

This handled the creation, storage, and restoration of full database backups. When the owner triggered a backup, this layer ran a complete export of the MySQL database, compressed it, and saved it to the backups folder on the server. When a restore was requested, it decompressed the chosen backup file and fed it back into the database. Old backups beyond the configured retention period were automatically cleaned up so the folder did not grow indefinitely.

Data Layer

This was where everything the shop needed to remember actually lived, built on a single MySQL database. Only the Application Layer was ever allowed to talk to this layer directly. A staff member using the interface could never reach into the database itself, accidentally or otherwise.

MySQL Database

This was the database engine that held every table listed below. It was chosen specifically because it could handle relationships between tables reliably, which mattered a lot once the team decided that a piece of information, like a part's price, should only ever live in one place instead of being copied across multiple files the way it was under the old spreadsheet system.

Users

This table stored login credentials and the role tied to each account. Passwords were never stored as plain text here. They were hashed, so even direct access to this table would not have exposed anyone's actual password.

Sales

This table held the header-level record of every transaction: the date, the total, and which staff member processed it. Every Sale Item, further down, tied back to a record in this table.

Sale Items

Where Sales held the overall transaction, this table held the individual line items inside it, meaning the specific parts or labor that made up that one sale. Breaking it apart this way was what made it possible to answer a question like "how many of this specific part have we sold this month" without having to dig through every full transaction by hand.

Products

This was the shop's full parts catalog, with one single, authoritative record for each part's name, price, and details. Every other table that needed to reference a part pointed back here instead of storing its own copy, which was exactly the fix for the mismatch problem the old spreadsheet files used to have.

Categories

This table organized the Products table into groups, making it easier to browse or search the catalog by type instead of scrolling through every part the shop carried.

Inventory Transactions

This was a running log of every single stock change, whether it came from a sale, a restock, or a manual adjustment. It was what made the current stock number trustworthy, since every change to it was recorded somewhere instead of just being overwritten silently.

Expenses

This table held the operational cost entries logged through the Expenses screen, kept separate from sales data so the two categories of spending and revenue never got tangled together.

Audit Log

This was where the Audit Logging service actually stored its records: a persisted history of notable actions taken across the system, kept for accountability if something ever needed to be traced back.

Password Reset Tokens / OTP Codes

This table stored the 6-digit one-time codes generated when a user requested a password reset through the Forgot Password screen. Each code was tied to a specific user account and expired after ten minutes. Once a code was successfully used to verify identity and reset a password, it was marked as used and could not be submitted again.

Product Stock

This was a computed view derived from the inventory_transactions table that always reflected the current on-hand quantity for each product. Rather than storing a stock number directly on the product record, the system calculated it dynamically from the full movement history, which made the number trustworthy because every change that affected it had a corresponding transaction record to back it up.

The Connections Between Layers

HTTP Requests/Responses: This was the back-and-forth between the Presentation Layer and the Application Layer. Every time a staff member clicked a button or submitted a form, a request traveled down to the application logic, and a response traveled back up with whatever the screen needed to display.

DB Query/Results: This was the back-and-forth between the Application Layer and the Data Layer. The application logic sent a query down to MySQL, and the database sent the matching results back up, which the application layer then shaped into something the presentation layer could actually show.

Keeping these three layers separate was really the core idea behind the whole design. The presentation layer dealt with what the user saw and touched, the application layer dealt with what needed to happen behind that, and the data layer dealt with remembering all of it accurately over time. A change or a mistake in one layer did not automatically put the other two at risk, which was exactly the kind of stability a small shop still growing into its own systems needed from something it was going to depend on every day [1].

4.5 Hardware and Software Requirements

The development and implementation of the Smart Inventory and Parts Planning System required suitable hardware and software to ensure that the system functioned properly during daily operations. These resources supported the system in processing sales transactions, updating inventory records, generating reports, monitoring stock levels, and producing inventory forecasts. Selecting the appropriate hardware and software also helped maintain the system's speed, accuracy, and overall reliability while it was being used by the staff and owner of D Speedway.

The recommended specifications were chosen to provide a stable working environment throughout the development and testing phases of the project. They also ensured that the system could handle its intended functions efficiently without requiring high-end computer equipment, making it practical and appropriate for the operational needs of D Speedway.

4.5.1 Hardware Requirements

Component	**Minimum Specification**	**Recommended Specification**
Processor	Intel Core I3/AMD Ryzen 3 (dual-core, 2.0 GHz)	Intel Core i5/ AMD Ryzen 5 or higher (quad-core, 2.5 GHz or higher)
Memory (RAM)	4 GB	8 GB
Storage	128 GB HDD/SSD	256 GB SSD or higher
Monitor	1366 x 768 Resolution	Ful HD (1920 x 1080) or hgher)
Internet Connection	Not Required(runs on localhost	10 Mpbs or higher (for Gmail SMTP email delivery)
Printer (Optional)	Any USB or Network Printer	Inkjet or Laser Printer for receipt Printing

`The hardware and software requirements were identified to ensure that the Smart Inventory and Parts Planning System operated efficiently and reliably during development and daily use. These specifications supported the system's core functions, including sales recording, inventory management, report generation, and forecasting. By using the recommended hardware and software, the system was able to provide stable performance while remaining practical and cost-effective for D Speedway's operational needs.

4.5.2 Software Requirements

Software	Purpose
Windows 10 / 11	Operating system used to run XAMPP, the development environment, and the web browser.
XAMPP	Local development environment that bundled Apache HTTP Server, PHP, and MySQL.
Apache HTTP Server	Web server that hosted and served the web application during development and testing.
PHP	Server-side scripting language used to develop the system's core functions and business logic.
MySQL	Relational database management system used to store and manage all system data.
PHPMailer	Library used to send OTP verification and password recovery emails through Gmail SMTP.
Composer	Dependency manager used to install and maintain PHP libraries, including PHPMailer.
Bootstrap 5	Front-end CSS framework used to create a responsive and user-friendly interface.
Bootstrap Icons	Icon library used to enhance the system's graphical user interface.
Chart.js	JavaScript library used to generate charts for inventory trends and seasonal demand forecasting.
SweetAlert2	JavaScript library used to display interactive alerts, notifications, and confirmation dialogs.
Google Fonts	Collection of web fonts used to improve the readability and appearance of the user interface.
Visual Studio Code	Source code editor used for developing, editing, and maintaining the system.

Overall, the selected software and technologies provided the necessary tools for designing, developing, and testing the Smart Inventory and Parts Planning System. Each application and framework played a specific role in supporting the system's functionality, from managing the database and processing business logic to creating a responsive user interface and generating reports. Together, these technologies contributed to the successful implementation of the system and ensured that it met the operational requirements of D Speedway.

4.6 Database Design

The database served as the core component of the **Smart Inventory and Parts Planning System** by storing and organizing information related to sales transactions, inventory records, user accounts, forecasting data, and reorder recommendations. It allowed the system to manage and retrieve data efficiently, ensuring that information remained accurate, consistent, and readily available whenever needed during the shop's daily

operations. Since the system relied on real-time inventory updates and automated report generation, a well-structured database was essential in supporting these processes.

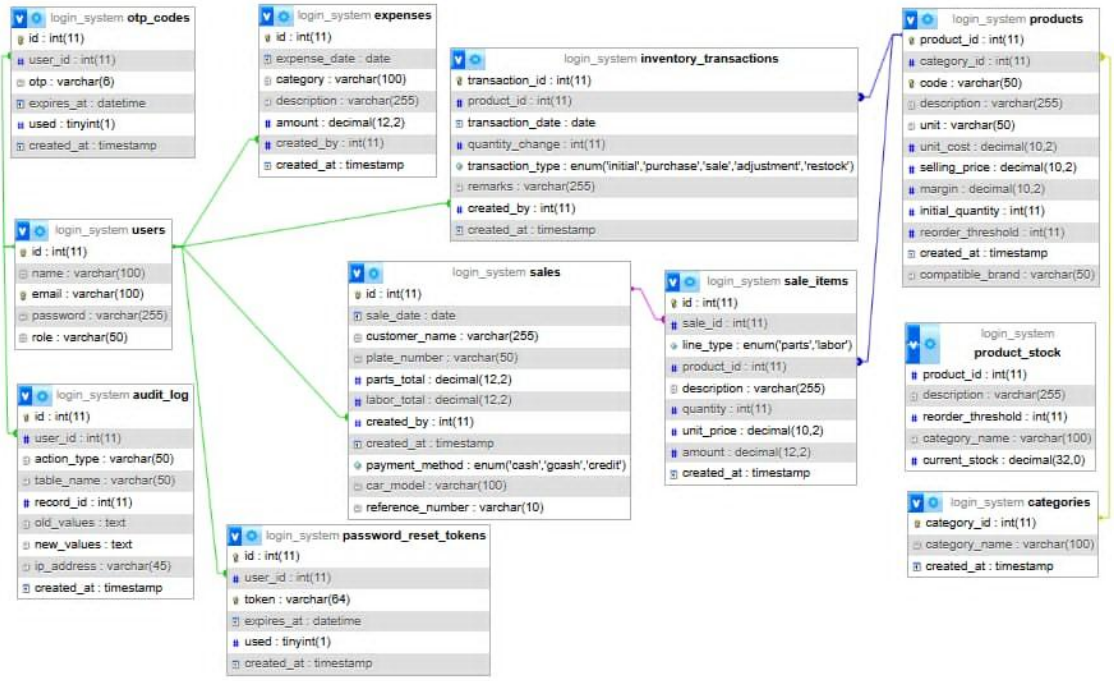


Figure 4.4 Entity Relationship Diagram

Users Table

It stored information about system users such as names, emails, passwords, roles and account creation dates. It was used to manage user accounts and access permissions.

Field	Description
id	Unique identifier assigned to each user account.
name	Full name of the user.
email	User's login email address. Each email must be unique.
password	Encrypted password secured using the bcrypt hashing algorithm (cost factor 12).
role	Specifies the user's access level and permissions within the system.
created_at	Date and time when the user account was created.

Categories Table

It stored the different categories of products offered in the system. It was used to group products into categories like Engine Oils and Spark Plugs.

Field	Description
category_id	Unique identifier assigned to each product category.

Field	Description
category_name	Name of the product category (e.g., Engine Oils, Spark Plugs).

Products Table

It held detailed information for every product including category, code, description, unit, cost, selling price, quantity and reorder level. It was used to monitor inventory and spot items that required restocking.

Field	Description
product_id	Unique identifier assigned to each product.
category_id	References the category to which the product belongs.
code	Optional internal product code or Stock Keeping Unit (SKU).
description	Full name and description of the product.
unit	Unit of measurement (e.g., Gallon, Liter, Piece, Box).
unit_cost	Purchase cost of the product per unit.
selling_price	Selling price of the product charged to customers.
initial_quantity	Initial stock quantity recorded when the product was first added to the inventory.
reorder_threshold	Minimum stock level that triggers a low-stock alert and adds the product to the reorder list.

Transactions Table

It logged all inventory movements such as stock additions, sales and deductions. It was used to track quantity changes and keep inventory records accurate.

Field	Description
id	Unique identifier assigned to each inventory movement record.
product_id	References the product associated with the inventory transaction.
transaction_date	Date when the inventory transaction occurred.
quantity_change	Quantity added to or deducted from the inventory. Positive values represent stock additions (e.g., initial stock or restocking), while negative values represent stock deductions (e.g., sales).
transaction_type	Indicates the type of inventory transaction or stock movement.
remarks	Optional notes or additional information related to the transaction (e.g., purchase, sale, or adjustment reference).
created_by	Identifies the user who recorded the inventory transaction.
created_at	Date and time when the inventory transaction record was created.

Sales Table

It kept details of completed sales including customer information, vehicle plate number, total parts cost, labor cost, payment method and the user who processed the sale. It was used to maintain complete sales records.

Field	Description
id	Unique identifier assigned to each sales transaction.
sale_date	Date when the sales transaction was completed.
customer_name	Name of the customer involved in the transaction.
plate_number	Vehicle plate number associated with the service or sale.
parts_total	Total amount charged for the purchased parts.
labor_total	Total amount charged for labor services.
payment_method	Payment method used to complete the transaction.
created_by	Identifies the user who recorded the sales transaction.
created_at	Date and time when the sales transaction was recorded in the system.

Sales Item Table

It listed each product, labor service or work included in every sale. It recorded the quantity, unit price and total amount for every line entry.

Field	Description
id	Unique identifier assigned to each sales line item.
sale_id	References the sales transaction to which the line item belongs.
product_id	References the inventory product for parts items; this field is **NULL** for labor or service items.
line_type	Indicates whether the line item represents a product (part) or a labor/service charge.
description	Name or description of the product or service included in the transaction.
quantity	Number of product units sold or hours of labor/service provided.
unit_price	Price charged per product unit or per hour of labor/service.
amount	Total amount for the line item, calculated by multiplying the quantity by the unit price.

Expense Table

It noted down all business outlays such as utilities, supplies and staff salaries. It included details like date, category, description, amount and the user who entered the record.

Field	Description
id	Unique identifier for each expense record.
expense_date	Date the expense was incurred.
category	Expense category (e.g., Utilities, Supplies, Salaries).
description	Brief description of what the expense was for.
amount	Amount of the expense in Philippine Peso.
created_by	User who logged the expense.
created_at	Timestamp when the expense was recorded.

OTP Codes Table

It saved one-time passwords issued for account or password recovery. It recorded the OTP value, assigned user, expiry time, usage status and creation time.

Field	Description
id	Unique identifier for each OTP record.
user_id	The user who requested the password reset.
otp	The 6-digit one-time password.
expires_at	Expiry timestamp; OTP is invalid after this time.
used	Flag set to 1 after the OTP has been verified.
created_at	Timestamp when the OTP was generated.

Audit Log

It tracked important database operations carried out by system users. It stored action type, affected table and record, old and new values, IP address and timestamp for reference.

Field	Description
id	Unique identifier for each audit record.

Field	Description
user_id	The user who performed the action.
action_type	Type of database action performed.
table_name	The database table that was affected.
record_id	The primary key of the affected record.
old_values	Snapshot of the record before the change (for UPDATE and DELETE).
new_values	Snapshot of the record after the change (for CREATE and UPDATE).
ip_address	IP address of the client that performed the action.
created_at	Timestamp when the audit entry was recorded.

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CHAPTER V

CONCLUSION AND RECOMMENDATION

5.1 Introduction

This chapter presents the summary of findings, conclusions, recommendations, impact of the study, and overall summary of the research entitled **"D Speedway: Automotive Repair: A Smart Inventory and Parts Planning System."** The study focused on the design, development, and evaluation of a web-based inventory and parts planning system intended to improve the inventory management practices of D Speedway. The system was developed in response to the challenges experienced by the business in managing inventory through manual spreadsheet-based recording, monitoring stock availability, forecasting future inventory requirements, and planning timely replenishment of automotive parts.

The primary objective of the study was to develop a reliable and efficient information system that could automate inventory-related processes while providing accurate and up-to-date information to support operational and managerial decision-making. Through the implementation of computerized inventory management, the system aimed to reduce manual work, minimize recording errors, improve inventory visibility, and provide business owners with a more organized approach to monitoring inventory movement. In addition, the system introduced forecasting and reorder planning features that allowed users to make informed purchasing decisions based on historical sales data rather than relying solely on experience or estimation.

Throughout the development process, the researchers carefully analyzed the existing workflow of D Speedway to identify the specific problems encountered in daily operations. The findings of this analysis served as the foundation for designing a system that directly addressed the business's operational needs. Every module included in the system was developed with the intention of simplifying inventory management while maintaining data accuracy and consistency. The researchers ensured that the system remained practical and appropriate for the scale and nature of the business while incorporating features that are commonly found in modern inventory management systems.

The implementation of the Smart Inventory and Parts Planning System demonstrated how information technology can contribute to improving the efficiency of small businesses. By replacing manual recording procedures with an integrated database, the system enabled inventory transactions to be recorded more accurately and efficiently. The automation of inventory updates, sales recording, and reorder planning reduced the amount of time required to perform routine tasks and allowed employees to focus more on customer service and daily business operations. These improvements also helped minimize the occurrence of stock discrepancies and delayed purchasing decisions that often resulted from outdated inventory records.

Aside from improving operational efficiency, the developed system also emphasized the importance of data-driven decision-making. The forecasting module analyzed previous sales transactions to estimate future inventory demand, enabling management to prepare for expected purchasing requirements. Rather than depending on assumptions or manually reviewing historical records, the forecasting feature generated useful information that could assist the business in maintaining adequate stock levels while avoiding excessive

inventory accumulation. The reorder recommendation feature further supported this objective by suggesting appropriate reorder quantities based on inventory status and forecasting results, thereby making procurement activities more systematic and organized.

To determine whether the objectives of the study were achieved, the completed system underwent evaluation using established software quality criteria. The evaluation assessed the system in terms of usability, functionality, accuracy, and performance to determine whether it satisfied the operational requirements of D Speedway. The evaluation involved intended users of the system, allowing the researchers to obtain practical feedback regarding the effectiveness, ease of use, and overall reliability of the developed application. The results of this evaluation provided evidence that the system was capable of supporting the daily inventory operations of the business while maintaining satisfactory levels of performance and user acceptance.

This chapter discusses the major findings obtained during the development and evaluation of the system. It explains how the objectives of the research were accomplished and how the implemented features contributed to improving inventory management and operational efficiency. It also presents the conclusions drawn from the findings, identifies recommendations for future enhancement and implementation, highlights the significance and impact of the study, and provides an overall summary of the research. Through these discussions, the researchers demonstrate that the Smart Inventory and Parts Planning System serves as a practical technological solution for addressing the inventory management challenges experienced by D Speedway while also providing a valuable reference for future studies involving intelligent inventory management systems.

5.2 Summary of Findings

The development and evaluation of the **Smart Inventory and Parts Planning System** produced results that demonstrated the successful achievement of the objectives established at the beginning of the study. The system addressed the major operational challenges encountered by D Speedway through the implementation of an integrated web-based inventory management solution that automated inventory monitoring, sales recording, forecasting, and reorder planning. Based on the design, development, implementation, and evaluation phases of the research, several significant findings were identified.

Centralized Inventory and Sales Management

One of the primary findings of the study was the successful replacement of the existing manual inventory management process with a centralized database-driven system. Prior to the implementation of the developed application, D Speedway relied on Microsoft Excel spreadsheets to record inventory transactions and sales activities. Although spreadsheets provided a basic method for storing records, they required manual updating and were prone to inconsistencies caused by human error. Maintaining separate files also made it difficult to ensure that inventory information remained current and synchronized across all business operations. [2], [4]

The developed **Smart Inventory and Parts Planning System** successfully consolidated inventory records, sales transactions, customer records, and product details into a single centralized database. This integration eliminated the need to maintain multiple spreadsheet files and significantly reduced the complexity of managing inventory information. Users were able to retrieve records more quickly, locate product information efficiently, and perform inventory-related tasks within a single application.

The centralized database also improved data consistency by ensuring that all users accessed the same updated information. Every inventory transaction performed within the system immediately reflected in the database, minimizing the possibility of duplicate entries and conflicting records. As a result, the business gained a more organized and reliable method of storing operational data while improving the overall efficiency of inventory management.

Furthermore, the availability of centralized information enhanced managerial decision-making. Since inventory records, sales data, customer records, and reorder information were stored in one location, management could easily review business performance and inventory status without manually compiling information from multiple sources. This improvement simplified daily operations and reduced the time spent searching for necessary records.

Automated Inventory Monitoring

Another significant finding of the study was the successful implementation of automated inventory monitoring. Under the previous manual process, inventory quantities had to be updated individually after every sale or purchase. This approach increased the likelihood of delayed updates and inaccurate stock records, particularly during busy business operations when multiple transactions occurred within a short period. [1], [2], [6]

The developed system automatically adjusted inventory quantities whenever inventory-related transactions were completed. Sales transactions immediately deducted the corresponding quantity from available inventory, while newly received products automatically increased stock levels. This automation ensured that inventory information remained current without requiring additional manual computation or record updating.

The implementation of automated inventory monitoring significantly improved inventory accuracy. Since stock levels were updated in real time, employees could easily determine product availability before accepting customer requests or placing purchase orders. The possibility of overselling unavailable products or overlooking depleted inventory was greatly reduced because current stock information was consistently maintained by the system.

The researchers also observed that automated monitoring minimized the occurrence of inventory discrepancies caused by forgotten updates, duplicate entries, or mathematical errors. By reducing dependence on manual inventory recording, the system provided more accurate information that supported daily operational activities and purchasing decisions.

Another advantage identified during the implementation was the reduction of administrative workload. Employees no longer needed to manually calculate remaining inventory after every transaction, allowing them to complete transactions more efficiently while dedicating more attention to customer service and other operational responsibilities.

Low-Stock Notification Feature

The implementation of the low-stock notification feature was another important accomplishment of the study. One of the major problems previously experienced by D Speedway was the delayed identification of products

that were already approaching depletion. Because inventory monitoring relied on manual inspection of spreadsheets, employees often became aware of low inventory only after customers requested unavailable parts.

The developed system addressed this issue by continuously monitoring inventory quantities and comparing them against predefined minimum stock levels. Whenever the available quantity of an item reached its assigned threshold, the system automatically generated a low-stock notification that alerted the user to replenish inventory.

This feature enabled management to identify inventory shortages before they became critical. Early notification provided sufficient time to perform restocking activities and replenish inventory before complete stock depletion occurred. Consequently, the business experienced fewer interruptions caused by unavailable automotive parts and was better prepared to meet customer demand.

The researchers found that the low-stock notification feature also improved inventory planning by allowing management to prioritize purchases according to inventory urgency. Instead of manually reviewing every product record, users could immediately identify items requiring immediate attention. This reduced the amount of time spent monitoring inventory while ensuring that purchasing decisions were based on accurate and updated inventory information.

In addition, the notification system supported better customer service by increasing product availability. Maintaining sufficient inventory reduced waiting times for customers and minimized delays in repair services that previously resulted from unavailable replacement parts. This improvement contributed to smoother business operations and increased operational reliability.

5.3 Conclusion

The findings of the study demonstrated that the **Smart Inventory and Parts Planning System** successfully achieved the objectives established by the researchers at the beginning of the project. The developed web-based system addressed the inventory management problems experienced by D Speedway by replacing the existing manual spreadsheet-based process with a centralized and automated information system. Through the integration of inventory monitoring, sales recording, forecasting, low-stock notifications, and reorder planning, the system provided a more organized and efficient approach to managing the business's daily operations.

Before the implementation of the developed system, D Speedway relied on manually updating inventory records using Microsoft Excel. Although this method allowed the business to maintain basic inventory information, it became increasingly difficult to manage as the number of transactions grew. Manual encoding required additional time and effort, increased the possibility of recording errors, and made it difficult to monitor inventory in real time. As a result, inventory discrepancies occasionally occurred, and management had limited access to accurate information when making purchasing decisions.

The implementation of the Smart Inventory and Parts Planning System successfully addressed these operational concerns by introducing automation into the inventory management process. Instead of manually updating spreadsheets after every transaction, inventory quantities were automatically adjusted whenever sales or stock replenishment activities were recorded. This automation ensured that inventory information remained current,

reduced the occurrence of human error, and improved the reliability of business records. The centralized database also provided a single source of information, making it easier for employees and management to retrieve inventory records, customer information, sales data, forecasting records, reorder records, and transaction histories whenever needed.

Another important achievement of the study was the successful integration of forecasting and reorder planning into the inventory management process. These features transformed historical sales data into useful information that supported purchasing decisions. Rather than relying solely on personal experience or estimating future inventory needs, management could review forecasted demand and reorder recommendations generated by the system. This allowed purchasing decisions to become more systematic and based on actual business data, reducing the likelihood of both stock shortages and excessive inventory accumulation.

The low-stock notification feature also contributed significantly to improving inventory control. By continuously monitoring stock quantities and comparing them with predefined minimum inventory levels, the system enabled management to identify products requiring replenishment before shortages affected business operations. This proactive approach reduced delays caused by unavailable automotive parts and improved the business's ability to provide timely repair services to customers. Maintaining adequate inventory levels also strengthened customer confidence by increasing the availability of commonly requested products.

The researchers also concluded that the automation of inventory-related activities contributed to greater operational efficiency. Routine tasks that previously required manual computation and repeated checking of spreadsheet records could now be completed more quickly using the developed application. Employees spent less time updating inventory records and searching for product information, allowing them to focus more on customer service and other essential business responsibilities. The reduction in administrative workload demonstrated how the application could improve productivity without increasing operational complexity.

The evaluation conducted during the study further confirmed the effectiveness of the developed system. Based on the assessment of usability, functionality, accuracy, and performance, the respondents found that the application met the operational needs of D Speedway. Users were able to perform inventory transactions, generate reports, monitor stock levels, and access forecasting information through an interface that was easy to understand and operate. The successful evaluation indicated that the system was capable of supporting actual business operations while providing reliable and consistent performance.

In terms of functionality, the developed application successfully integrated the major processes involved in inventory management into a single system. Inventory monitoring, sales management, customer records, forecasting, and reorder planning operated together within one centralized database. This integration minimized data duplication, improved consistency across business records, and simplified the overall workflow of inventory management. The researchers concluded that the complete integration of these modules contributed significantly to the effectiveness of the system.

The study also demonstrated the value of using information technology to improve the operational practices of small and medium-sized enterprises. Many small businesses continue to depend on manual recording methods because they are inexpensive and familiar. However, the findings of this study showed that even a relatively simple web-based information system can provide substantial improvements in efficiency, data accuracy,

inventory visibility, and decision-making. By automating routine processes and organizing business information within a centralized database, the developed system illustrated how digital technologies can address common operational challenges without requiring highly complex or expensive solutions.

Furthermore, the research confirmed that inventory management extends beyond simply recording the movement of products. Effective inventory management also involves monitoring stock availability, identifying purchasing priorities, analyzing sales trends, forecasting future demand, and supporting timely procurement decisions. The Smart Inventory and Parts Planning System addressed each of these aspects by combining automation with decision-support features that enabled management to make more informed and consistent inventory-related decisions. This comprehensive approach strengthened the overall inventory management process and provided greater control over business operations.

The researchers therefore conclude that the Smart Inventory and Parts Planning System is a practical, reliable, and effective solution for addressing the inventory management challenges encountered by D Speedway. The successful implementation of automated inventory monitoring, forecasting, reorder recommendations, and centralized record management demonstrates that the objectives of the study were fully achieved. The developed application not only improved inventory accuracy and operational efficiency but also provided valuable information that supports sound business decision-making.

Finally, the study concludes that the developed system has the potential to contribute to the long-term growth and sustainability of D Speedway. Accurate inventory information, timely purchasing decisions, and efficient operational processes are essential factors that influence the quality of customer service and the overall performance of an automotive repair business. By providing these capabilities through an integrated web-based application, the Smart Inventory and Parts Planning System offers a practical technological solution that can support the continued development of the business. Moreover, the study serves as a useful reference for future researchers and developers who intend to design similar inventory management systems for small and medium-sized enterprises in other industries.

5.4 Recommendations

To further improve the effectiveness and sustainability of the Smart Inventory and Parts Planning System, the researchers proposed the following recommendations:

System Improvements

- A mobile application version should be developed to provide easier access for shop owners and staff.
- Barcode or QR code scanning should be integrated to speed up inventory transactions.
- Automated email or push notifications should be added for low-stock alerts, so the owner and manager would be informed immediately when a product fell below its reorder threshold without needing to check the system manually.
- Data security should be enhanced through the implementation of Two-Factor Authentication (2FA).
- Advanced sales analytics and graphical dashboards should be included to improve business monitoring.

Future Research Directions

- Future researchers might integrate Artificial Intelligence (AI) to improve inventory demand forecasting.
- A cloud-based version might be developed to allow remote access and data synchronization.
- Long-term studies might be conducted to evaluate forecasting accuracy over several years.
- Future systems might be expanded to support multiple repair shop branches and warehouse management.
- Further enhancements to the reorder recommendation feature might also be explored to improve the accuracy of inventory replenishment decisions through more advanced forecasting techniques.

Implementation Strategies

- User training should be conducted before full system implementation.
- Regular database backup and preventive system maintenance should be performed.
- System performance should be monitored, and user feedback should be collected for continuous improvement.
- Inventory records should be updated regularly to maintain forecasting accuracy.
- Policies on user access and data management should be established to ensure proper system utilization.

5.5 Impact of the Study

The completion of the **Smart Inventory and Parts Planning System** made a meaningful contribution to the field of Information Technology by demonstrating how a web-based information system can be utilized to address the operational challenges of a small automotive repair business. The study showed that information systems are not only applicable to large organizations but can also provide practical and cost-effective solutions for small and medium-sized enterprises that continue to rely on manual methods of managing business operations. Through the development and implementation of the system, the researchers demonstrated that digital technologies can improve efficiency, accuracy, and decision-making without requiring highly complex or expensive software.

One of the major contributions of the study was the successful application of information technology in automating inventory management processes. Prior to the implementation of the system, D Speedway managed inventory records using manually updated spreadsheets. Although this approach was sufficient during the early stages of the business, it became increasingly difficult to maintain accurate and up-to-date records as daily transactions increased. The developed system replaced these manual procedures with a centralized database that automatically recorded sales transactions, updated inventory quantities, monitored stock availability, and generated reports. This automation significantly reduced repetitive manual tasks while improving the consistency and reliability of business records.

The study also highlighted the practical value of integrating several inventory management functions into a single application. Rather than using separate tools for recording inventory, monitoring stock levels, and reviewing sales information, the Smart Inventory and Parts Planning System combined these processes within one web-based platform. This integration simplified daily operations, minimized duplicate data entry, and ensured that all information remained synchronized. As a result, management gained immediate access to accurate inventory data, allowing business decisions to be made using current and reliable information.

For **D Speedway**, the developed system provided several operational benefits that directly addressed the challenges previously encountered by the business. The automation of inventory monitoring improved stock accuracy by ensuring that inventory quantities were updated immediately after every completed transaction. This reduced the occurrence of discrepancies between recorded and actual inventory levels and minimized errors associated with manual data entry. Employees were able to determine product availability more quickly, enabling them to respond to customer inquiries with greater confidence and accuracy.

The implementation of the forecasting module also contributed positively to the business by improving inventory planning. Instead of relying primarily on observation or previous experience, management could analyze historical sales records and use the generated forecasts as a basis for purchasing decisions. This feature allowed the business to prepare for future inventory requirements more effectively, reducing the likelihood of stock shortages while preventing unnecessary overstocking of slow-moving products. Consequently, inventory resources could be managed more efficiently, supporting both operational continuity and financial planning.

The low-stock notification and reorder recommendation features further strengthened inventory management by enabling timely replenishment of automotive parts. Automatic notifications informed users whenever inventory levels approached predefined minimum quantities, allowing management to coordinate purchases before shortages affected repair services. The reorder recommendation feature provided suggested reorder quantities based on inventory status and historical sales information, making purchasing activities more organized and systematic. Together, these features reduced delays caused by unavailable parts and improved the overall availability of products required for vehicle repair and maintenance.

Another important impact of the study was the improvement in overall business productivity. Because many routine inventory tasks became automated, employees no longer needed to spend significant amounts of time updating spreadsheets, manually calculating stock balances, or reviewing inventory records. Administrative work was simplified, allowing personnel to focus more on customer service, repair operations, and other productive activities. The reduction in manual workload contributed to a more efficient working environment while decreasing the possibility of operational errors.

The study also demonstrated the importance of data-driven decision-making within small businesses. By collecting, organizing, and analyzing sales and inventory information, the developed system transformed routine operational data into meaningful information that supported managerial decisions. Features such as forecasting, inventory reports, and reorder recommendations enabled management to make purchasing decisions based on recorded business activities rather than relying solely on assumptions or personal judgment. This strengthened the decision-making process and promoted more consistent inventory management practices.

Beyond its contribution to D Speedway, the study served as a valuable reference for students, software developers, researchers, and future practitioners in the field of Information Technology. The research presented a practical example of how systems analysis, database management, web development, and inventory planning concepts could be integrated into a complete business information system. Future researchers might use this study as a foundation for developing similar systems for other industries or for introducing additional features such as mobile applications, cloud computing, machine learning, advanced forecasting techniques, or enhanced inventory analytics. The documentation of the development process, system architecture, implementation procedures, and evaluation results provided useful guidance for future academic and software development projects.

The study also contributes to educational institutions by providing an example of how classroom knowledge can be applied to solve real-world business problems. It demonstrates the practical application of concepts learned in systems analysis and design, software engineering, database management, and web application development. As such, the study may serve as a useful instructional reference for future Information Technology students undertaking capstone projects or conducting research involving business information systems.

Furthermore, the research supports the continuing digital transformation of local businesses by encouraging the adoption of information technology in everyday operations. Many small enterprises continue to depend on traditional methods because of limited financial resources or familiarity with manual processes. This study illustrates that practical, affordable, and customized information systems can significantly improve operational efficiency without requiring major technological investments. By demonstrating the benefits of automation in inventory management, the study encourages other small businesses to consider implementing similar technological solutions that improve accuracy, productivity, and service quality.

Ultimately, the impact of the study extends beyond the successful development of a single software application. It demonstrates that carefully designed information systems can improve business operations, strengthen inventory management practices, support informed decision-making, and contribute to the sustainable growth of small enterprises. Through the Smart Inventory and Parts Planning System, the researchers were able to show how Information Technology can provide practical solutions that address real operational challenges while promoting greater efficiency, reliability, and organizational effectiveness.

5.6 Summary

This study focused on the design, development, implementation, and evaluation of the **Smart Inventory and Parts Planning System** for D Speedway, an automotive repair business that experienced challenges in managing inventory using manual spreadsheet-based methods. The primary objective of the research was to develop a web-based information system capable of improving inventory management by automating inventory monitoring, sales recording, forecasting, and reorder planning while providing accurate information to support daily business operations and management decisions.

Throughout the study, the researchers carefully examined the existing inventory management process of D Speedway and identified several operational issues that affected the efficiency of the business. These included manual inventory updates, inconsistent record management, difficulty in monitoring stock availability, delayed

identification of low inventory, and the absence of forecasting tools to support purchasing decisions. These findings served as the basis for designing a system that directly addressed the operational needs of the business through the integration of automated inventory management features.

The Smart Inventory and Parts Planning System successfully automated the recording of sales transactions and inventory updates by utilizing a centralized database that maintained accurate and consistent business records. Every completed transaction immediately updated inventory quantities, eliminating the need for manual stock computation and reducing the occurrence of human errors. The centralized storage of inventory records, customer information, sales data, forecasting records, and reorder records also improved data organization and allowed users to retrieve business records quickly whenever needed.

One of the major accomplishments of the developed system was the integration of intelligent inventory planning features. The forecasting module analyzed historical sales records to estimate future product demand, allowing management to anticipate inventory requirements more effectively. Complementing this functionality, the reorder recommendation feature generated suggested reorder quantities based on current stock levels and sales trends, enabling purchasing decisions to become more systematic and data-driven. The low-stock notification feature further enhanced inventory control by informing users whenever inventory levels reached predefined minimum quantities, allowing timely replenishment before shortages affected business operations.

The evaluation conducted during the study confirmed that the developed application successfully satisfied its intended objectives. Based on assessments of usability, functionality, accuracy, and performance, the respondents found that the system was capable of supporting the operational requirements of D Speedway. Users were able to perform routine inventory tasks efficiently through an interface that was clear, organized, and easy to navigate. The evaluation results further demonstrated that the application produced reliable information, maintained accurate inventory records, and performed consistently during normal business operations.

The successful implementation of the Smart Inventory and Parts Planning System demonstrated the value of adopting information technology within small and medium-sized enterprises. The study showed that replacing manual inventory management with an automated information system can significantly improve operational efficiency, inventory accuracy, and managerial decision-making. By reducing repetitive administrative tasks and providing real-time access to inventory information, the system enabled employees and management to perform their responsibilities more effectively while improving the overall quality of business operations.

The researchers conclude that all objectives established for the study were successfully achieved. The developed application effectively addressed the inventory and sales management challenges experienced by D Speedway while providing practical features that improved inventory monitoring, forecasting, purchasing decisions, and record management. The positive evaluation results confirmed that the system is functional, reliable, accurate, and user-friendly, making it suitable for actual implementation within the business.

In addition to its practical contribution to D Speedway, the study also provided valuable insights for future researchers, software developers, and Information Technology students interested in developing business information systems. The methodologies, design principles, implementation strategies, and evaluation procedures presented throughout the research might serve as useful references for future studies involving inventory management, forecasting, decision support systems, and web-based business applications. Future

developments might build upon this research by incorporating emerging technologies such as mobile platforms, cloud-based deployment, artificial intelligence, advanced analytics, and enhanced inventory forecasting capabilities.

Overall, the **Smart Inventory and Parts Planning System** demonstrated that the effective application of Information Technology can significantly improve the operational performance of a small automotive repair business. By automating essential inventory processes, improving data accuracy, supporting informed decision-making, and enhancing operational efficiency, the developed system provides a practical solution to the challenges faced by D Speedway. The researchers believe that the continued implementation and future enhancement of the system will further strengthen inventory management practices, improve customer service, support sustainable business growth, and encourage greater adoption of digital technologies among small and medium-sized enterprises.

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